DEEP LEARNING CA3 · VIVA AND PRESENTATION PREP

CryptoGraph Viva Guide

Everything about CryptoGraph-Secure, built up from zero: what it does, how every part works, why each choice was made, what the numbers mean, and how to answer the hard questions. Plain words first; the technical words the professor will use are in **bold** so you recognise them, and every one is explained in Part 0.

30 coins	**1.46 M**	**TGN**	**0.205**
hourly Binance prices, 2020 to 2026	shock events the model learns from	main model, 70,021 parameters	test AUPRC at 1 hour (random: 0.105)

START HERE

How to use this guide

Read it in order once: each part builds on the one before. Then drill the question bank.

The guide is a staircase. **Part 0** explains every money word and every deep-learning word in plain language; start there if any word trips you up. Part 1 tells the whole story in a minute. Parts 2 to 5 explain the problem, the data and how we score a model, with no maths beyond fractions. Part 6 teaches the building blocks of neural networks from scratch, so that Part 7, the main model, makes sense. Parts 8 and 9 cover the other models and the results. Part 10 walks through the six notebooks, Part 11 explains every technique, and Part 12 goes through the report section by section, so nothing the professor points at is new. Then come the software, the dashboard and the limits. Part 17 is a bank of viva questions with model answers, and Part 18 is the one-page list of numbers to learn by heart.

Four kinds of boxes appear throughout:

SIMPLE VERSION

The idea as you would explain it to a 12-year-old. Start every answer like this.

SAY IT IN THE VIVA

A ready answer in the professor's language. Learn the meaning, not the exact words.

CAREFUL

A trap, a weak spot, or a question where an honest answer beats a confident one.

NUMBERS TO KNOW

Figures worth memorising. They are all collected again in Part 18.

A study plan for the next day

45 min	Part 0, the words. Read it slowly and say each example out loud.
1.5 hours	Parts 1 to 4: the story, the problem, the data, the question. Say the one-minute pitch out loud until it flows.
1.5 hours	Parts 5 and 6: how we score models (AUPRC, random level, bootstrap) and the building blocks (parameters, training, graphs, memory, attention).
2.5 hours	Part 7, the TGN. Draw its diagram from memory until you can explain each box, the parameter count and why we chose it.
1 hour	Part 8: every other model, and the answer to “why these models only?”.
1.5 hours	Part 9: results. Learn the main table and the five findings. Then Part 18.
2.5 hours	Part 10: the six notebooks. Open each notebook beside it; for each, be ready to say its question, three findings and one decision it led to.
1 hour	Part 11: read the “plain words” line of every technique; the rest only when a question needs it.
2 hours	Part 12: the report. Keep the PDF open beside it; practise equations 3, 4 and 7 and Tables III, V and VI out loud.

1 hour	Parts 13 to 15: the pipeline, the dashboard and the past-crises replays. Click through the deployed dashboard while reading Part 14.
2 hours	Part 17 in quiz mode, twice: groups A to I first, then J, K and L. Better: your teammate asks, you answer, then swap.
1 hour	Rehearse the demo flow at the end of Part 14 twice with a timer. Then sleep at least six hours: a rested brain answers better than a crammed one.

Short on time? Part 0, Part 1, the TGN diagram in Part 7, Part 9, the report tables in Part 12 and Part 18; then quiz yourself on question groups A, D, F and L.

PART 0

The words you need first

Read this before anything else (about 45 minutes). Every word is explained with an everyday example, in the order you need them: each word only uses words explained above it.

SIMPLE VERSION

This project sits where two worlds meet: **money** (crypto markets) and **machines that learn** (deep learning). Each world has its own jargon, and the professor will use both. Once these words feel normal, the rest of the guide reads like a story.

Four groups: A. Money and markets, B. Statistics, C. Machine learning and D. Deep learning. The grey line under a word says where it shows up in our project. Part 19 repeats every word from A to Z for quick lookup.

A. Money and markets

Coin (cryptocurrency)

Digital money bought and sold on the internet, such as Bitcoin (BTC) or Ethereum (ETH).

We follow 30 coins, plus LUNC and FTT for two famous crashes.

Exchange

A marketplace where coins are traded, like a stock market that never closes: open 24 hours a day, 7 days a week.

Binance, the biggest; we use its free public price records.

Stablecoin, USDT, peg and depeg

A stablecoin is a coin built to always be worth one US dollar; USDT ("Tether") is the biggest. "Pegged" means held at $1; a **depeg** is when it slips away from $1, a sign of panic.

Binance prices are in USDT; we flag the 4 days USDT slipped more than 0.5%.

Candle (OHLCV)

One hour of trading squeezed into five numbers: the price at the start (**open**), the highest (**high**), the lowest (**low**), the price at the end (**close**) and how much was traded (**volume**). Charts draw it like a little candle. Binance calls it a "kline".

59,088 hours of candles, from January 2020 to September 2026.

Return

How much the price changed, as a percentage: $100 to $103 is a return of +3%.

Log return

The same idea written with logarithms: ln(103 ÷ 100) = 0.0296, almost the same as 3%. Its advantage: hour-by-hour log returns simply add up to the day's log return, which plain percentages do not (+10% then −10% leaves you at −1%, not 0).

Every input starts from hourly log returns.

Volume and relative volume

Volume is how much was traded in an hour. Relative volume compares it with the coin's normal over the past week: "three times busier than usual".

The biggest single reason behind our alerts (about 25%).

Volatility

How jumpy a price is. A calm coin moves 0.3% in a typical hour, a jumpy one 2%. Measured as the typical size of recent returns (their standard deviation, Part B), which is called **realised volatility**.

Features rv_24h and rv_72h: jumpiness over the last 24 and 72 hours.

Liquidity, thinly traded

How easily you can trade without moving the price. In a thinly traded coin few trades happen, so its price updates late and can look as if it "follows" others.

QNT, the one coin that seemed to follow ETH.

Market capitalisation

Price × number of coins in existence: the coin's total size.

Our 30 coins are the biggest by market cap (CoinGecko's list) that trade on Binance with enough history.

Crash, pump and shock

A crash is a sudden big fall, a pump a sudden big rise. A **shock** is our word for an unusually big move in either direction *for that coin*, judged against its own recent jumpiness.

A move bigger than 2.29 times the coin's typical hourly move.

Correlation, co-movement

Whether two coins go up and down together. Crypto coins are strongly linked: when Bitcoin falls, most coins fall.

Average hourly correlation about 0.58 (1 = always together).

Market factor, systematic and idiosyncratic

The part of a move that all coins share ("the whole market moved") is **systematic**; the part that belongs to one coin alone ("its own news") is **idiosyncratic**.

The shared part is 64% of hourly movement.

Diversification

Spreading money over many coins so one bad coin hurts less. It only works if coins do not all fall together, and in crises they do.

Beta, "high beta"

How strongly a coin swings when the market swings. A high-beta coin rises and falls more than the market.

Contagion and interdependence

Contagion: trouble spreading from one coin to others, beyond their usual link, like a yawn going round a room. **Interdependence**: they always move together anyway. Forbes and Rigobon (2002) warned that the two are easy to confuse.

So we judge shocks against each coin's normal and only count shocks that come strictly later.

Spillover, connectedness, transmitter and receiver

Spillover: how much of one coin's surprise comes from other coins. Connectedness: how much spillover there is overall. A **transmitter** sends more than it receives; a **receiver** the opposite.

Measured with the Diebold–Yilmaz method (Part 11).

Systemic risk

The risk that trouble in one place brings down the whole system, like one bank's failure causing queues at every bank.

Leverage

Trading with borrowed money. With 10× leverage, a 10% move doubles your stake or wipes it out.

Futures, perpetual futures, funding rate, premium

A future is a contract to buy or sell later at a price agreed now. **Perpetual** futures never expire; to keep their price close to the real price, one side pays the other a small fee every few hours, the **funding rate**. The **premium** is how far the futures price sits above the real price.

We tried these as inputs; they made results worse.

Liquidation and liquidation cascade

When a leveraged trader's losses get too big, the exchange force-sells their position: a liquidation. Forced selling pushes prices down, which forces more selling, like dominoes: a cascade.

10 October 2025.

ETF

An exchange-traded fund: a fund sold on the ordinary stock market that holds an asset. The US approved spot Bitcoin ETFs on 10 January 2024, so ordinary investors could buy Bitcoin through their usual broker.

Sectors

Groups of coins by what they do. **Payments** (XRP, XLM, LTC: sending money); **smart-contract platforms** (ETH, SOL, ADA: shared computers that run apps); **DeFi** (AAVE, UNI: banking apps with no bank); **oracle** (LINK: brings outside data such as prices onto blockchains); **layer 2** (ARB, POL: faster, cheaper lanes built on top of Ethereum); **meme coins** (DOGE, SHIB, PEPE: driven by jokes and hype); **exchange token** (BNB, tied to Binance); **privacy** (ZEC: hides who paid whom); **identity / AI** (WLD).

A hand-made list of 9 sectors in sectors.csv.

Listing, delisting, ticker change

Listing: an exchange starts trading a coin; delisting: it stops. A ticker is the coin's short name; MATIC was renamed POL in 2024, swapped one for one.

FTT was delisted after FTX failed; XMR was delisted in February 2024, so we left it out.

Survivorship bias

Studying only the winners. We picked today's top 30, so coins that died along the way are missing, which can make the market look healthier than it was.

Risk desk, regulator, early warning

The people who would use this tool: an exchange's team that watches risk, a market supervisor, or a trader who wants a warning before trouble spreads.

UTC and hour of the week

UTC is world standard time (London time without summer time). An hour of the week is one of 168 slots, Monday 00:00 to Sunday 23:00.

Wednesday 12:00 UTC is the jumpiest slot, Sunday 06:00 the calmest.

The nine crises in the data, in one line each

Date	Crisis	What happened, simply
12 Mar 2020	COVID "Black Thursday"	Pandemic panic: everything was sold at once; Bitcoin lost roughly a third of its value in a day.
19 May 2021	China ban	China barred banks and payment firms from crypto services; prices crashed.
9 May 2022	Terra / LUNA	UST, a stablecoin held at $1 by an algorithm instead of cash, lost its peg; its sister coin LUNA went almost to zero within days.
13 Jun 2022	Celsius / Three Arrows	A big crypto lender froze customers' withdrawals, and a large crypto fund collapsed.
8 Nov 2022	FTX	The exchange FTX could not pay customers back; its own coin FTT collapsed and the exchange went bankrupt.
10 Mar 2023	SVB / USDC	Silicon Valley Bank failed; the stablecoin USDC kept cash there and briefly fell below $1.
10 Jan 2024	Bitcoin ETF approval	Good news, not a crash: the US allowed spot Bitcoin ETFs.
5 Aug 2024	Yen carry-trade unwind	Investors who had borrowed cheap Japanese yen to buy other assets rushed to pay it back; markets fell worldwide.
10 Oct 2025	Liquidation cascade	A tariff threat set off forced selling of borrowed-money positions, one domino after another.

B. Statistics

Mean and median

Mean: add everything up and divide. Median: the middle value when sorted; one crazy value cannot drag it around.

"Back to normal after a median of 8 to 10 hours".

Standard deviation

The typical distance from the average: how spread out numbers are. For returns, it *is* the volatility.

z-score

How many standard deviations a value is from normal. z = 3 means "three typical moves big".

Our shock: |z| above 2.29 (|z| means ignore the sign).

Percentile

The value below which a given share of the data falls; the 99th percentile is bigger than 99% of values.

Distribution, bell curve, fat tails

A distribution says how often each size of move happens. In a bell curve (the "normal" distribution) extreme values are very rare. **Fat tails**: extreme values happen far more often than a bell curve says, as in crypto.

Pearson and Spearman correlation

Pearson uses the actual values; Spearman uses only the order (1st, 2nd, 3rd…), so it is safe from extreme values and good for comparing two rankings. Both run from −1 to +1.

Regression and R^2

Regression fits a formula that predicts one number from others (often a straight line). R^2 is the share of the ups and downs the formula explains: 0 = nothing, 1 = everything.

Null hypothesis and p-value

A test starts from a boring assumption, the **null hypothesis** ("there is no effect"). The **p-value** says how surprising our data would be if the boring assumption were true. A small p (below 0.05) means "hard to explain by luck". It is not the chance that the effect is real.

Significant, one-sided, two-sided

Significant: unlikely to be luck. One-sided asks "is A better than B?"; two-sided asks "are A and B different, either way?".

Confidence interval

A range of believable values for the true answer. If the 95% interval for "TGN minus cascade" is 0.019 to 0.074, it does not include 0, so the TGN is reliably better.

Bootstrap

Checking how much a result could wobble by luck: draw the data again many times *with replacement* (like drawing marbles and putting each one back), recompute each time, and look at the spread.

We draw whole days, 1,000 times.

Null model

Fake data built to have no real effect (shuffled or shifted in time), to see what chance alone produces.

Multiple testing

Run many tests and some will look significant by luck, like a streak of heads in many coin flips. Corrections such as Benjamini–Hochberg raise the bar.

Base rate, positive rate, class imbalance

How often the answer is "yes". At 1 hour only 10.5% of our events are "yes": the classes are imbalanced, so "always say no" is 89.5% accurate and useless.

Stationary

Behaves the same way over time. Returns are stationary (they bounce around zero); prices are not (they drift and grow).

Lag and autocorrelation

A lag is a step back in time: "lag 1" is one hour earlier. Autocorrelation is a series' correlation with its own past.

C. Machine learning

Model

A recipe that turns inputs into a prediction. Learning means adjusting the recipe from examples.

Feature (input) and label (target)

Features are the facts the model sees (a coin's volatility, its volume…). The label is the right answer we want it to predict.

Label: did coin j shock within k hours? 1 = yes, 0 = no.

Classification and regression

Classification predicts a category or a chance (shock or not); regression predicts a number (next hour's volatility).

Training, validation and test sets

Practice questions (training), a mock exam used to choose settings (validation), and the final exam, taken once (test).

Split by date: to June 2022, July 2022 to December 2023, 2024 onwards.

Data leakage and embargo

Leakage: the model accidentally sees the future, which is cheating and gives fake good scores. An embargo is a gap between periods so nothing seeps across the border.

7 days.

Parameter and hyperparameter

Parameters are the knobs the model sets itself while learning; hyperparameters are settings we choose (memory size, learning rate, number of epochs).

The TGN has 70,021 parameters.

Loss

One number that says how wrong the model is. Training tries to make it smaller.

Gradient and gradient descent

The gradient says which way to turn each knob to reduce the loss, like feeling which way is downhill in fog. Gradient descent takes small steps downhill.

Learning rate

The size of each step: too big and you overshoot the valley, too small and you never arrive.

0.0005 for the TGN.

Optimizer (Adam)

The rule that turns gradients into knob changes. Adam gives each knob its own step size that adapts as it learns.

Batch and epoch

A batch is a handful of examples processed together; an epoch is one full pass through all the training data.

About 200 events per batch; up to 100 epochs.

Overfitting, underfitting, generalisation

Overfitting: memorising the practice questions; underfitting: not even learning those; generalisation: doing well on questions never seen.

Regularisation

Tricks that stop memorising, such as dropout and weight decay (Part D).

Early stopping and patience

Stop training when the mock-exam score has not improved for a while, and keep the best version. Patience is how long you wait.

10 epochs.

Seed

The number that fixes a run's random start. Training with several seeds shows whether a result is luck.

5 seeds, numbered 0 to 4.

Baseline

A simpler competitor that the main model must beat to be worth it.

Ablation

Removing one part of a model to see what that part was worth.

The TGN without memory.

Ensemble

Combining the answers of several models, like asking several doctors.

Decision tree and gradient boosting

A decision tree reaches an answer through yes/no questions ("is volume unusually high?"). Gradient boosting adds hundreds of small trees, each fixing the mistakes of the ones before.

Logistic regression

The simplest learned model for chances: one weighted sum of the inputs, squeezed into a probability.

Precision, recall, AUPRC, AUROC, lift

Precision: of the alarms, how many were right. Recall: of the real events, how many we caught. AUPRC and AUROC sum up the whole trade-off in one number; lift is a score divided by the random score. All explained slowly in Part 5.

Threshold and calibration

A threshold is the cut-off where a chance becomes an alert. Calibration makes chances honest: things given 30% should happen about 30% of the time.

D. Deep learning

Neuron and neural network

A neuron takes some numbers, multiplies each by a weight, adds them up, and bends the result. A neural network is many neurons in layers, each layer feeding the next.

Weights and bias

The knobs (parameters). A weight says how much an input matters; the bias shifts the result up or down.

Layer, linear layer, MLP

A layer is a row of neurons. A linear (or "fully connected") layer links every input to every neuron. An MLP (multi-layer perceptron) is a small stack of such layers.

The TGN's prediction head is an MLP: 161 numbers in, 64 in the middle, 5 out.

Activation function, ReLU

The bend after the sum. Without bends, many layers collapse into one straight line. ReLU is the simplest bend: keep positive numbers, turn negatives into 0.

Logit and sigmoid

The logit is the raw score from the last layer, any number from very negative to very positive. The sigmoid squeezes it into a chance between 0 and 1 (a logit of 0 means 50%).

The TGN outputs 5 logits, one per window.

Vector, dimension, tensor

A vector is a list of numbers; "64-dimensional" means a list of 64 numbers. A tensor is a block of numbers in any number of directions, such as 200 events × 5 numbers.

Concatenation (||)

Gluing lists end to end: [1, 2] ‖ [3] = [1, 2, 3]. The report's equations use the ‖ sign.

Backpropagation

How the network works out each knob's share of the blame for an error, passing the blame backwards from the output to the first layer, so gradient descent knows which knobs to turn.

Embedding

A short list of numbers that sums up something complicated, such as "this coin's situation right now", in a form the network can compare and combine.

RNN, LSTM and GRU

Networks that read a sequence one step at a time and keep a running memory. LSTM and GRU add **gates**: valves that decide what to keep, what to forget and what to overwrite.

A GRU updates each coin's memory in the TGN; the LSTM baseline reads one coin's last week.

Attention and heads

Attention gives each input a weight, all the weights adding up to 1, so the important ones count more; the network learns the weights. Several **heads** do this side by side, each free to look for something different.

The TGN has 2 heads looking at each coin's 10 latest events.

Transformer

A network built mostly from attention ("Attention is all you need", 2017), the basis of ChatGPT-style models.

DyGFormer is a small Transformer.

Graph: node and edge

Dots and lines. Nodes are the things (coins); edges are relationships. Directed edges have arrows (who affects whom); weighted edges have strengths.

GNN and message passing

A graph neural network: each node updates itself by collecting "messages" from its neighbours. A GCN averages its neighbours; a GAT uses attention over them.

Temporal graph and event stream

A graph whose links happen at particular times. The data is a list of time-stamped events.

"Coin i shocked at hour t; coin j is the one we ask about."

Memory (TGN)

A 64-number diary for each coin, rewritten by the GRU every time the coin takes part in an event.

Message (TGN)

The bundle of numbers used to update a coin's diary: both coins' diaries, the event's 5 numbers and a time code.

Time encoding

Turning "how long since this coin was last updated" into 16 numbers made from waves (cosines), so the network can tell an hour from a day.

Link prediction

The usual job of TGN-style models: will these two nodes interact? We changed the question to "will coin j shock after coin i?".

Focal loss

A loss that pays less attention to easy, obvious examples and more to the rare, hard ones. $\gamma = 2$ sets how strongly easy cases are quietened; $\alpha = 0.25$ sets the weight on "yes" cases.

Dropout and weight decay

Dropout switches off a random 20% of neurons at each training step so the network cannot lean on any one of them. Weight decay gently pulls every knob towards zero so none becomes extreme.

Autoencoder

A network trained to rebuild its own input through a narrow middle. Things it rebuilds badly are unusual: anomalies.

The T-GAE, our failed stress detector.

Attribution (explainability)

Splitting a prediction into how much each input pushed it up or down.

Integrated gradients and TempME.

Adversarial attack, white-box, ε

Changing inputs a tiny bit on purpose to fool a model. White-box means the attacker knows the model inside out; ε (epsilon) is the size of the change.

FGSM and PGD.

Quantisation, fp32, int8

fp32 stores each knob as a precise 32-bit decimal; int8 as an 8-bit whole number: four times less space, but rounded.

CPU and GPU

A CPU is a computer's normal processor; a GPU is a graphics chip that does many multiplications at once, usually faster for deep learning.

All our final runs used a CPU; no GPU was needed.

PyTorch, PyTorch Geometric, scikit-learn, Streamlit

The software libraries: PyTorch for neural networks, PyTorch Geometric (PyG) for graph models, scikit-learn for classic machine learning such as boosting, and Streamlit to turn Python into a web page (the dashboard).

Checkpoint, commit, run registry

A checkpoint is a saved copy of a trained model's knobs. A git commit is a saved snapshot of the code. The run registry logs every training run with its commit, so any number can be traced back.

SAY IT IN THE VIVA

If a word in a question is unclear, it is fine to restate it: "By volatility you mean how jumpy the price is; we measure it as the standard deviation of recent hourly returns." Restating shows understanding and buys a few seconds to think.

PART 1

The whole project in one minute

Three versions of the same story, from a child to the professor.

SIMPLE VERSION

Think of 30 cryptocurrencies as 30 kids in a classroom. Every so often one kid suddenly does something wild. Often, within a few hours, other kids start doing wild things too, the way a yawn spreads around a room.

Our program watches all 30 coins hour by hour. When one coin makes a sudden big move, it guesses which of the other coins will make a big move of their own in the next 1, 3, 6, 12 or 24 hours, and how likely that is. It explains its guesses in plain words and shows them on a website, together with how it did in past crises.

For a friend

We built a system that predicts **shock contagion** in crypto: when one coin has an unusually big hourly price move, which other coins will have one soon after. We used six and a half years of hourly prices for 30 coins from Binance, turned every big move into a set of "will this coin be next?" questions, and trained a **Temporal Graph Network** (TGN) to answer them. We compared it with simpler models, explained what drives its answers, tested how easily it breaks, and built a plain-language dashboard that shows its latest predictions and replays past crises.

For the professor: the 60-second pitch

SAY IT IN THE VIVA

"CryptoGraph-Secure predicts hourly shock contagion between cryptocurrencies. We built a leakage-controlled dataset of 30 coins from Binance, January 2020 to September 2026, with fixed date splits and a 7-day embargo. Whenever a coin has an abnormal move, we create an event towards every other coin, labelled by whether that coin also shocks within 1 to 24 hours: 1.46 million events. Our main model is a Temporal Graph Network, which keeps a memory for each coin and reads events strictly in time order. On the 2024 to 2026 test period it reaches an AUPRC of 0.205 at one hour, about twice the random level of 0.105, and beats a fitted independent-cascade model at every horizon, confirmed with a day-block bootstrap. Its riskiest 1% of alerts come true three to five times as often as random picks. The honest finding is that the signal lives mostly in the receiving coin's own state, and the graph memory adds value mainly at 6 to 24 hours. We also built explanations, robustness tests and a plain-language Streamlit dashboard."

The full title in the report is *"CryptoGraph-Secure: Predicting Cryptocurrency Shock Contagion with Temporal Graph Networks"*. The work was done from late September to October 2026 by Hardik Gulati and Dakshesh Chauhan (AIML-A, SIT Pune).

PART 2

The problem: shocks and how they spread

After this part you can say exactly what the project predicts, and why it is useful and hard.

A few words first

- **Cryptocurrency (coin):** digital money such as Bitcoin (BTC) or Ethereum (ETH). Its price changes all day, every day, with no closing time.
- **Exchange:** a marketplace where coins are bought and sold. We use **Binance**, one of the largest.
- **USDT (Tether):** a coin designed to stay worth one US dollar. On Binance, prices are quoted in USDT, so "BTC at 80,000" means 80,000 USDT.
- **Hourly bar (candle):** a summary of one hour of trading: the opening price, highest, lowest, closing price and how much was traded (**volume**).

What is a shock?

SIMPLE VERSION

Every coin has its own normal amount of wobble. Bitcoin might normally move 0.5% in an hour, a meme coin 2%. A **shock** is an hour when a coin moves much more than its own normal wobble: about 2.3 times its usual hourly move or more, up or down. It is like a usually calm kid suddenly shouting.

"Normal" is measured from the coin's own previous week (168 hours), so a jumpy coin needs a bigger move to count. Both crashes and sudden jumps count, because both are risky. Shocks happen in about 4% of hours.

What is contagion?

Contagion means a shock spreading from one coin to others, like a sneeze going round a class or aftershocks after an earthquake. In crypto this happens fast: when the Terra/LUNA coin collapsed in May 2022 or the FTX exchange failed in November 2022, most big coins dropped within hours. On 10 October 2025 a tariff headline set off about 19 billion dollars of forced selling in roughly a day.

The exact question we answer

SAY IT IN THE VIVA

"Coin *i* has a shock at hour *t*. For every other coin *j*, we predict whether *j* will have a shock of its own within the next *k* hours, for *k* = 1, 3, 6, 12 and 24. Only strictly later hours count: if *j* moves in the same hour, that is moving together, not spreading." This is task **T1**, the main task.

Two side tasks were also built, mainly as baselines:

- **T2:** will a given coin have a shock in the next hour? (an LSTM model)
- **T3:** how much will a coin's price wobble over the next 1, 6 and 24 hours (future **volatility**)? (an LSTM against a classic formula called HAR-RV)

We never predict whether a price goes up or down. Our early analysis showed hourly direction is close to unpredictable (returns barely relate to past returns), while the *size* of moves clearly clusters in time (**volatility clustering**). So we predict the part that has signal.

Why it matters

- **Traders and funds:** cut risk on the coins most likely to be hit next.
- **Exchanges:** raise safety margins before a wave of forced selling.
- **Regulators and researchers:** understand how linked the market is (systemic risk).

Classic finance research measures how strongly markets are linked on average (correlation, spillover indices). It does not say *who will be hit next, and how soon*. That gap is our contribution.

Why it is hard

- **Rare:** after a shock, only about 10% of other coins shock in the next hour.
- **All at once:** most big moves are news-driven and hit many coins in the *same* hour, which leaves little "first this coin, then that one" order to learn.
- **Noisy:** crypto prices are jumpy, and the market changes over the years.
- **Easy to cheat by accident:** any information from the future that slips into the inputs makes results look great but impossible in real life (**data leakage**).

CAREFUL

Why "Secure" in the name? The original plan stressed security and robustness: attacks on the model, anomaly detection and trustworthy data. In the final project this means the robustness tests (noise, missing data, deliberate attacks), the anomaly-detection experiment, checksum-verified data and safe model loading. If asked, say that, and do not claim it detects hacks or fraud.

PART 3

The data: what we collected and how we cleaned it

After this part you can say where every number came from and why the data can be trusted.

Where it comes from

Prices come from the **Binance public data archive** (data.binance.vision): free, official files of hourly bars. Each file comes with a **SHA-256 checksum**, a digital fingerprint. We checked every downloaded file against its fingerprint: `3,049` files, zero failures. If a single byte had been corrupted, the fingerprint would not match.

The period runs from **1 January 2020 to 27 September 2026**, every hour, in UTC time: 59,088 hours for each coin, about 1.6 million coin-hours in total. A separate daily price dataset (Yahoo Finance and Coin Metrics) was used only to double-check the hourly prices; they agreed to about 0.1% on a typical day.

Why hourly, not daily?

SIMPLE VERSION

If you only look at the class once a day, you see that ten kids were wild today, but not who started it. Looking every hour shows the order.

The project first used daily data. The professor's own review pointed out that contagion in crypto happens within hours, so we moved to hourly data. We also tested 2-hour bars: they were worse (TGN 0.165 against 0.205 at the first bar), so 1 hour stayed the main resolution.

Which 30 coins, and why those?

We took the largest coins by market value (a CoinGecko ranking from 27 September 2026) and kept the first 30 that passed four rules:

- not a stablecoin (like USDT), wrapped or staked copy of another coin, or a token of a real-world asset;
- traded against USDT on Binance;
- at least 2 years (730 days) of hourly history;
- data for at least 90% of the test-period hours.

Sector	Coins	Count
Smart-contract platforms	ETH, SOL, TRX, ADA, NEAR, AVAX, SUI, HBAR, DOT, ICP, ETC	11
Payments	BTC, XRP, XLM, BCH, LTC	5
DeFi / oracle	LINK, UNI, ENA, AAVE, JUP	5
Meme coins	DOGE, SHIB, PEPE	3
Layer 2	ARB, POL	2
Exchange token	BNB	1
Privacy	ZEC	1
Infrastructure	QNT	1
Identity / AI	WLD	1

Two more coins, **LUNC** (the old Terra LUNA) and **FTT** (the FTX exchange's token), are kept only for crisis case studies, not as model coins. Monero (XMR) was left out because Binance delisted it in February 2024, so it has no test data. Every exclusion is written down with its reason.

Traps we had to clean up

- **A recycled name:** the ticker LUNAUSDT belonged to the old Terra LUNA until its collapse in May 2022, then was reused for a brand-new coin. Gluing them together would create a fake crash, so they are never joined.
- **A renamed coin:** MATIC became POL in September 2024 (one-for-one). The two histories are joined and the switch-over gap is masked.
- **Missing hours** (exchange maintenance): gaps of 1 or 2 hours are filled with the last price and flagged; longer gaps stay empty and the coin counts as inactive there. Nothing is ever invented.
- **A format change:** Binance timestamps switched from milliseconds to microseconds in 2025; both are converted to the same clock.
- **The dollar peg:** prices are in USDT. If USDT itself slipped away from one dollar, every coin would seem to move together. We flag days when USDT was off its peg by more than 0.5% (checked with Coin Metrics data until it ends in May 2026).
- **Time of day:** trading volume and wobble follow the clock (busier when US and Asian markets overlap). Wobble ranges from 0.69 to 2.17 times its average depending on the hour of the week. We divide out this daily pattern, measured on training data only, so the model does not just learn "it is 14:00".

What the model sees about each coin each hour

Six numbers per coin per hour (**node features**), each scaled using only training data:

Feature	In plain words
`log_return`	how much the price changed in the last hour
`rv_24h`	how jumpy the price was over the last day
`rv_72h`	how jumpy the price was over the last 3 days
`log_volume_ds`	how much was traded, with the time-of-day pattern removed
`volume_rel_168h`	trading volume compared with its own past week
`log_hours_listed`	how long the coin has been trading (capped at 2 years)

Two more candidates (7-day jumpiness and a high-low range measure) were dropped because they repeated others almost exactly (correlation above 0.9 on training data).

Plus eight **network features** per coin, recomputed once a day from the past week or month, describing the coin's position in the market's web of relationships:

Network feature	In plain words
strength	how strongly it moves together with all the other coins
eigenvector centrality	how central it is: linked to coins that are themselves well linked
MST degree	how many direct links it has in the "skeleton" of strongest links (the minimum spanning tree)
community size	how big its group of closely moving coins is (Louvain communities)
spillover PageRank	how important it is as a spreader of price swings (Diebold-Yilmaz network)
spillover NET	whether it gives out more price swings than it receives
shock PageRank	how important its shocks are in the "whose shocks are followed by whose" network
shock out-lift	how much more often other coins shock right after this coin does, compared with usual

So each coin is described by 6 + 8 = **14 numbers** at every hour.

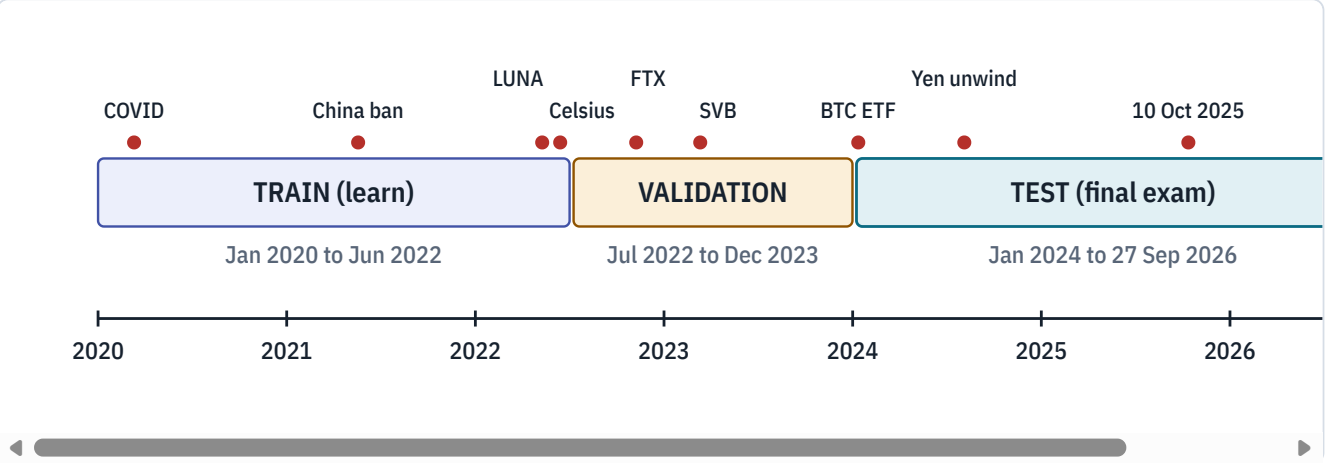
The golden rule: no peeking at the future

SIMPLE VERSION

Studying with last year's exam answers taped inside your pencil case gives a great score that means nothing. Our model must never see anything from the future, even by accident.

- Everything the model sees at hour t uses only data up to hour t.
- Every setting fitted from data (scaling, time-of-day pattern, the shock threshold, the network threshold) is fitted on the **training years only**.
- The answer for hour t only looks at later hours.
- Automated tests check these rules, for example that changing data after t never changes anything computed at t.

Splitting time into practice, mock exam and final exam



The three periods, with the catalogued crises inside the hourly data. A 7-day gap (the embargo) sits at the start of validation and of test. The data ends on 27 September 2026.

SIMPLE VERSION

Training years are the practice papers the model learns from. Validation years are the mock exam we use to pick the best settings. Test years are the final exam: sealed until the end, opened once per final model.

- **Train:** January 2020 to June 2022 (includes COVID, the China ban, the LUNA collapse).
- **Validation:** July 2022 to December 2023 (includes FTX and the SVB bank failure).

- **Test:** January 2024 to 27 September 2026 (includes the bitcoin ETF approval, the yen unwind and the 10 October 2025 crash).
- Each period is a block of **whole dates**, never a random shuffle: shuffling time would let the model learn from the future.
- A **7-day embargo** (gap) at the start of validation and test, and no example whose 24-hour answer window crosses into the next period.

SAY IT IN THE VIVA

"We use fixed date splits with a seven-day embargo, so no sample's input or label window crosses a boundary. All fitted quantities are fitted on train only. Model choices were made on validation, and each final model scored the test period once."

PART 4

Turning prices into a question a computer can answer

After this part you can explain a shock, an event and a label with a worked example.

Step 1: spot the shocks

For each coin and hour we compute a **z-score**: this hour's move divided by the coin's normal hourly move over the previous 168 hours (not counting this hour, so no peeking).

```
z = (this hour's return, time-of-day adjusted) ÷ (typical hourly move over the previous week)
```

If |z| is above **2.29**, the hour is a shock. Why 2.29? It is the level at which about **4%** of training hours count as shocks, matching the 3.9% rate of the earlier daily data. It was set on training data only, so the test period could not influence it.

Step 2: turn each shock into events

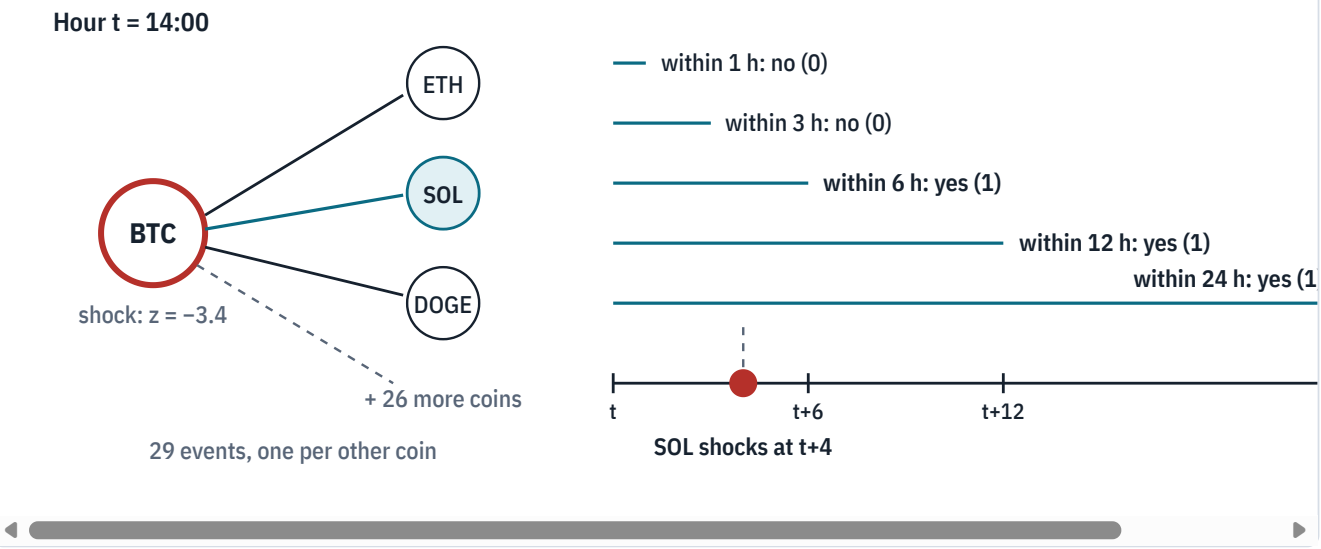
When coin *i* shocks at hour *t*, we create one **event** for every other coin *j* that is trading at that hour: "*i* just shocked; what about *j*?" With 30 coins, one shock makes up to 29 events. If five coins shock in the same hour, that is up to 5 × 29 = 145 events.

Each event carries five numbers, called the **message**:

1. how often *j* followed *i* one hour later over the past week (**lag-1 correlation**);
2. how closely *i* and *j* move in the same hour (**same-hour correlation**);
3. the size of *i*'s move now;
4. how jumpy *i* is now;
5. a "this is a shock" flag. Every event is a shock, so this one is always 1 and carries no information. It is a harmless leftover; mention it if asked about weaknesses.

Step 3: write down the answers (labels)

For every event we record five yes/no answers: does *j* shock within the next 1, 3, 6, 12 and 24 hours, counting only hours *t*+1 onwards? One event, five **labels**. The model gives five probabilities at once.



A worked example (made-up numbers): BTC shocks at 14:00, so every other coin gets an event. SOL's own shock four hours later makes its labels 0, 0, 1, 1, 1.

NUMBERS TO KNOW

- **1.46 million events** in total: 360,590 train, 300,998 validation, 801,559 test.
- Share of "yes" answers on the test period (the **positive rate**): 10.5% within 1 hour, 20.6% within 3, 31.0% within 6, 44.7% within 12, 62.7% within 24.
- Each event: 5 message numbers; each coin: 14 feature numbers.

CAREFUL

The answer for coin *j* depends only on *j* and the hour, not on which coin triggered the event. If BTC and ETH both shock at 14:00, SOL gets two events with the same five answers. So in practice the model answers "after a shock somewhere, which coins move next?". This is why the dashboard shows one chance per coin (the highest over the events pointing at it), and why the explanations find that the receiving coin's own state matters most. If asked, say it plainly: it is a known property of the label, and a direction-aware label ("does *j* crash after *i* crashes?") is listed as future work.

PART 5

How we judge a model

After this part you can explain AUPRC, why random scores 0.105, and why we resample whole days.

The accuracy trap

SIMPLE VERSION

Imagine a weather app in a desert that always says "no rain". It is right 95% of the time and completely useless. Accuracy is a bad score when the thing you care about is rare.

Within 1 hour, 89.5% of our events are "no". A lazy model that always says "no" would be 89.5% accurate and would never warn anyone. So we never report accuracy.

Precision and recall

Think of a smoke alarm.

- **Precision:** of the times it beeped, how often was there really a fire? If we flag 10 coins and 4 of them move, precision is 4 out of 10 = 40%.
- **Recall:** of all the real fires, how many did it catch? If 20 coins moved and we flagged 4 of them, recall is 4 out of 20 = 20%.

Flagging more coins raises recall but usually lowers precision. Every model has to trade one for the other.

AUPRC: one score for the whole trade-off

Our model does not just say yes or no; it gives each event a probability. Sort all events from most to least risky. Walk down the list. Every time you reach a coin that really moved, write down your precision so far. The average of those numbers is the **average precision**, also called **AUPRC** (area under the precision-recall curve).

- A perfect model puts every real move at the top of the list: AUPRC = 1.0.
- Random guessing scores **the positive rate**: 0.105 within 1 hour, because a random list is right 10.5% of the time at every depth.
- So the TGN's **0.205** at 1 hour means it is about **twice as good as random**. That ratio is the **lift**.

CAREFUL

Never compare AUPRC across horizons directly. Within 24 hours, 62.7% of coins move anyway, so random already scores 0.627, and our 0.739 is a smaller lift (about 1.2×) than 0.205 is at 1 hour (about 2×). Always quote AUPRC next to the random level.

We also report **AUROC**: the chance that a randomly picked coin that moved was ranked above a randomly picked coin that did not. 0.5 is random, 1.0 is perfect. The TGN gets 0.64 to 0.69: clearly better than random, but modest. Most shocks are news-driven and arrive across the whole market at once, which caps how predictable they are.

Two scores people understand instantly

- **Top-1% precision:** of the 1% of alerts the model is most sure about, how many came true? For the TGN on the test period: 32% within 1 hour (random 10.5%), 61% within 3 hours (random 21%), 76% within 6 hours (random 31%) and 89% within 24 hours (random 63%).
- **Calibration:** when the model says 30%, does it happen about 30% of the time? Raw model scores are not true chances, so we convert them with **isotonic regression** (an always-rising curve) fitted on the validation years. The report lists checking this on the test years as still open; our final review did check it, and the chances the dashboard shows matched what happened within a few percentage points.

Making comparisons fair

1. **Five seeds.** A neural network starts from random knob settings, so two trainings give slightly different results. We train every final neural model five times (seeds 0 to 4) and report the average. The spread between seeds is about ±0.005 AUPRC, so smaller differences mean nothing.
2. **Same events for everyone.** Every model is scored on exactly the same test events with the same answers.
3. **A day-block bootstrap.** To ask "is model A really better than B, or was it luck?", we rebuild the test period 1,000 times by drawing its 971 days at random with replacement, and compute A's score minus B's score each time. The middle 95% of those 1,000 differences is the **95% interval**. If the whole interval is above zero, A is reliably better.
4. **Why whole days?** One crash creates hundreds of events that are all related. Resampling single events would pretend they are independent and make us overconfident. Resampling days keeps each day's events together.
5. **Test once.** All tuning used validation. Each final model scored the test period once, and every version that ever touched the test period is listed in the report (Table VI), including the ones we did not adopt.

SAY IT IN THE VIVA

"The task is imbalanced, about 10% positives at one hour, so accuracy is meaningless. We use average precision, whose random baseline equals the positive rate, plus AUROC and the precision of the top 1% of alerts. Differences between models are tested with a paired day-block bootstrap over 971 test days, because events within a day are strongly dependent."

PART 6

The building blocks, from scratch

Everything you need to understand the TGN: parameters, training, graphs, memory and attention.

A neural network is a recipe with adjustable knobs

SIMPLE VERSION

Imagine a huge mixing desk with thousands of knobs. Numbers go in one side (a coin's recent moves), pass through the desk, and one number comes out (the chance of a shock). At first the knobs are random and the answers are nonsense. Training means showing the desk thousands of past examples and, after each one, turning every knob a tiny bit in the direction that would have made the answer less wrong.

- **Parameters (weights):** the knobs. Our TGN has 70,021 of them. Each is a single number the training adjusts.
- **Layer:** one stage of the desk. A simple layer multiplies its inputs by weights, adds them up, and passes the total through a small bend (**ReLU**: keep it if positive, else zero), which lets the network learn curved, not just straight-line, patterns.
- **Sigmoid:** the last step squeezes any number into a value between 0 and 1, read as a probability. Our model has five outputs, one per time window, each with its own sigmoid.

How training works

1. **Loss:** a number that measures how wrong the answers are. Lower is better.
2. **Gradient:** for every knob, which way to turn it to lower the loss. Computed automatically (**backpropagation**).
3. **Optimizer (Adam):** the rule for how far to turn each knob. Adam (Kingma and Ba, 2015) adapts the step size for every knob on its own. Our **learning rate**, the overall step size, is 0.0005.
4. **Batch:** we learn from about 200 events at a time, not one by one.
5. **Epoch:** one full pass over all training events. The TGN trains for up to 100 epochs.
6. **Early stopping:** after every epoch we check the validation score. If it has not improved for 10 epochs in a row (the **patience**), we stop and keep the best version.

Overfitting is memorising the practice papers instead of learning the subject: great on training data, poor on new data. Three guards against it:

- **Dropout:** during training, 20% of the head's neurons are switched off at random, so no single neuron becomes a crutch.
- **Weight decay:** a small pull of every knob back towards zero (0.0001), which keeps the model simple.
- **Early stopping** on validation, and a deliberately **small model**.

Focal loss: paying attention to the hard cases

Because "no" is nine times more common than "yes", a normal loss lets the model coast on easy "no" answers. **Focal loss** (Lin and colleagues at Facebook AI, 2017) turns down the weight of examples the model already gets right and turns up the weight of hard, rare ones. Settings: focusing strength $\gamma = 2$, positive weight $\alpha = 0.25$. It was chosen over a weighted ordinary loss by validation AUPRC.

Graphs: dots and lines

SIMPLE VERSION

A **graph** is a drawing of dots joined by lines. In a friendship graph, the dots are people and the lines are friendships. In ours, the dots (**nodes**) are the 30 coins and the lines (**edges**) are relationships: two coins that move together, or one coin's shock pointing at another coin.

A **graph neural network (GNN)** lets each dot update what it knows by listening to its neighbours, a step called **message passing**. A coin linked to many jittery coins learns that it lives in a jittery neighbourhood.

Temporal graphs: when the lines appear matters

In our data, lines appear at particular moments: "BTC shocked at 14:00, pointing at SOL". The order matters (a shock at 13:00 can explain one at 15:00, never the reverse). A graph whose lines carry timestamps is a **temporal** (or **dynamic**) graph. Ours is a stream of timed events, a **continuous-time dynamic graph**. Older methods instead take a snapshot of the graph once a day, which loses the order of events inside the day.

Memory: a diary for every coin (the GRU)

SIMPLE VERSION

Each coin keeps a little diary, 64 numbers long. Every time something happens to the coin, the diary is updated: some old notes are kept, some are overwritten. The diary lets the model remember "this coin has been nervous for the last two days" without re-reading all of history.

The diary is updated by a **GRU** (gated recurrent unit; Cho and colleagues, 2014). It has two "gates", small learned switches that decide how much of the old memory to keep and how much new information to let in. Its cousin, the LSTM, does the same job with three gates.

Attention: deciding what to focus on

SIMPLE VERSION

When you read a sentence you do not give every word equal weight. **Attention** does the same for the model: it looks at a coin's 10 most recent events and gives each a weight (all weights add up to 1), so the important ones count more.

Technically, the coin asks a **query** ("what am I looking for?"), each past event offers a **key** ("what I'm about") and a **value** ("what I'd contribute"); the weights come from how well the query matches each key. This is the idea behind the **Transformer** (2017). Our TGN uses two attention "heads" in parallel, each able to focus on something different.

Time encoding and embeddings

- **Time encoding:** a model cannot use "3 hours ago" well as a single raw number, so the time gap is turned into 16 numbers built from waves of different speeds (cosines), which let it tell very recent events from old ones.
- **Embedding:** a short list of numbers that summarises something complicated. The TGN turns each coin's memory plus its recent events into a 64-number embedding.

PART 7

The main model: the Temporal Graph Network (TGN)

After this part you can say what it is, who made it and when, how it works box by box, how many parameters it has and why we chose it.

SIMPLE VERSION

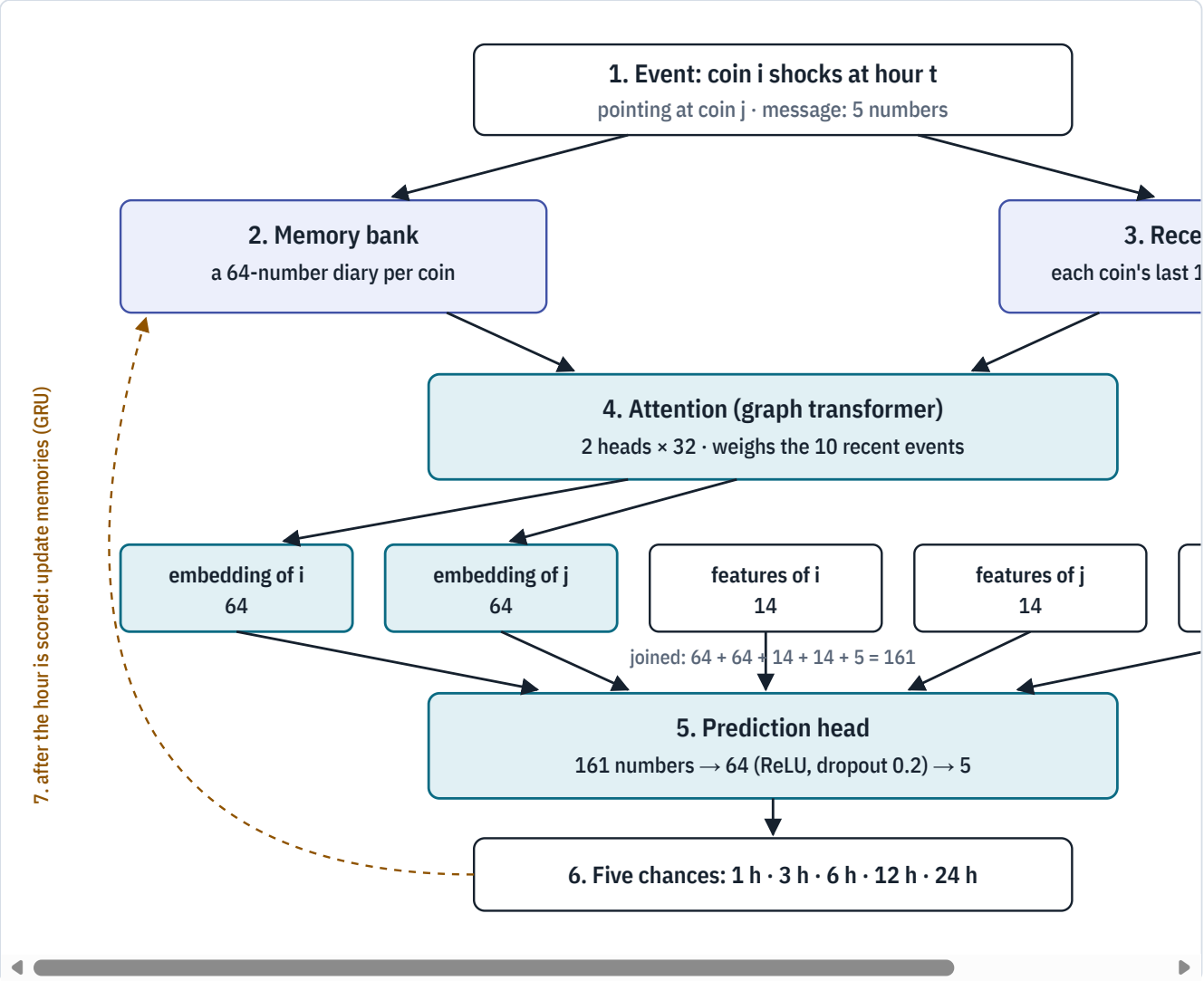
The TGN keeps a diary for every coin. When a coin has a big move, the TGN reads the diaries of that coin and of each other coin, glances at what happened to each of them recently, looks at how both are behaving right now, and gives five chances: will the other coin have a big move within 1, 3, 6, 12 or 24 hours? Only after it has answered for the whole hour does it write the hour's events into the diaries. It can never use the future to answer.

Who made it, and when

The TGN was introduced in the paper *"Temporal Graph Networks for Deep Learning on Dynamic Graphs"* by Emanuele Rossi, Ben Chamberlain, Fabrizio Frasca, Davide Eynard, Federico Monti and Michael Bronstein, researchers at **Twitter** working with Imperial College London. It was first published in **June 2020** (arXiv 2006.10637) and presented at an ICML 2020 workshop. It was designed for streams of interactions such as users replying on Twitter, editing Wikipedia or posting on Reddit, and it became a standard model for this kind of data.

It combines older ideas: memory updated by a GRU (2014), attention over recent neighbours as in TGAT (Xu and colleagues, Walmart Labs, 2020) and the Transformer (2017), and learnable time encoding.

Our version, called `ContagionTGN` in `src/cryptograph/models/tgn.py`, is built with **PyTorch Geometric** (PyG), a widely used graph library for PyTorch (Fey and Lenssen, 2019). We use PyG's memory module (`TGNMemory`), its recent-neighbour store (`LastNeighborLoader`) and its graph-transformer attention layer (`TransformerConv`), and we wrote our own prediction head for the five time windows and our own training loop (`src/cryptograph/training/tgn_task.py`).



How the TGN scores one event, in seven steps. The dashed amber loop is the only way the future reaches the memory: an hour's events are written in only after the whole hour has been predicted.

Step by step

1. **An event arrives:** coin *i* shocked at hour *t*; the event points at coin *j* and carries its 5-number message.
2. **Read the diaries:** fetch the 64-number memories of *i* and *j*.
3. **Look at recent events:** for each of the two coins, fetch its 10 most recent events, with the time since each one, turned into a 16-number time code.
4. **Attention:** a graph-transformer layer with 2 heads weighs those 10 events and turns each coin's memory plus its recent events into a 64-number **embedding**.
5. **Join everything:** both embeddings (64 + 64), both coins' current 14 features (14 + 14) and the event's 5 numbers: 161 numbers in total. A small two-layer network (the **head**) turns them into 64 numbers and then into 5.

6 **Five chances:** a sigmoid turns each of the 5 outputs into a probability, one per time window. Each window gets its own probability (not a softmax), because "within 6 hours" includes "within 3 hours".

7 **Update the diaries:** only after every event of hour *t* has been scored, both coins' memories are updated with the event by the GRU. Events are processed in time order in batches of about 200, and a batch never splits an hour, so events in the same hour cannot see each other.

Our exact settings

Setting	Value	In plain words
Memory size	64	numbers in each coin's diary
Time code	16	numbers describing "how long ago"
Embedding size	64	summary of a coin's memory and recent events
Recent events read	10	per coin, the latest ones
Attention	2 × 32	two heads of 32 numbers each
Head	161 → 64 → 5	two dense layers, ReLU in between
Dropout	0.2	20% of head neurons switched off in training
Loss	focal (γ 2, α 0.25)	focus on rare, hard cases
Optimizer	Adam	learning rate 0.0005, weight decay 0.0001
Batch	about 200	events, never splitting an hour
Epochs	up to 100	stop after 10 epochs without improvement
Model choice	validation AUPRC at 1 h	the best epoch is kept
Seeds	5	trained five times; results averaged

How many parameters, and where they are

The TGN has **70,021 trainable parameters**. You can rebuild that number box by box:

Part	How it is counted	Parameters	Share
Memory updater (GRU)	input 149 = 64 + 64 + 5 + 16; 3 gates × (149×64 + 64×64 + 64 + 64)	41,280	59%
Attention layer	query, key, value and skip: 4 × (64×64 + 64); event projection 21×64	17,984	26%
Prediction head	(161×64 + 64) + (64×5 + 5)	10,693	15%
Two time encoders	2 × (16 + 16)	64	0.1%
Total		**70,021**	**100%**

The diaries themselves are **not** parameters: 30 coins × 64 numbers = 1,920 numbers of **state** that change as events stream in and are reset at the start of every epoch. Parameters are learned; state is remembered.

> **SAY IT IN THE VIVA**
>
> "About 70 thousand parameters, 70,021 exactly: 59% in the GRU that updates memory, 26% in the attention layer and 15% in the prediction head. That is tiny compared with language models, on purpose: we have 30 nodes and 360 thousand training events, so a small model overfits less and trains on a laptop CPU in 30 to 50 minutes per seed."

For comparison: DyGFormer has 24,021 parameters and the LSTM 5,665. A large language model has billions.

How it was trained

- Each epoch replays the training events in time order with memory starting from zero; validation continues straight after, so memory carries over in time, as it would in real life.
- For the final scores, the model replays training (without learning), then validation, then test, so memory is warmed up exactly as it would be in real use.
- On a laptop CPU (no GPU), each of the five seeds took about 30 to 50 minutes; the best epochs were between 45 and 98. One pass over the full 1.46 million events takes one to two minutes.
- The first final version (up to 30 epochs, patience 4) stopped while its validation curve was still rising (0.138). Looking only at validation curves, we raised the budget to 100 epochs with patience 10 (version 2, validation 0.153), and that became the final model.
- Variants tried on validation and not adopted: a learning-rate schedule (0.151), extra context features (0.151), stronger regularisation (0.143), a smaller 32-number model (0.133) and futures-

market data (0.131). None beat version 2 by more than the seed-to-seed noise (±0.005), so we kept the simplest well-trained model.

Why we chose the TGN

1. **The data is a stream of timed events.** Shocks happen at specific hours and point from one coin to others. The TGN was built exactly for such continuous-time event streams; most other graph models expect fixed snapshots.
2. **Memory fits the problem.** A coin that has been nervous for two days is more likely to move again. The memory tracks that without re-reading history, and our test (memory on against off) shows it helps for the 6 to 24 hour windows.
3. **It uses both sides of the story:** each coin's own state (like an LSTM) and its relationships with other coins (like a graph model).
4. **It is honest about time.** Predict first, update memory after, which matches our no-peeking rule naturally.
5. **It is small and fast:** 70 thousand parameters, CPU training, a full prediction in under a minute.
6. **It is well established** as a standard model for dynamic graphs, with a maintained implementation in PyTorch Geometric, so we were not building on untested code.
7. **It can be explained:** TempME, a recent explainer for temporal graph models, works on TGN-style models, and we needed explanations.

Why not the alternatives?

- **An LSTM alone** reads one coin's history and ignores the other coins. We used it as a baseline for the single-coin tasks.
- **A static graph network** (GCN or GAT on one graph) ignores time entirely.
- **A snapshot graph network** (one graph per day, plus a GRU over the days) loses the order of events inside each day. It was planned and dropped, because the TGN, DyGFormer and boosting comparison already answers whether the graph matters.
- **A big Transformer on raw price series** needs far more data than 30 coins provide. DyGFormer is the Transformer-style dynamic-graph model, and we did test it as a second opinion.
- **The official DyGLib library** trains with random fake links and splits time by fractions, which would break our leakage rules, so we re-implemented DyGFormer compactly instead.

CAREFUL

The TGN is not a price predictor (it never says up or down), not a trading bot, and not a language model. And memory does not help within 1 hour: the version without memory scores the same there (0.208 against 0.205). Say this before the professor finds it: memory pays off for the slower, multi-hour spread.

PART 8

The other models, and why each one is there

After this part you can answer "why these models only?" with a reason for every model.

SIMPLE VERSION

To show the main model is good, you race it against others. Each opponent was picked to test one specific claim: is there any signal at all, does a simple formula already do it, do we need deep learning, does the graph matter, does memory matter, does it depend on one design?

Model	What it is, simply	Question it answers	Test AUPRC 1 h
Random guessing	picks coins at random	What is the floor?	0.105
Independent cascade	an epidemic model: each shocked coin gets one chance to "infect" each other coin, with a probability learned from training data (Kempe, Kleinberg and Tardos, 2003)	Does a classic network-science model already do it?	0.159
Gradient boosting	hundreds of small decision trees, each fixing the last one's mistakes (scikit-learn); same inputs as the TGN head, without memory or attention, plus coin IDs and market context	Do we need deep learning at all?	0.178
TGN without memory	the same TGN with memory and attention switched off	Does memory matter?	0.208
DyGFormer	a Transformer over each coin's last 16 events, cut into pairs, plus how often the two coins appeared together (Yu and colleagues, NeurIPS 2023); our compact version has 24,021 parameters	Does the result depend on one architecture?	0.208
TGN (main)	**Part 7**	**The main model**	**0.205**
Ensemble	average of TGN, DyGFormer and boosting, weighted on validation	Does combining help?	0.199

Models for the side tasks and other questions:

- **LSTM** (long short-term memory; Hochreiter and Schmidhuber, 1997): reads one coin's last 168 hours. 5,665 parameters (1 layer, 32 hidden units). Used for next-hour shock risk (AUPRC 0.109, 2.8 times random, AUROC 0.71) and future volatility.
- **HAR-RV** (Corsi, 2009): a simple formula that forecasts volatility from the last hour, day and week. It beat the LSTM at forecasting volatility at every horizon (error 5.0 against 5.5, in units of 10^{-5}, at

1 hour). Deep learning earns its place on rare events and networks, not on smooth volatility.

- **Simple rules** (a logistic regression on 4 rules: how unusual the move is now, shock now, shocks in the last day, how much of the market is shocking): the next-hour baseline (AUPRC 0.095).
- **Diebold-Yilmaz spillover** (2012, 2014): a classic finance method (a VAR model and its variance decomposition) that measures how much of each coin's volatility comes from the others. Used to rank coins and as network features, not as a predictor.
- **T-GAE** (temporal graph autoencoder): learns what a normal day of the coin network looks like, predicts the next day, and flags days it predicts badly (3,646 parameters). It was a negative result (it separated stress days barely better than a coin flip: AUROC 0.51, against 0.89 for simply measuring turbulence), although it did rank 10 October 2025 as the top anomaly.
- **Integrated gradients** (Sundararajan and colleagues, Google, 2017) and **TempME** (Chen and Ying, NeurIPS 2023): explainers, covered in Part 9.

"Why these models only?"

SAY IT IN THE VIVA

"Each model answers one question in the experimental design. Random and simple rules show there is signal. The independent cascade is the network-science baseline a learned model must beat. Gradient boosting tests whether plain machine learning on the same inputs is enough. The LSTM tests a single coin's history without the graph. DyGFormer checks that our result does not hinge on one architecture. Removing memory isolates what the memory adds. More models would not answer a new question."

Things we tried and rejected

- **Futures-market data** (funding rates and premiums): made validation worse for both the TGN (about 0.13) and boosting (0.140), probably because leverage behaved differently across the years. Reported as a negative result.
- **Retraining on train plus validation**: helped boosting (0.178 to 0.185) but hurt the TGN (0.205 to 0.188), so the final TGN stays trained on train only.
- **2-hour bars:** worse than 1-hour (TGN 0.165 against the cascade's 0.148 at the first bar).
- **The snapshot graph network:** not built, for the reason in Part 7.

PART 9

What we found, told honestly

After this part you can present the main table and the five findings, including the weak spots.

The main table (test period, January 2024 to September 2026)

AUPRC by time window. Higher is better; the last row is random guessing. Neural models are the average of five seeds.

Model	1 h	3 h	6 h	12 h	24 h
TGN (main)	**0.205**	**0.352**	**0.464**	**0.588**	**0.739**
DyGFormer	0.208	0.351	0.463	0.587	0.736
TGN without memory	0.208	0.344	0.454	0.576	0.729
Gradient boosting	0.178	0.330	0.451	0.581	0.736
Ensemble (TGN + DyGFormer + boosting)	0.199	0.330	0.465	0.596	0.748
Independent cascade	0.159	0.279	0.381	0.507	0.672
Random (positive rate)	0.105	0.206	0.310	0.447	0.627

The five findings in plain words

1. **Every model beats random.** There is real, if modest, signal. The TGN is about twice as good as random within 1 hour.

2. **The TGN clearly beats the network-science model.** Against the independent cascade it is better at every window: +0.046 at 1 hour (95% interval 0.019 to 0.074) and +0.084 at 6 hours (0.051 to 0.116). In none of the 1,000 bootstrap rebuilds did the cascade come out ahead.

3. **The TGN edges out gradient boosting only at short windows, and only just.** 0.205 against 0.178 at 1 hour and 0.352 against 0.330 at 3 hours (one-sided p about 0.03), but the 95% intervals just touch zero, so call it borderline. From 6 hours on they are tied. Boosting had matched the TGN on validation but memorised its training years (training score 0.64 against 0.155 on validation), so it generalised worse.

4. **DyGFormer ties the TGN everywhere.** Two very different temporal graph designs land in the same place, so the result is not a fluke of one architecture.

5 **Memory helps the slow spread, not the next hour.** Without memory: the same at 1 hour (−0.003, interval −0.014 to +0.010), but 0.010 to 0.013 worse from 6 to 24 hours (p at most 0.025).

Where the graph helps: big crashes, not lone coins

We split test events by how widely the triggering shock spread: **market-wide** (30% or more of coins shocked in the same hour), **partial** (10 to 30%) and **isolated** (under 10%, a coin moving on its own news).

Source shock	Next-hour positive rate	TGN	Boosting
Market-wide crash	0.163	0.245	0.205
Market-wide jump	0.118	0.225	0.201
Partial crash	0.088	0.130	0.103
Isolated crash	0.058	0.114	0.117
Isolated jump	0.043	0.086	0.088

When one coin moves on its own news, the network adds nothing beyond that coin's own state. After broad sell-offs, the TGN's view of the whole market pays off.

How good are the top alerts?

Of the 1% of events the TGN was most sure about, this share came true (five-seed average, test period):

	1 h	3 h	6 h	12 h	24 h
TGN's riskiest 1%	**32%**	**61%**	**76%**	**83%**	**89%**
Random pick	10.5%	21%	31%	45%	63%

Counted per coin and hour instead (each coin once per hour, the way the dashboard shows chances), the single model the dashboard serves gets 28% against 6% at 1 hour and 60% against 23% at 6 hours. These figures come from our final review, not from the report.

What drives an alert (explanations)

Integrated gradients measure how much each input pushed a score up or down compared with an average situation; the pushes add up exactly to the score (we check this "completeness" every run). Averaged over alerts:

- about **65%** comes from the **receiving coin's own state**: its volume compared with its past week (25%), its jumpiness over the last day (12%) and its last move (9%);
- about **25%** from the coin that shocked;
- about **10%** from the event itself; memory and recent events about 7%.

> **SIMPLE VERSION**
>
> The model mostly flags coins that are already stretched (unusually busy and jumpy) at the moment another coin crashes. That is the same pattern in the LUNA, FTX and October 2025 crashes.

TempME looks for small chains of past events ("motifs") that explain an alert. We ported it to our TGN. It turned out barely better than picking events at random (a fidelity score of 0.157 against 0.152), worse than simply taking the most recent events (0.192), and far from the best possible choice (0.248). The reason: the TGN's attention over recent events is almost perfectly even, and shocks spread across the market within the same hour rather than along chains, so there are no chains to find. That is a finding about the market, not a bug.

What we learned about the market

- One common factor (the whole market moving together) explains **64%** of hourly price movement.
- "Coin A moves, then coin B an hour later" is very weak (a typical lagged correlation of 0.035). Coins move together within the hour, which is why the graph has no fixed "who leads whom" lines.
- Groups of coins that move together partly match their sectors, but contagion ignores sectors: after a shock, another coin's chance of shocking is about the same inside and outside the sector. Meme coins are the exception.
- In all nine catalogued crises, correlation, spillover and shock spreading rise and the network's skeleton (MST) shrinks.

How long shocks last (notebook 05)

- Across 18,171 "fresh" shocks (no shock of the same coin in the previous day), a coin's jumpiness was back to normal after a median of **8 to 10 hours** (28% within 6 hours, 87% within 24, 99% within 72).
- 55 to 65% of shocks are followed by another shock of the same coin within 24 hours.
- After a market-wide crash about 8 other coins shock within 6 hours, against 4.5 to 5 after an isolated shock.
- The extra jumpiness compared with a matched calm hour halves in about 2.5 hours in every sector.

How easily it breaks (robustness, notebook 06)

- Removing up to half of the past events: the score changes by under 0.5%. Missing data is not a problem.
- Random noise of half a standard deviation on every input: 5.7% lower (boosting: 4.7%).
- A deliberate attack (**FGSM**, Goodfellow and colleagues, 2015), which nudges inputs in exactly the direction that hurts the model most, by only 0.1 standard deviations: about **35% lower**. Random noise of the same size costs under 1%. A stronger 10-step attack (PGD) was barely worse than FGSM.
- The explanations are stable: under small input noise, the top reason for an alert stays the same for 97 to 99% of alerts.

> **SAY IT IN THE VIVA**
>
> "The model is robust to missing and noisy data but vulnerable to targeted manipulation: a gradient attack of 0.1 standard deviations cuts one-hour AUPRC by about 35%, while random noise of the same size costs under 1%. That is the 'secure' part: it tells you what a deployed system must defend against, such as multi-source input checks or adversarial training."

Making it run cheaply (compression)

We tried shrinking the TGN to 8-bit numbers (**int8 quantisation**): 3.1 times smaller (94 kB against 296 kB) but 1.8 times slower on CPU, because the time goes into looking up recent events and updating memory, not into the arithmetic, and it cost 0.004 to 0.007 AUPRC. So the dashboard serves the normal full-size model: about 6 microseconds per event.

A check from the final code review: a much simpler model

CAREFUL

In our last review we also tried a **logistic regression** (one weighted sum, no deep learning) on just four numbers: the receiving coin's volume against its past week, hours since its last shock, how much of the market shocked in the last day, and how big its current move is. Fitted on training data only, it matched the TGN on the test period (0.212 against 0.206 at 1 hour; the bootstrap intervals include zero at every window).

If asked "did you compare with a simple model?", answer honestly: "Yes. The independent cascade and boosting are in the report, and in our final review a four-feature logistic regression matched the TGN. It confirms what our explanations show: at hourly resolution, a coin's own turbulence and the market's breadth carry most of the signal, because shocks spread across the market within the same hour. The TGN learns this automatically from the raw event stream, its memory captures what 'hours since the last shock' measures, and it clearly beats the network-diffusion baseline. Finer data, such as order books or minute bars, is where a graph model could pull ahead."

PART 10

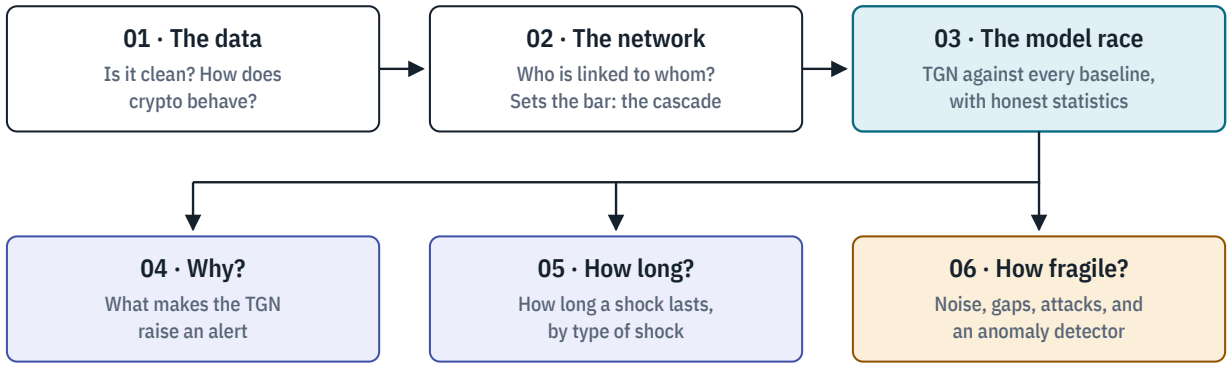
The six notebooks, one by one

After this part you can open any notebook, point at any chart, and say what it checks, what it found and which decision it led to.

SIMPLE VERSION

A notebook is a lab book: code, charts and notes on one page. The pipeline (Part 13) is the kitchen that cooks everything; the notebooks are where we taste the food and write down what we learned. They read the saved outputs; they do not train models.

The six notebooks live in `notebooks/` and follow the story of the project: check the data, study the network of coins, race the models, explain the winner, measure how long shocks last, and try to break it. Each section of a notebook ends with a short "What this shows" note; this part retells those notes in plain words. Section numbers below match the headings inside each notebook. Techniques in Part 11 are linked where they first appear.



Each notebook answers one question; 04 to 06 all start from the models of 03.

Notebook	Question	Headline answer	Number to remember
01 · Data	Is the data trustworthy, and how does crypto behave?	Use returns, not prices; jumpiness is predictable, direction is not; contagion is fast	`2.3× in the next hour`
02 · Network	How are coins connected, and how do shocks travel?	One market factor dominates; shocks hit many coins in the same hour	`64% · cascade 0.159`
03 · Models	**Does the TGN beat simpler methods, honestly?**	**Beats the cascade at every window; ties DyGFormer; edges boosting at 1 to 3 h**	**`0.205 vs 0.159`**
04 · Explanations	Why does the TGN raise an alert?	The receiving coin's own state, not particular past events	`65 / 25 / 10%`
05 · Persistence	How long does a shock last; does its type matter?	Back to normal in 8 to 10 h; broad shocks pull in more coins	`8 to 10 h`
06 · Robustness	Can it be fooled; can an autoencoder spot stress?	Fine with gaps and noise, weak against targeted attacks; the autoencoder fails	`−35% at FGSM 0.1`

Notebook 01 · The data and a first look at the market

`01_market_dynamics_and_spillover_baseline.ipynb` · 61 cells · 18 sections in two halves

Before any model, prove the data is clean and learn how crypto prices behave. Almost every later choice points back to a chart here. **Part I** (sections 1 to 12) uses the project's first dataset: daily prices for 20 coins over ten years (2016 to 2026) from Yahoo Finance, cross-checked against Coin Metrics. **Part II** (sections 13 to 18) uses the hourly Binance data the models are actually trained on.

SIMPLE VERSION

Before cooking, check the ingredients are fresh, then taste them. Part I tastes ten years of daily prices; Part II checks the hourly ingredients the recipe really uses.

CAREFUL

Why the odd file name? The notebook was planned to hold the spillover baseline too. That analysis moved to notebook 02, section 2b, next to the hourly spillover network, and notebook 01 was rebuilt on the audited data with the layout of the archived first draft. The name stayed.

Why a daily part at all? The project started on daily data. After the professor's review the models moved to hourly data, because contagion happens within hours (section 17). Part I is kept because it shows the basic facts of crypto returns on the longest history, and its daily spillover analysis is the cross-check in notebook 02.

Part I · The daily data

1. **Quality audit.** Every coin-day goes through the same checks: duplicate dates, missing days, zero or negative prices, a high below the low, open or close outside the day's range, zero volume, volume spikes, stale prices and moves above 40%. Result: 90,643 active coin-days, 28 bad prices corrected, 156 moves above 40% of which 123 were confirmed by a second source or a known crisis and 33 kept but flagged. The only missing stretch (AVAX, 69 days) was left empty, never invented. The two sources disagree by more than 2% on 23% of days in 2017 but on about 0.2% from 2022, so 2017 and 2018 are the least reliable years.

2. **Coins born at different times, and the active mask.** BTC, ETH, XRP, LTC, XLM and XMR have data from 2016; SOL starts in 2020 and WBT only in 2022. Instead of inventing prices before a coin existed, every coin carries an **active mask**: 1 when it exists and has data, 0 otherwise, and the models ignore the zeros. Full daily candles (open, high, low, close, volume) exist from 9 November 2017; before that, closing prices only.

3. **Ten years of prices and crises.** Growth on a log scale with the catalogued crises as red lines. Every crisis hits all coins at once, and prices trend for years, so raw prices cannot go into a model.

4. **Stationarity (ADF test).** Log prices have no stable level for 15 of 20 coins; returns are stable for all 20. So every input is built from returns and volatility, never from prices.

5. **Fat tails (kurtosis and Jarque–Bera).** Median excess kurtosis 18 (a bell curve has 0); the bell curve is rejected for all 20 coins; Bitcoin's worst day was −37% (12 March 2020). So: Huber loss for volatility, inputs clipped at ±5 after scaling, and scores built for rare events.

6 **Direction vs volatility (autocorrelation).** Yesterday's direction says nothing about today's (lag-1 correlation −0.024; the same sign only 47% of days), but calm follows calm and storms follow storms (+0.184 for squared returns). So the project predicts shocks and volatility, not up or down.

7 **Shock and contagion labels, daily version.** A shock is |z| > 2.5 against the previous 30 days' volatility, measured up to the day before; contagion is coin j shocking within 1 to 3 days after coin i. Shocks are about 4% of days. For a random pair of coins a follow-on shock is rare (under 1%), but right after coin i shocks, coin j shocks within 1 to 3 days 6 to 14% of the time. That jump is the signal the graph model must learn.

8 **Scaling on training data only (robust scaling).** Each feature is centred on the coin's median and divided by its interquartile range, fitted on training dates only. BTC's and HBAR's raw volumes sit at completely different levels; after scaling they look alike, so the model sees "unusually busy for this coin" and nothing from the future leaks in.

9 **Splits, embargo and a calmer future.** Fixed dates chosen so each period contains a big crash: training to June 2022 (2018 crash, COVID, Terra), validation to December 2023 (FTX), test from 2024 (the October 2025 liquidations), with a 30-day embargo on daily data (7 days on hourly). Training was far more volatile (107% a year against 63% and 72%), yet shocks stay at about 4% of days in every period, because each move is judged against the coin's own recent volatility.

10 **Correlation: why a graph model.** Every pair of coins moves together to some degree (correlation 0.39 to 0.80, average 0.57), and more so after crises (0.61 in the 30 days after, against 0.54). A one-coin model cannot use this shared movement; a graph model can.

11 **The daily graph and its threshold.** Two coins are linked when their 30-day correlation is at least 0.6, plus each coin's two strongest links. Lower thresholds (0.3 to 0.4) link almost every pair, which carries no information. "Coin i yesterday, coin j today" links appear no more often than chance (1.6 to 4.3% against 5.4%), so the TGN gets direction from which coin shocked first, not from fixed arrows.

12 **Data loaders check.** Each model's data loads through its real loader with no missing or infinite values: 28,880 / 9,725 / 18,820 LSTM windows and 16,266 / 8,241 / 14,877 TGN events (training / validation / test). About 2,000 training days is small for deep learning, so the models must be small, regularised and stopped early. (The hourly data gives 22 times more training events: 360,590.)

Part II · The hourly data the models use

13 **Coverage and quality.** 59,088 hours × 32 coins (30 model coins plus LUNC and FTT). Coins enter when Binance lists them: 15 from January 2020, while ARB, SUI, PEPE, WLD, JUP and ENA only arrive in 2023 and 2024, so they have no training data. Every model coin covers at least 90% of the test period (lowest: ENA, 90.8%). Gaps are exchange maintenance of at most 5 hours: gaps of up to 2 hours are filled and flagged (13 hours in all), longer ones masked. **MATIC → POL:** a 79-hour gap and a 0.0% price jump (a 1:1 swap), so the two are joined. **LUNA → LUNC:** the 2,863-hour gap after the May 2022 collapse stays masked, and the new LUNA is never spliced in. FTT's 7,467-hour gap is its delisting after FTX. Hourly prices added up to days match the daily dataset to a median of about 0.1%; SHIB (1.9%) is off only because the daily source rounds its price of about $0.00001 to six decimals, and PEPE, also priced in tiny fractions of a cent, shows a similar gap. USDT left its $1 peg by more than 0.5% on four days (March 2020 and the March 2023 USDC scare); those days are flagged, not removed.

14 **Rush hours (seasonality).** Fitted on training data only: the busiest hour (Wednesday 12:00 UTC) is 2.17 times as jumpy as an average hour, the quietest (Sunday 06:00 UTC) 0.69 times; weekend volume is 0.77 times the weekday level. The busy hours match the US morning, when US economic data come out. The pipeline divides this pattern out before computing features and shocks; otherwise a normal Wednesday afternoon would count as a shock.

15 **Does BTC or ETH move first?** Three checks: cross-correlation against a circular-shift null, Granger tests with Benjamini–Hochberg correction, and replication on validation. Coins move together within the same hour (correlation 0.65 with BTC, 0.68 with ETH) but at every other lag the correlation is tiny (median of the largest: 0.035), and its peak sits at +24 hours, a daily rhythm rather than a leader. Only 8 of 45 leader-to-coin and 10 of 45 coin-to-leader Granger tests survive correction; the largest extra explained share is 0.47% (ETH to QNT, a thinly traded coin), the median 0.02%. The training pattern does not repeat on validation (−0.11 for BTC, −0.40 for ETH). **Decision:** no lead-lag arrows in the graphs. The TGN keeps the trailing lag-1 correlation as one of its 5 message numbers, without expecting much from it.

16 **The Epps effect and the graph threshold.** Correlation barely depends on the interval (0.58 at 1 hour, 0.61 to 0.62 at 4 to 12 hours, 0.59 at 1 day), so the Epps effect is weak. But over 168-hour windows almost every pair clears 0.5: the old draft threshold made a graph with 75% of all possible links. The threshold is therefore fitted on training data to match the daily graph's 30% density: **0.75 at 1 hour, 0.77 at 2 hours**. With each coin's two strongest links added, density is 30 to 39%.

17 **Hourly labels and the contagion jump.** Shock thresholds 2.29 (1-hour bars) and 2.33 (2-hour bars), set on training data for about 4% of hours. In the hour after coin i shocks, coin j shocks **2.3 times** more often than in a random hour (9.3% against 4.0% in training). The jump is the same on validation (2.4 times) and test (2.7 times), and fades to about 1.3 times over 24 hours. Contagion is fast, which is why 1 hour is the main resolution.

18 **Hourly data loaders.** At 1 hour: 427,130 / 333,094 / 711,740 LSTM windows (168 hours × 6 features), 905 / 541 / 993 daily graph snapshots, and 360,590 / 300,998 / 801,559 TGN events with 5 message numbers and 5 labels each. No missing or infinite inputs; events in time order. The 2-hour stream (752,333 events) is the robustness check.

NUMBERS TO KNOW

Prices wander for 15 of 20 coins, returns are stable for 20 of 20 · kurtosis 18 · direction −0.024, jumpiness +0.184 · rush hour 2.17×, quiet hour 0.69× · lead-lag at most 0.035 · threshold 0.75 · contagion jump 2.3× (train), 2.4× (validation), 2.7× (test).

Notebook 02 · The network of coins

`02_network_analysis.ipynb` · 37 cells · 10 sections

Treat the market as a network: who is connected, who sits in the middle, which groups exist, how crises change the shape, and how shocks travel. It also builds the contagion baseline the TGN must beat and the 8 network features the models use. Every network is built from past data only: 2,456 daily snapshots from January 2020 to September 2026. Rankings use the test window from 2 April 2024, when ENA, the last of the 30 coins, was listed, so every coin is ranked on the same hours. Sectors come from a hand-made list: 11 smart-contract platforms, 5 payment coins, 5 DeFi and oracle tokens, 3 meme coins, 2 layer-2s and 4 one-coin sectors.

Network	Built from	Arrows?	Used for
Correlation	168-hour correlations of time-of-day-adjusted returns	no	the market mode, communities, the MST, network shape
Spillover (Diebold–Yilmaz)	variance decomposition of a VAR, 30-day windows	yes	spillover index, NET ranking, PageRank
Shock transmission	how often coin j shocks within 1 hour after coin i, 90-day windows	yes, in time order	PageRank, shock lift, the cascade baseline

1 **The market mode (eigenvalues).** The largest eigenvalue of the 30-coin correlation matrix is 19.3 out of 30: one "whole market" factor explains **64%** of hourly movement, and it is the only eigenvalue above the random-noise bound (1.08). Most of what any coin does in an hour is "the market moved".

2 **Spillover (Diebold–Yilmaz).** The hourly total spillover index is 91.6%: over 24 hours, more than nine-tenths of each coin's forecast uncertainty comes from other coins. Biggest net senders: LINK, DOT, AVAX; biggest receivers: TRX, ZEC, QNT. **The catch:** the hourly NET ranking is identical to eigenvector centrality (rank correlation 1.00), because with one lag (chosen by BIC) and tiny lagged effects the decomposition is driven by same-hour correlation. So hourly NET measures how tightly a coin is tied to the market, not who moves first. The index rose from 86% in training to 90 to 91% later, jumps after crises, and fell as low as 66% in a calm 2020 window.

2b **Daily cross-check.** The corrected daily analysis (20 coins with full data from 9 November 2017, a VAR with 2 lags, 10-day horizon, 3,244 days): index 86.0%, ETH the biggest net sender. On the 13 coins both share, daily and hourly NET agree only weakly (0.21): days leave room for one- and two-day delays, hours do not. So "who spreads shocks" depends on the time scale, and explanations are later checked against several rankings rather than one truth.

3 **Who is central (centrality, PageRank).** Five rankings: DY NET, PageRank on the spillover network and eigenvector centrality are the same ranking (1.00); MST degree agrees moderately (about 0.8); PageRank on the shock-transmission network, the only one built from time order, is related but different (about 0.73). It puts ETC, XRP, POL, AVAX and ADA on top and Bitcoin only 23rd: Bitcoin's shocks are not followed by unusually many others. Differences are small (about 0.6 to 1.3 around an average of 1) and drift (SOL rose from 0.57 in 2020 to about 1.0 to 1.2 from 2022), which is why the models get them as features that change over time, not as fixed labels.

4 **Communities vs sectors (Louvain, NMI).** With the market mode removed, Louvain finds 5 groups on the test window: the meme coins (DOGE, SHIB, PEPE); BTC, ETH, the Ethereum DeFi tokens (LINK, AAVE, UNI) and BNB; newer platforms that swing harder than the market (SOL, AVAX, NEAR, SUI, with JUP, WLD, ENA and, less obviously, ZEC); older payment coins (XRP, XLM, ADA, HBAR); and a group led by DOT and ETC. The match with sectors is real but partial: NMI 0.47 against 0.32 for shuffled labels ($p = 0.003$). Groups reflect how coins trade (age, listings, investors) as much as what they do. Modularity is 0.41 to 0.46 and falls after crises, when the groups merge into one market move.

5 **The skeleton (minimum spanning tree).** The 29 strongest links that connect all 30 coins. Strongest: DOGE–SHIB and BTC–ETH; the meme coins form one branch and ETH is the hub of the big coins. DOT has the most links (11), but to loosely attached coins (TRX, ZEC, QNT, HBAR, ICP) whose nearest neighbour happens to be DOT: a meeting point for outsiders, not a leader. The tree's average link length is 0.66 and shrinks in every crisis (0.43 after COVID, lowest 0.36 in April 2024): when the tree is short, all coins move as one and spreading money across coins stops protecting you.

6 **Network shape around crises (density, clustering, k-core).** Two weeks before against two weeks after each of the 9 crises in the hourly period: correlation rises (+0.03 to +0.24), density rises (+0.08 to +0.54), the tree shrinks, the spillover index rises (+0.2 to +7.7 points) and shock lift rises; modularity falls in 8 of 9. The biggest shifts: COVID and the May 2021 China ban; the SVB and USDC scare barely moved the network. Overall the test-period network is denser and more clustered than training (density 0.35 against 0.30, clustering 0.57 against 0.47).

7 **Sectors and contagion (assortativity).** Strong links are not more common within sectors (assortativity −0.14 in training, −0.05 in test). After a shock, another coin shocks within the hour equally often inside and across sectors (training 9.3% against 9.3%, test 10.8% against 10.4%, base rate 4%). The one exception is meme coins, which pass shocks to each other more (13.7% against 10.9%). So the sector stays as a coin feature, but the graph is not built around sectors.

8 **Case study: the FTX collapse, hour by hour.** FTT is an event coin, not a model input. On 6 November 2022 at 15:00 UTC, the hour Binance's CEO said Binance would sell its FTT, FTT fell 14 standard deviations and no model coin shocked that hour or the next. The market reacted on 8 November: at 02:00 FTT fell 10 standard deviations and 18 of 30 coins shocked in the *same* hour; at 16:00, around Binance's announced takeover, 20 coins jumped together; at 17:00 to 18:00, as the deal fell apart, FTT fell 17 standard deviations and all 24 coins trading then shocked down. In a random hour about 1.2 coins shock. Contagion was simultaneous and driven by news, not a chain from one coin to the next.

9 **The bar to beat (independent cascade).** Each shocked coin gets one chance to "infect" each other coin, with chances learned on training data; scored on exactly the TGN's events. Test AUPRC at 1 hour: 0.159 against 0.106 for each coin's base rate (validation 0.117 against 0.087). The edge shrinks with the window and is gone on validation by 24 hours. Pairwise chances alone are no better than the base rate: the cascade's gain comes from counting how many coins shocked at once. This is the bar the TGN must clear, and it should do so by learning *who* is at risk.

10 **Network features for the models.** Eight numbers per coin per daily snapshot, from past data only (a test checks that changing later data leaves them unchanged): strength, eigenvector centrality, MST degree, community size, spillover PageRank, DY NET, shock PageRank and shock out-lift. Coverage is 100% for the correlation-based ones and 95 to 100% for the others, which need 30 or 90 days of history and at least 5 shocks. No pair is redundant (highest rank correlation 0.78). The test-window rankings in `rankings.csv` are only used to check explanations, never as model inputs.

CAREFUL

The cascade table saved inside notebook 02 shows 0.160 at 1 hour (base rate 0.107). That output dates from 1 October; on 2 October the cascade was switched to score exactly the TGN's evaluation events (commit `07f4c27`). The current figures, used in notebook 03 and the report, are 0.159 and 0.106. If asked, that is the whole story: same model, final event set.

Notebook 03 · The model race

`03_models_vs_baselines.ipynb` · 43 cells · the final results

The results for the three tasks: **T1** contagion (the main task), **T2** next-hour shock of one coin, **T3** future volatility. The rules: every model is scored on exactly the same events (after the 7-day embargo, with the whole label window inside the period); all tuning used validation; the test period was scored once per final run.

> **CAREFUL**
>
> The contents list at the top of notebook 03 numbers some sections differently from their headings (for example "5b" for the DyGFormer section headed "4b"), because sections were added later. The headings are right, and the numbers below follow them.

0. **Run registry (top of the notebook, before section 1).** A table of every run with its git commit. All final runs (names ending in `_final`) ran on a clean commit and scored the test period once. Development runs (`_dev`) were scored on validation only. The first final TGN (`tgn_1h_final`, version 1) is kept for the record; it was under-trained.

1. **TGN against the cascade.** Test AUPRC at 1 hour: TGN 0.205, cascade 0.159, random 0.105, so about twice random and 1.3 times the cascade. The lead is largest in the first hours and narrower at 24 hours (0.739 against 0.672). Test scores are higher than validation scores for every model, the cascade included, because 2024 to 2026 had more shocks and more moving together than 2022 to 2023: a property of the period, not of the models.

2. **Seed ensemble.** Averaging the 5 seeds' probabilities adds about +0.001 on test; the seeds already agree within about ±0.003. AUROC is 0.64 to 0.69: clearly better than chance, far from perfect, as expected for hourly, news-driven shocks.

3. **What memory adds (ablation).** The same model with memory and neighbour attention switched off: just as good at 1 hour (0.208 against 0.205) but 0.008 to 0.013 worse from 3 to 24 hours on test. This version uses a minimum of 30 epochs, because in an earlier version four seeds stopped on a lucky first-epoch peak. Best epochs: TGN 98, 45, 76, 55, 86; without memory 46, 47, 45, 49, 59.

4. **Is it luck? (day-block bootstrap).** 1,000 rebuilds from the 971 test days (795,787 events at 1 hour). TGN against the cascade: +0.046 (0.019 to 0.074) at 1 hour and +0.084 (0.051 to 0.116) at 6 hours; no rebuild favoured the cascade. TGN against no memory: no difference at 1 hour (−0.003, interval −0.014 to +0.010), borderline at 3 hours ($p = 0.049$), significant from 6 to 24 hours (+0.011 to +0.013, p at most 0.025).

4b **A second graph model, DyGFormer.** No memory; it rereads the last 16 interactions of both coins each time. Tied with the TGN on test at every window (0.208 against 0.205 at 1 hour; every interval includes zero), and also beats the cascade everywhere (+0.053 at 1 hour). On validation the TGN was ahead (0.153 against 0.142). Two different designs landing on the same score suggests this is close to what these inputs allow. One DyGFormer seed peaked at epoch 4 and still scored normally on test (0.205): the minimum-epochs rule did its job.

4c **Does the graph matter? (gradient boosting).** Boosting sees what the TGN's prediction head sees apart from the graph parts (both coins' features, the event's 5 numbers, the coin ids) plus the shock state of both coins and how much of the market is shocking; trees chosen per window on validation. On validation it matched the TGN (0.155 against 0.153 at 1 hour). On test it fell behind at 1 hour (0.178 against 0.205) and 3 hours (0.330 against 0.352), p about 0.03 with intervals just touching zero, and tied from 6 hours. It had memorised its training years (training AUPRC 0.64 against 0.155 on validation). Lesson: choosing on validation alone would have picked the weaker model, which is why the test is scored only once, at the end.

4d **Combining the three (ensemble).** Scores turned into ranks and averaged, with weights picked per window on validation. Validation chose boosting for 1 to 3 hours, the TGN alone for 6 hours, mixes for 12 to 24 hours. On test the mix beats the TGN at 24 hours (0.748 against 0.740, +0.008, $p = 0.001$), is borderline at 12 hours and is worse at 1 to 3 hours. Lesson: combine only where validation and test agree.

4e **Retraining on training plus validation (refit).** Retrained on data up to December 2023 with the epoch count fixed in advance (77, the median best epoch plus 1): boosting improved (+0.005 to +0.012) but the TGN got worse (−0.017 to −0.040, significant from 3 hours). Without early stopping and with 1.8 times more data per epoch it over-trained. The epoch count was not re-tuned, because choosing it after seeing the test score would be tuning on the test. The main TGN stays the training-only model.

4f **The riskiest alerts (precision at the top).** Of the TGN's top 1% of test events, 32% were followed by contagion within 1 hour (random 10.5%), 61% within 3 hours and 76% within 6 hours. DyGFormer's top alerts are sharper at short windows (43% and 67%) despite the same overall score: the two models order the very top differently. Taking more alerts lowers precision gently: for the refitted TGN the top 10% still come true 23% of the time within 1 hour, 2.2 times random.

5 **Over- or underfitting? (lift ratio).** Compare how much better than random each model is on training and on validation. The cascade cannot memorise; its drop (ratio about 1.1) is just the market changing. The TGN drops by about 1.5 at 1 hour, so it overfits its training years somewhat at short windows; at 24 hours every model is close to the cascade (1.08 to 1.14). Nothing underfits: every model beats the cascade on training and on validation.

6 **What else was tried (validation only).** Training long enough mattered most: version 1 (30 epochs, patience 4) stopped while still improving (0.138, then 0.153 with version 2). After that nothing moved validation by more than about ±0.005, the size of seed noise: a learning-rate schedule (0.151), extra event context (0.151), context without memory (0.156). Heavier regularisation (0.143) and a smaller model (0.133) were clearly worse. Later, Binance futures data (funding rates and premiums, 4,069 checksum-verified files) made validation worse for both the TGN (0.131) and boosting (0.140). So the simplest well-trained model stayed.

7 **1 hour against 2 hours.** With 2-hour bars the TGN still beats the cascade (test 0.168 against 0.148 at one bar, 0.564 against 0.490 at 24 hours), significantly from 3 hours on; at one bar p = 0.10. Coarser bars blur which coin moved first, so 1 hour is the right resolution.

8 **The side tasks.** T2, next-hour shock of one coin: the LSTM reaches 0.109 ± 0.003 (AUROC 0.706) against 0.039 for random and 0.095 for the best simple rule, a logistic regression on 4 rules. T3, volatility over the next 1, 6 and 24 hours: HAR-RV beats the LSTM at every window, on validation and test (test error 5.0 against 5.5×10^{-5} at 1 hour, 1.7 against 2.0 at 24 hours); the LSTM beats "same as last period" at 1 and 6 hours and only matches it at 24.

SAY IT IN THE VIVA

"Notebook 03 is built to stop us fooling ourselves: one event set for every model, tuning on validation only, the test scored once, five seeds, and a bootstrap that resamples whole days. It also reports what did not work: the refit, the futures data and the ensemble at short windows."

Notebook 04 · Why does the TGN raise an alert?

`04_tempme_explanations.ipynb` · 25 cells · explains the final TGN, seed 0

Two levels of "why": which **past events** drive an alert (TempME), and which **inputs** drive it (integrated gradients). The explainer was trained on validation events; everything reported is on the test period (the riskiest 1% of alerts at 6 hours plus a random sample) and on three crashes.

1 **Reference rankings.** The coin rankings from notebook 02 that explanations are compared with. The correlation-based ones are one and the same ranking; the time-ordered shock PageRank is related but different (about 0.73). Computed on the test window and never used as model inputs.

2 **How the explanations are made, and two exactness checks.** For each event the TGN looks at the 10 most recent events of each coin (up to 20). Removing an event means giving it zero weight in the attention; the memory is left as it is. The captured inputs reproduce the TGN's own outputs to 2.4×10^{-7} (rebuild check), and the integrated-gradient pushes add up to the output difference within 0.009. TempME's explainer was trained on 10,000 validation events with the official settings; its score flattened after 3 epochs and training stopped at epoch 13: it found very little to learn.

3 **Do the chosen events carry the prediction? (fidelity and characterization).** With 3 events kept: TempME 0.157 on top alerts, random choice 0.152, the TGN's own attention 0.147, the 3 most recent events 0.192, and a mask tuned separately for every alert, the best any event-level explanation can do, 0.248. Removing all recent events moves a prediction by only about 0.013 in probability. The influence of past events is spread almost evenly; no small set explains an alert.

4 **Why event explanations are weak here.** The TGN's attention is perfectly even (normalised entropy 1.000; the largest of 10 weights averages 0.100). Event patterns around real alerts look like those in a time-shuffled stream: 83% against 86% are three-step walks through four different coins; the only excess is walks that come back to a coin (5.9% against 3.3%). Shocks spread across the market within the hour, not along chains, so there are no special chains for TempME to find.

5 **What drives an alert (integrated gradients, 6-hour window, baseline = the training average).** The receiving coin about 65%, the shocking coin 25%, the event itself 10%. Biggest single driver: the receiver's volume against its past week (25%), then its 24-hour volatility (12%) and its latest return (9%). Memory and recent events together about 7%. The split is the same for top alerts, random events and all three crashes: one consistent rule, "a coin already trading heavily and moving a lot is the one most likely to be hit next".

6 **Do explanations point at the central coins? (plausibility).** TempME's credit runs the other way: about −0.8 against the spillover and PageRank rankings, −0.65 against shock PageRank. It credits loners (TRX, WLD, ZEC, BCH, QNT) most and BTC and ETH least. The tuned mask points the same way but weaker (−0.2 to −0.4); the TGN's attention follows how often a coin appears (+0.3). Because TempME is barely better than random here, its credit is not evidence of who causes contagion; notebook 02's rankings are the better measure.

7 **Three crashes.** The five riskiest 6-hour alerts in each: LUNA (12 May 2022, training period) 4 of 5 came true (chance about 0.57); FTX (8 to 9 November 2022, validation) 4 of 5 (about 0.59); 10 October 2025 (test) 5 of 5 (about 0.65). **13 of 15** in all. The leading reasons are always the receiver's 24-hour volatility and volume surge, and TempME's top events are those of the previous 1 to 3 hours involving the same coins (for example AVAX to WLD at 20:00 before the 21:00 WLD alerts).

Notebook 05 · How long does a shock last?

`05_persistence_by_shock_type.ipynb` · 19 cells · hourly data, 30 coins, 2020 to 2026

Definitions first. A shock is the project's shock (|z| > 2.29). Only fresh shocks count (no shock of the same coin in the previous 24 hours), so one stormy episode counts once. Each shock gets a type by **breadth**, the share of other coins shocking in the same hour: **market-wide** 30% or more, **partial** 10 to 30%, **idiosyncratic** (a coin on its own) under 10%; and a direction, crash or pump. Intervals resample whole days, 500 times.

1. **Crises hour by hour (event windows).** The middle coin's path from 24 hours before to 72 hours after each crisis date (cumulative log returns): COVID −53% by hour 24, the China ban −52%, FTX −15%, the yen carry-trade unwind −16% already by hour 6, the bitcoin ETF approval +9% by hour 24. Hour 0 is midnight UTC of the date, so some moves start before it.

2. **Hours back to normal (persistence).** 18,171 fresh shocks out of 57,792 shock hours. A coin's volatility is back to its level before the shock after a median of **8 to 10 hours** for every type (isolated crashes fastest, 8 hours; isolated pumps and market-wide crashes 10 hours); 28% within 6 hours, 87% within 24, 99% within 72. Average |z|: about 0.62 in the day before (calmer than the 0.72 average hour), 3.5 at a market-wide shock (about 3.0 for others), 1.1 an hour later, 0.8 after 6 hours. The same in training, validation and test (9 to 10 hours): a stable fact about this market.

3. **By type (re-shocks, followers, continuation).** Other coins shocking within 6 hours: 8.1 after a market-wide crash, 7.2 after a market-wide pump, 6.2 after a partial shock, 4.5 to 5.1 after an isolated one. 55 to 65% of shocks are followed by another shock of the same coin within a day (isolated crashes least, 55%). Direction is mostly unpredictable; only pumps tend to keep going for a while (isolated pumps +0.34 standard deviations over 24 hours, market-wide pumps +0.42 over 6 hours).

4. **What it means for the models.** Test events split by the type of the triggering shock: after market-wide and partial shocks the TGN beats boosting at every window (market-wide crash 0.245 against 0.205 at 1 hour; partial crash 0.130 against 0.103); after isolated shocks boosting is level or slightly ahead (0.117 against 0.114). Contagion is far more likely after a market-wide crash (16% at 1 hour against 4 to 6% after isolated shocks). Three consequences: alerts matter most in the first 6 to 24 hours; a real system could use the TGN after broad shocks and a simpler model after isolated ones; no separate persistence model is needed, because the 1 to 24-hour labels already cover a shock's whole life.

5. **Against an ordinary hour, and each crisis's reach (matched null, blast radius).** Comparing the 72 hours after each fresh shock with those after a random calm hour of the same coin in the same month (17,591 shocks), the extra jumpiness **halves in about 2.5 hours** in every sector (2.35 to 2.56), with no differences between sectors and a small tail still visible at 72 hours. Every catalogued crisis was market-wide at its busiest hour (15 to 29 coins at once) and reached every active coin within 72 hours. They differ in strength: LUNA raised the shock rate most (+0.19 against the week before), then COVID (+0.10) and FTX (+0.07); the bitcoin ETF approval barely (+0.01). Peak spillover came 1 hour after the start for FTX and 21 hours for Celsius.

SIMPLE VERSION

A shock is like a stone thrown in a pond. The big splash is over in two or three hours, the ripples are gone in about half a day, and a big stone (a market-wide crash) rocks about twice as many boats as a pebble.

Notebook 06 · Can it be fooled, and can it spot stress?

`06_robustness_tgae.ipynb` · 17 cells · the "secure" part of CryptoGraph-Secure

Two questions. Can a temporal graph autoencoder (T-GAE) flag unusual market days? And how stable are the contagion models when their inputs are noisy, missing or deliberately manipulated? A stress day, the yardstick, has at least one hour in which 70% or more of active coins shocked (20.7% of days); it comes from the returns, not from the T-GAE. The stress tests use the final TGN (seed 0) and the final boosting model on the test events, after a clean replay reproduces their saved predictions (largest difference 0.0002).

1. **The T-GAE against stress days.** Its score for same-day stress reaches AUROC 0.511 on test (0.557 on validation), close to a coin flip, while the day's plain turbulence (average |z|) reaches 0.893. As a next-day warning: 0.490 against 0.600. Adding the T-GAE to turbulence changes almost nothing (0.893 to 0.898; next day 0.600 to 0.589). The five seeds agree within 0.01, so this is not bad luck. Why: an autoencoder learns to reproduce even extreme days, so its error measures unusual *structure*, not turbulence. Two designs were tried on validation; neither found a signal.

2. **Noise and missing events (noise, missing events).** With 10, 25 or even 50% of test events never reaching the TGN's memory, its score moves by less than 0.5%: predictions rest on the coins' current state. Noise of half a standard deviation costs the TGN 5.7% and boosting 4.7% at 1 hour; a full standard deviation costs 16% against 8%, yet the TGN still scores higher at 1 hour (0.171 against 0.164). The TGN's probabilities move about half as much under the same noise (0.026 against 0.065).

3. **Deliberate attacks (FGSM, transfer).** A white-box attack on the TGN of 0.05 standard deviations per input cuts its 1-hour score by 19%, 0.1 by 35%, 0.25 by 61% (0.203 to 0.080). Random noise of size 0.1 costs only 0.3%, so the danger is manipulation, not ordinary data errors. The same inputs fed to boosting, which the attacker never saw, do no harm at 1 hour but cost 11 to 33% at 6 hours, because they push the same features (volatility, volume) the same way. Defences (checking inputs across exchanges, mixing in a non-differentiable model, adversarial training) are future work.

4. **The strangest test days.** Number one is 10 October 2025, the liquidation cascade (score 21.3 against 14.4 for the next day). The next nine cluster in late February to early March 2024 and late July to mid August 2026; most were not stress days and none is near a catalogued event, so they are reported as **"unclassified"**: the T-GAE says a day was unusual, not why, and no manipulation type is claimed.

Three extra checks ran after the notebook, from the same code (their results are in `reports/robustness/` and `reports/compression/`): a 10-step PGD attack was within 0.002 of one-step FGSM at every size, so FGSM was already close to the worst case; driver stability: the top reason stayed the same for 99, 98 and 97% of the 2,000 riskiest alerts under noise of 0.05, 0.1 and 0.25 standard deviations; and int8 compression: 3.1 times smaller but 1.8 times slower, so the full model is served.

CAREFUL

The run table in notebook 06 shows `dirty = True`. The notebook itself explains it: the only uncommitted file at the time was the team's challenges log, kept out of version control on purpose; the code is exactly commit `eb04c49`.

The archive folder: our first drafts, and what was wrong with them

`notebooks/archive/` keeps the two first-draft notebooks (29 and 30 September 2026), run on the original pipeline (9 features, a slightly different coin list, daily data to 24 September 2026). They are kept unchanged, with a README listing every problem found, and they no longer run against the current code. Knowing them is useful: they show the project checked its own work and fixed it.

First draft did	Why that is a problem	Now
Computed statistics on the model input (returns clipped to ±2, missing days set to 0)	Wrong kurtosis (10 instead of 18), autocorrelation and correlations	Raw returns from the cleaned prices
Counted days before a coin existed as missing (12,973; "81.95% pass rate"), and marked 217 zero or negative prices as passed	A false picture of the data's quality	Active mask: not yet listed is not missing
Measured each day's shock against a rolling average that included that day	Leakage: each day judged partly against itself	Volatility up to the hour before only
Reported "multi-asset contagion probability 38.71%": days on which two or more coins shocked	That is co-occurrence, not contagion	Contagion needs a strictly later shock
Split by 70/15/15 fractions; drew the embargo but never applied it	No guarantee each period holds a crash; leakage at the edges	Fixed dates and an applied embargo
Described edges as "directed, $\|r\| \geq 0.60$" while building symmetric edges at 0.35 to 0.40	Text and data did not match	Threshold fitted to density; stated as undirected
Used the order-dependent (Cholesky) spillover method while describing the generalized one	Bitcoin looked like a giant (spillover index 54.2%, NET +731.5) only because it was the first column	Generalized method: index 86.0%, ETH on top
Read a result key that existed only on the author's laptop; stated a correlation claim not computed from the data; mislabelled one crisis (19 May 2021)	Could not be reproduced or checked	Every number printed from the repository's code; crisis dates with source links

SAY IT IN THE VIVA

"Our first draft measured shocks partly against themselves and counted same-day moves as contagion, which made things look better than they were. We archived it with a list of every error, rebuilt the pipeline with past-only windows and strictly-later labels, and the honest numbers are smaller but real."

PART 11

Every technique, explained

Each method used anywhere in the project, in three lines: plain words, how it works, and what it gave us. Learn the first line of each; read the rest when a question needs it.

SIMPLE VERSION

A project like this is a toolbox. Some tools look at one coin, some at pairs of coins, some at the whole network; some are the models, some judge the models, and some try to break them. Here is every tool, one at a time.

1 One coin's price

2 Coins moving together

3 The coin network

4 Contagion baselines

5 The learning models

6 Training tricks

7 Scoring and statistics

8 Explaining the model

9 The life of a shock

10 Stress tests and compression

1. Looking at one coin's price

Log returns

PLAIN WORDS The percentage change from one hour to the next, written so that changes add up neatly.

HOW IT WORKS r = ln(price now ÷ price an hour ago). For small moves it is almost the percentage; unlike percentages, a day's log return is exactly the sum of its 24 hourly ones.

IN OUR PROJECT Every input, the shock score and the volatility features start from hourly log returns (the feature `log_return`).

Stationarity and the ADF test

PLAIN WORDS A series is "stationary" if it behaves the same way at any time, like a heartbeat, not like a child's height. Prices drift and grow; returns bounce around zero. Patterns found in drifting series are often fake.

HOW IT WORKS The augmented Dickey–Fuller test starts from the boring assumption "this series wanders off and never comes back" (it has a **unit root**). A small p-value throws that assumption out, so the series is stationary.

IN OUR PROJECT Log prices: wandering for 15 of 20 coins. Returns: stationary for all 20. So no model ever sees raw prices (notebook 01, section 4).

Fat tails: kurtosis and the Jarque–Bera test

PLAIN WORDS In a bell curve, giant moves almost never happen. Crypto has far more giant moves than a bell curve allows.

HOW IT WORKS Kurtosis measures how heavy the tails are (a bell curve scores 0 "excess kurtosis"). Jarque–Bera tests whether the lopsidedness and tail weight of a sample could come from a bell curve.

IN OUR PROJECT Median excess kurtosis 18; the bell curve rejected for all 20 coins. So: Huber loss for volatility, inputs capped at ±5, and scores built for rare events (notebook 01, section 5).

Autocorrelation and volatility clustering

PLAIN WORDS Does this hour look like the last one? For direction (up or down): no, it is a coin flip. For jumpiness: yes; calm follows calm and storms follow storms.

HOW IT WORKS The autocorrelation function (ACF) correlates a series with itself 1, 2, 3… steps earlier. On returns it tests direction; on squared returns it tests jumpiness.

IN OUR PROJECT Direction −0.024 (the same sign as yesterday only 47% of days), jumpiness +0.184. That is why we predict shocks and volatility, never price direction (notebook 01, section 6).

Rush hours: removing seasonality

PLAIN WORDS Markets have rush hours. A move that is normal at 2 pm on a Wednesday is unusual at 6 am on a Sunday, so each hour is judged against its own usual level.

HOW IT WORKS On training data only, measure the typical jumpiness and volume of each of the 168 hours of the week, then divide every hour's return by its slot's typical size.

IN OUR PROJECT Wednesday 12:00 UTC is 2.17× an average hour, Sunday 06:00 is 0.69×; weekends trade 0.77× the weekday volume (notebook 01, section 14).

The z-score shock rule

PLAIN WORDS "How many normal moves big is this move?" Four times bigger than usual is a shock; as big as usual is not.

HOW IT WORKS z = this hour's (rush-hour-adjusted) return ÷ the coin's typical move over the previous 168 hours, ending the hour before. |z| above 2.29 is a shock; 2.29 was chosen on training data so that about 4% of hours are shocks.

IN OUR PROJECT Using only past hours keeps it honest; the first draft measured each day partly against itself (Part 10, archive table). Daily first version: 2.5 against 30 days.

Robust scaling (median and IQR, training data only)

PLAIN WORDS Put every coin on the same ruler, so the model sees "busy for this coin", not "busy in dollars".

HOW IT WORKS Subtract the coin's median and divide by its interquartile range (the spread of the middle half of its values), which ignores extreme days. Fitted on training dates only.

IN OUR PROJECT Coins listed after the training years (ARB, SUI, PEPE, WLD, JUP, ENA) use the scaler fitted on all training coins (notebook 01, sections 8 and 13).

Huber loss and clipping

PLAIN WORDS If a few huge errors decide how a model learns, one crash day can hijack training. Huber loss counts small errors normally but big ones gently; clipping caps extreme inputs.

HOW IT WORKS Squared error inside a small range, straight-line error outside it. Clipping limits each scaled input to between −5 and +5.

IN OUR PROJECT The LSTM volatility model trains with Huber loss; the shock models use focal loss instead.

Removing duplicate inputs

PLAIN WORDS Two inputs that say the same thing waste the model's effort.

HOW IT WORKS On training data, if two features have a rank correlation above 0.9, keep one.

IN OUR PROJECT At 1 hour the 7-day volatility repeated the 3-day one (0.90), so it went: 6 market features. All 8 network features stayed (highest pair 0.78).

2. How coins move together

Correlation

PLAIN WORDS Do two coins go up and down at the same time? +1 = always together, 0 = unrelated, −1 = always opposite.

HOW IT WORKS Pearson correlation of returns over a window: 30 days on daily data, 168 hours on hourly data.

IN OUR PROJECT Every daily pair is positive (0.39 to 0.80, average 0.57), and higher after crises (0.61 against 0.54). Shared movement is the reason for a graph model (notebook 01, section 10).

The Epps effect

PLAIN WORDS Measured over very short intervals, related prices can look less related, because one reacts a few minutes late and its move lands in the next interval.

HOW IT WORKS Compare the average correlation measured over 1, 2, 4, 12 and 24-hour intervals (Epps, 1979).

IN OUR PROJECT Weak in crypto: 0.58 at 1 hour, 0.60 at 2, 0.61 at 4, 0.62 at 12, 0.59 at a day. Coins trade around the clock on busy exchanges, so prices catch up within the hour (notebook 01, section 16; report section III-C).

Cross-correlation and lead-lag

PLAIN WORDS If Bitcoin moves now, does Ethereum move an hour later? If so, Bitcoin "leads".

HOW IT WORKS Correlate the leader at hour t with the other coin at hour t + k, for every k from −24 to +24, on returns divided by recent volatility so crash hours do not dominate.

IN OUR PROJECT Strong in the same hour (0.65 with BTC, 0.68 with ETH), tiny at every other lag (0.035), with the biggest values at +24 hours: a daily rhythm, not a leader (notebook 01, section 15).

The circular-shift null

PLAIN WORDS To know what counts as "more than chance", build fake data in which the timing link is broken on purpose, and see how big correlations get there.

HOW IT WORKS Rotate one coin's series by a random offset of at least 30 days (the end wraps round to the start). Each coin keeps its own habits, such as calm and stormy spells, but any real timing link is destroyed. Repeat 500 times; the 99th percentile of the largest correlation is the chance line.

IN OUR PROJECT Most pairs beat the line, but only through the 24-hour rhythm, which survives a shift. So the line is only an upper limit, and Granger tests settle the question.

Granger causality

PLAIN WORDS "Does Bitcoin's last few hours help predict Ethereum, beyond what Ethereum's own last few hours already tell us?" It tests extra predictive power, not true cause and effect.

HOW IT WORKS Predict the coin from its own past 3 hours; then add the leader's past 3 hours. If the prediction improves by more than luck allows, the leader "Granger-causes" the coin. The gain in R^2 is the size of the effect.

IN OUR PROJECT After correction only 8 of 45 leader-to-coin and 10 of 45 coin-to-leader tests pass, so neither direction leads. The biggest gain is 0.47% (ETH to QNT), the median 0.02%.

Many tests at once: Benjamini–Hochberg

PLAIN WORDS Run 90 tests and a few will look "significant" by pure luck, like a streak of heads.

HOW IT WORKS Sort the p-values and raise the bar so that, of the results you call significant, at most about 5% are flukes (the "false discovery rate"). It is milder than Bonferroni, which guards against even one fluke and would hide real effects.

IN OUR PROJECT Applied to the 90 Granger tests.

Replication and the lead-lag decision rule

PLAIN WORDS A real pattern should show up again in new data.

HOW IT WORKS Measure the one-hour-later pattern on training data, measure it again on validation, and correlate the two.

IN OUR PROJECT It does not repeat (−0.11 for BTC, −0.40 for ETH). Our rule for adding "who leads" arrows to the graph: beat the null, go one way only, add at least 1% of R^2, and replicate. It failed three of the four, so the graph has no such arrows.

Eigenvalues, the market mode, and the Marchenko–Pastur bound

PLAIN WORDS Look for the main "directions" in which 30 coins move. One direction dominates: everything moving together, "the market".

HOW IT WORKS The 30 eigenvalues of the correlation table add up to 30; each says how much of all the movement one direction explains. Random-matrix theory (the Marchenko–Pastur law) gives the biggest eigenvalue that pure noise could produce for this many coins and hours: here 1.08.

IN OUR PROJECT The biggest is 19.3, so the market explains 64% of hourly movement, and it is the only one above the noise line (notebook 02, section 1; report section IV-D, citing Zarifian and colleagues).

VAR models and BIC

PLAIN WORDS Predict each coin's next return from the recent returns of all coins at once.

HOW IT WORKS A vector autoregression (VAR) is a set of straight-line formulas, one per coin, each using the last p hours of every coin. BIC picks p by rewarding a good fit and charging for every extra number.

IN OUR PROJECT BIC chose 1 hour back on hourly data; the daily cross-check uses 2 days.

Diebold–Yilmaz spillover: TO, FROM, NET and the spillover index

PLAIN WORDS "How much of coin A's surprise comes from coin B?" Add it up over all pairs and you know who mostly sends shocks and who mostly receives them.

HOW IT WORKS From the VAR, a forecast-error variance decomposition (FEVD) splits each coin's 24-hour forecast uncertainty into shares caused by each coin. TO = what a coin sends, FROM = what it receives, NET = TO minus FROM, and the total spillover index (TSI) = the average share coming from other coins. "Generalized" (Pesaran and Shin, 1998) means the answer does not depend on the order the coins are listed in.

IN OUR PROJECT Hourly TSI 91.6%, but hourly NET ranks coins exactly like eigenvector centrality (1.00): at hourly speed it measures how tied a coin is to the market. Daily: TSI 86.0%, ETH the top sender; daily and hourly rankings agree only weakly (0.21) (notebook 02, sections 2 and 2b).

Cholesky against generalized (the first draft's mistake)

PLAIN WORDS The older recipe gives a different answer depending on which coin you list first, like a race in which lane 1 always gets a head start.

HOW IT WORKS Cholesky assumes the first coin can affect the others within the same period but not the other way round; the generalized version makes no such assumption.

IN OUR PROJECT The first draft used Cholesky with Bitcoin in column 1: Bitcoin kept 97% of its own variance and looked like a giant sender (index 54.2%, NET +731.5). Generalized: index 86.0%, ETH on top.

3. The coin network

Correlation network with a fitted threshold

PLAIN WORDS Draw a line between two coins when their last week of returns moved strongly together. Too many lines and everything is joined to everything, which tells you nothing.

HOW IT WORKS Join coins whose 168-hour correlation is at or above a threshold chosen on training data so that about 30% of all possible lines exist: 0.75 at 1 hour, 0.77 at 2 hours. Each coin also keeps its 2 strongest lines, so none is left alone.

IN OUR PROJECT The old draft value of 0.5 joined 75% of pairs, a near-complete "hairball". Final density 30 to 39% (notebook 01, section 16).

Network shape: density, clustering, path length, k-core

PLAIN WORDS Simple ways to describe a network: how many lines there are, whether a coin's friends are also friends with each other, how many hops separate coins, and how big the tightest inner circle is.

HOW IT WORKS Density = lines present ÷ lines possible. Clustering = share of a coin's neighbour pairs that are joined. Shortest path = average hops between coins. Largest k-core = the biggest group in which every coin has at least k lines inside the group.

IN OUR PROJECT The test-period network is denser and more clustered than training (0.35 against 0.30; 0.57 against 0.47), and both rise after crises (notebook 02, section 6).

Minimum spanning tree and tree length

PLAIN WORDS The market's skeleton: the fewest, strongest lines that still connect all 30 coins, with no loops.

HOW IT WORKS Turn correlation into a distance, $d = \sqrt{(2(1 - \rho))}$: 0 for coins that move identically, 1.41 for unrelated ones, 2 for opposites (Mantegna, 1999). Keep the 29 shortest lines that join everything. The average line length (Onnela and colleagues, 2003) shrinks when the market moves as one block.

IN OUR PROJECT Strongest lines DOGE–SHIB and BTC–ETH. DOT has the most lines (11), but to loners. The tree averages 0.66 and shrinks in every crisis (0.36 at its lowest), exactly when spreading money across coins stops protecting you (notebook 02, section 5).

Strength and eigenvector centrality

PLAIN WORDS Strength: how strongly a coin is tied to everyone on average. Eigenvector centrality: you are important if you are tied to important coins.

HOW IT WORKS Strength = average correlation with the other coins. Eigenvector centrality = the coin's share of the biggest eigenvalue's direction in the correlation network; both scaled so the average coin is 1.

IN OUR PROJECT Top on the test window: LINK, DOT, AVAX, ADA, ETH. Both are model inputs.

PageRank on a "depends on" network

PLAIN WORDS Google's old idea: a random surfer clicks from page to page; the pages where the surfer spends most time matter most.

HOW IT WORKS Used only on networks with arrows. The arrow points from coin i to coin j when i is driven by j, so importance collects at the coins others depend on. Scaled so the average is 1.

IN OUR PROJECT On the spillover network it copies the NET ranking (1.00). On the shock-transmission network it differs (0.73): ETC, XRP, POL, AVAX and ADA on top, Bitcoin only 23rd (notebook 02, section 3).

Shock-transmission network and shock lift

PLAIN WORDS Count how often coin j has a shock within an hour after coin i has one, and compare with how often j shocks anyway. The extra is the "lift".

HOW IT WORKS Over the previous 90 days: the chance that j shocks within 1 hour after i did, minus j's normal rate. Needs at least 5 shocks of i. The only network built from time order.

IN OUR PROJECT Gives two model inputs, shock PageRank and shock out-lift (the average extra chance that other coins shock right after this one).

Louvain communities and modularity

PLAIN WORDS Find groups of coins that are more tied to each other than to the rest, like friend groups in a class. First take out "the whole market moved", or everything is one big group.

HOW IT WORKS Remove the market direction from the correlations, then let the Louvain algorithm (Blondel and colleagues, 2008) merge coins into groups to maximise **modularity**: lines inside groups minus what chance would give (0 = no group structure). A fixed seed makes it repeatable.

IN OUR PROJECT Five groups: meme coins; BTC, ETH, DeFi and BNB; newer high-beta platforms; old payment coins; a group led by DOT and ETC. Modularity 0.41 to 0.46, falling in crises (notebook 02, section 4).

NMI and the shuffle test

PLAIN WORDS How much do two ways of sorting the same coins agree, and could that much agreement happen by chance?

HOW IT WORKS Normalised mutual information is 0 for unrelated sortings and 1 for identical ones. Shuffle the sector labels 2,000 times; the share of shuffles that agree as much is the p-value.

IN OUR PROJECT 0.47 against 0.32 for shuffles, p = 0.003: real but partial. Coins group by how they trade as much as by what they do.

Assortativity

PLAIN WORDS Do coins mostly link to their own kind (the same sector), or to others?

HOW IT WORKS A correlation between the sectors at the two ends of each line: positive = mostly same sector, negative = mostly different.

IN OUR PROJECT Slightly negative (−0.14, then −0.05). Contagion is the same inside and across sectors (9.3% against 9.3%), except meme coins (13.7% against 10.9%) (notebook 02, section 7).

Event studies and case studies

PLAIN WORDS Line up the hours around a known event and look closely at what happened, in order.

HOW IT WORKS Cut a window around each crisis (24 hours before to 72 after) and follow returns, shock counts or network measures hour by hour, compared with the weeks before.

IN OUR PROJECT FTX: FTT fell 14 standard deviations on 6 November with nobody following; on 8 November 18 to 24 coins shocked in the same hours as FTT's biggest falls (notebook 02, section 8; notebook 05, section 1).

4. Contagion baselines

Independent cascade

PLAIN WORDS Like a cold going round a class: each sick child gets one chance to infect each other child, with a chance that depends on the pair. A child stays healthy only if every sick child misses.

HOW IT WORKS Kempe, Kleinberg and Tardos (2003). For each pair, p_{ij} = how often j shocked within k hours after i shocked, on training data, pulled towards j's normal rate as if we had 10 extra observations at that rate (so rare pairs get no wild numbers). With several coins shocked at once: chance(j) = $1 - (1 - p_{1j}) \times (1 - p_{2j}) \times \ldots$

IN OUR PROJECT Test AUPRC 0.159 at 1 hour against 0.105 for random. Its pair numbers alone do no better than the base rate, so its gain comes from counting how many coins shocked at once. About 900 numbers, so it cannot memorise (notebook 02, section 9).

Base rate

PLAIN WORDS The laziest sensible guess: each coin's usual chance of shocking.

HOW IT WORKS Give every event the receiving coin's training-period shock rate. It scores about the positive rate, which is why the positive rate is the "random" line.

IN OUR PROJECT 0.106 at 1 hour on test.

5. The learning models

TGN (the main model)

PLAIN WORDS A diary for every coin, a habit of looking at each coin's 10 latest events, and a small judge that reads both coins' situation and gives 5 chances.

HOW IT WORKS Rossi and colleagues (2020): GRU memory of 64 numbers per coin, attention with 2 heads over 10 recent events, prediction head 161 → 64 → 5. 70,021 parameters.

IN OUR PROJECT Everything about it is in Part 7.

DyGFormer

PLAIN WORDS A second opinion with no diary: every time, it rereads each coin's latest history from scratch.

HOW IT WORKS Yu and colleagues (NeurIPS 2023). For each event it takes the last 16 interactions of both coins, notes who they were with, when, and how often the same partners appear on both sides (**neighbour co-occurrence**), cuts the sequence into pieces of 2 and runs a small Transformer (2 layers, 2 heads, width 32). Our compact rebuild: 24,021 parameters. The official library was not used because its random "negative" examples and fraction-based splits would break our no-peeking rules.

IN OUR PROJECT Tied with the TGN on test at every window (0.208 against 0.205 at 1 hour) (notebook 03, section 4b).

Gradient boosting (histogram trees)

PLAIN WORDS A team of small decision trees ("if volume is unusual and jumpiness is high, risk goes up"); each new tree fixes the mistakes of the trees before it.

HOW IT WORKS scikit-learn's histogram gradient boosting, which sorts each input into ranges to train fast. Learning rate 0.05, at most 31 leaves per tree, at least 200 events per leaf; trees added 50 at a time (up to 1,000) until validation stops improving. Trees per window: 100, 150, 50, 200, 150.

IN OUR PROJECT Same inputs as the TGN's head plus shock state and market breadth. 0.178 on test at 1 hour; it memorised its training years (0.64 on training against 0.155 on validation).

LSTM

PLAIN WORDS Reads one coin's last week hour by hour, keeping a running memory. It never sees other coins.

HOW IT WORKS Hochreiter and Schmidhuber (1997): 1 layer, 32 memory units, dropout 0.2, 5,665 parameters; 168-hour windows of 6 features; focal loss for shocks, Huber loss for volatility.

IN OUR PROJECT Next-hour shock: 0.109, 2.8 times random. Volatility: beaten by HAR-RV.

HAR-RV

PLAIN WORDS Future jumpiness is roughly a weighted mix of the last hour's, the last day's and the last week's jumpiness. Simple and very hard to beat.

HOW IT WORKS Corsi (2009): a straight-line formula, future volatility = a + b × last hour + c × last 24 hours + d × last 168 hours, with a to d fitted by least squares on training data.

IN OUR PROJECT Beats the LSTM at every window (test error 5.0 against 5.5×10^{-5} at 1 hour).

Logistic regression on simple rules

PLAIN WORDS One weighted sum of a few numbers, squeezed into a chance.

HOW IT WORKS Fitted on training data; each weight says how much a rule pushes the risk.

IN OUR PROJECT Next-hour shock with 4 rules (how unusual the move is now, shock now, own shocks in the last day, market breadth): 0.095. In the final review a 4-feature version matched the TGN on contagion (Part 9).

Graph convolution (GCN)

PLAIN WORDS Each coin updates its numbers by listening to its neighbours, louder for stronger links, then reshapes the result with learned knobs.

HOW IT WORKS One layer computes, for every node, a weighted average of its own and its neighbours' numbers, then applies a learned linear map. Two layers let news travel two hops.

IN OUR PROJECT The first two layers of the T-GAE (14 → 32 → 16 numbers per coin).

Temporal graph autoencoder (T-GAE)

PLAIN WORDS Learns what a normal day of the coin network looks like and tries to predict tomorrow. If tomorrow looks very different from what it expected, the day is "surprising": an anomaly.

HOW IT WORKS For each daily network: two GCN layers, a GRU carrying each coin's state from day to day, and two decoders, one guessing tomorrow's links and one guessing each coin's 14 activity numbers. The anomaly score is the prediction error. 3,646 parameters, trained on 2020 to 2022, 5 seeds.

IN OUR PROJECT Barely separates stress days (0.511 against 0.893 for plain turbulence); it did rank 10 October 2025 as the strangest test day (notebook 06).

Ensembles: of seeds and of models

PLAIN WORDS Ask several experts and combine their opinions.

HOW IT WORKS Seed ensemble: average the 5 runs' chances. Model ensemble: turn each model's scores into ranks (so they share one scale), then a weighted average with weights picked per window on validation.

IN OUR PROJECT Seeds: +0.001. Models: better than the TGN at 24 hours (+0.008), worse at 1 to 3 hours.

6. Training tricks

Focal loss

PLAIN WORDS A teacher who spends little time on questions everyone gets right and more on the hard, rare ones.

HOW IT WORKS Lin and colleagues (2017): loss = $-\alpha\,(1-p)^{\gamma}\log p$, where p is the chance given to the right answer. When p is close to 1 (easy), $(1-p)^{\gamma}$ is tiny, so the example barely counts. $\gamma = 2$; $\alpha = 0.25$ for "yes" events and 0.75 for "no" events.

IN OUR PROJECT Used by the TGN, DyGFormer and the shock LSTM (report equation 8).

Early stopping, patience and a minimum number of epochs

PLAIN WORDS Stop studying when mock-exam scores stop improving, keep your best version, and do not quit after one lucky early test.

HOW IT WORKS After each epoch, score validation AUPRC at 1 hour; keep the best; stop after 10 epochs without improvement; for DyGFormer and the no-memory model, never before epoch 30.

IN OUR PROJECT The first final runs (30 epochs, patience 4) stopped while still improving: 0.138, then 0.153 with the longer budget.

Weight decay and dropout

PLAIN WORDS Two ways to stop memorising: keep the knobs small, and switch random neurons off while learning so the model cannot lean on any one.

HOW IT WORKS Weight decay adds a small pull towards zero on every knob (0.0001); dropout zeroes a random 20% of the numbers at each step.

IN OUR PROJECT More of both (dropout 0.4, weight decay 0.001) or a smaller network only made validation worse (0.143 and 0.133 against 0.153).

Seeds and reproducibility

PLAIN WORDS Training starts from random numbers; the seed fixes them so a run can be repeated exactly.

HOW IT WORKS 5 seeds (0 to 4) per final model, deterministic settings, every run logged with its git commit.

IN OUR PROJECT Seeds agree within about ±0.003 (±0.005 in development runs); a replay of the final TGN reproduces its stored test score within 0.0007.

Refitting on training plus validation

PLAIN WORDS After choosing the settings, train once more on more data before the final exam.

HOW IT WORKS Retrain on everything up to December 2023, with the number of epochs (77) or trees fixed in advance, because there is no mock exam left to stop on.

IN OUR PROJECT Helped boosting (0.178 to 0.185), hurt the TGN (0.205 to 0.188). Not re-tuned, because that would be tuning on the test.

Calibration (binned isotonic)

PLAIN WORDS Making chances honest: of all the times the page says 30%, about 30% should come true.

HOW IT WORKS On validation data only, group events by score (at least 500 per group) and fit an always-rising curve from score to real frequency; chances are also forced to grow with the window (1 hour ≤ 3 hours ≤ … ≤ 24 hours).

IN OUR PROJECT Used by the dashboard (Part 14). The report notes calibration was not checked on the test period.

7. Scoring and statistics

AUPRC, AUROC, positive rate and lift

PLAIN WORDS AUPRC: how well real "yes" events are pushed to the top of the list; random scores the share of "yes" events. AUROC: the chance a random "yes" is ranked above a random "no" (0.5 = coin flip). Lift: a score divided by the random score.

HOW IT WORKS AUPRC is computed as average precision (report equation 9): walk down the list, and at each new "yes" found, add (gain in recall) × (precision so far).

IN OUR PROJECT TGN 0.205 against 0.105 at 1 hour (lift 1.95); AUROC 0.64 to 0.69. Slowly explained in Part 5.

Precision at the top

PLAIN WORDS Of the alerts the model was most sure about, how many came true?

HOW IT WORKS Sort test events by predicted risk, take the top 1% (or 5%, 10%), and count the share that really were followed by contagion.

IN OUR PROJECT Top 1%: 32% within 1 hour (random 10.5%), 61% within 3, 76% within 6.

Day-block bootstrap and p-values

PLAIN WORDS Could the win be luck? Rebuild the test period 1,000 times from randomly drawn whole days and see how often the winner still wins.

HOW IT WORKS Draw 971 test days with replacement; recompute both models' AUPRC on the same events; repeat 1,000 times (report equation 10). The middle 95% of the differences is the interval; the share of rebuilds where the difference is zero or below is the one-sided p-value. Whole days, because one crash creates hundreds of linked events.

IN OUR PROJECT TGN against cascade: +0.046 (0.019 to 0.074) at 1 hour; no rebuild favoured the cascade.

Training-to-validation lift ratio (an overfitting check)

PLAIN WORDS If one student aces the homework but slips much more on new questions than a student who cannot memorise, the first one memorised.

HOW IT WORKS Lift on training ÷ lift on validation. The cascade cannot memorise, so its ratio (about 1.1) is just the market changing.

IN OUR PROJECT TGN about 1.5 at 1 hour: some overfitting at short windows, none at 24 hours (notebook 03, section 5).

Mean squared error

PLAIN WORDS The average of the squared misses; lower is better. Squaring makes big misses count extra.

HOW IT WORKS Compare predicted and real volatility over the next 1, 6 and 24 hours, against HAR-RV, "same as last period", the coin's average and zero.

IN OUR PROJECT HAR-RV lowest at every window.

8. Explaining the model

Integrated gradients and completeness

PLAIN WORDS Start from an "average situation" and slide step by step to the real one, writing down how much each input pushes the score along the way. The pushes add up to the whole change.

HOW IT WORKS Sundararajan and colleagues (2017): add up the model's gradient along a straight path from the baseline (the training average) to the real input. **Completeness**: the pushes sum exactly to the difference between the real score and the baseline score; we check it every run (largest gap 0.009).

IN OUR PROJECT Receiver 65%, source 25%, event 10%; the top single reason is the receiver's volume against its past week (25%).

TempME

PLAIN WORDS Looks for small patterns of past events ("motifs", such as a → b then b → c) around the two coins of an alert, and scores which recent events mattered.

HOW IT WORKS Chen and Ying (NeurIPS 2023): sample motifs of up to 3 steps around both coins, learn an importance score for each with an information bottleneck (below), and apply the scores as weights on the TGN's attention. Trained on 10,000 validation events with the official settings.

IN OUR PROJECT Barely better than random (0.157 against 0.152), because there are no special chains to find.

Information bottleneck

PLAIN WORDS Explain with as little as possible: an explanation that keeps everything explains nothing.

HOW IT WORKS The explainer is rewarded for keeping the prediction and charged for how much it keeps, pulled towards keeping about 30% of events ("prior sparsity 0.3"); $\beta = 0.5$ sets the balance and a temperature of 0.07 makes keep-or-drop almost a yes/no choice.

IN OUR PROJECT Part of TempME.

Fidelity and the characterization score

PLAIN WORDS A good explanation passes two tests: the chosen events alone are enough (keep only them and the prediction stays), and they are needed (remove them and it changes).

HOW IT WORKS Fidelity− = change when keeping only the chosen events (lower is better); fidelity+ = change when removing them (higher is better); both divided by the change from removing all recent events. Characterization combines them (a harmonic mean, as in GraphFramEx, Amara and colleagues, 2022).

IN OUR PROJECT With 3 events kept: TempME 0.157, random 0.152, attention 0.147, most recent 0.192, a mask tuned for every single alert 0.248.

Attention entropy

PLAIN WORDS Is the model's attention focused on a few events, or spread evenly like butter on toast?

HOW IT WORKS Normalised entropy of the attention weights: 0 = all weight on one event, 1 = perfectly even.

IN OUR PROJECT 1.000: the TGN averages its 10 recent events instead of picking some.

Time-shuffled null for motifs

PLAIN WORDS If real event patterns look like patterns in a stream with shuffled times, there is no special pattern.

HOW IT WORKS Shuffle the event times to break their order, count the motif types in both streams, compare.

IN OUR PROJECT 83% against 86% of motifs are three-step walks through four different coins; only walks that return to a coin are more common in real data (5.9% against 3.3%).

Plausibility with Spearman correlation

PLAIN WORDS Do the coins an explanation credits match the coins network analysis calls central?

HOW IT WORKS Each top alert gives credit to the coins of its 3 most important events; compare the credit ranking with the network rankings by rank correlation, also after dividing by how often each coin appears at all ("lift over exposure").

IN OUR PROJECT TempME runs against the rankings (about −0.8), so its credit is not evidence of who spreads contagion.

The rebuild check

PLAIN WORDS Before explaining a model, prove you are explaining the real one.

HOW IT WORKS Capture the inputs the TGN used for each event, recompute its output from them, and compare with the stored output.

IN OUR PROJECT Largest difference 2.4×10^{-7}.

9. The life of a shock

Fresh shocks and breadth types

PLAIN WORDS Count each storm once, and sort storms by how many coins they hit at once.

HOW IT WORKS A fresh shock has no shock of the same coin in the previous 24 hours. Breadth = share of the other coins shocking in the same hour: market-wide 30% or more, partial 10 to 30%, isolated (idiosyncratic) under 10%; plus crash or pump.

IN OUR PROJECT 18,171 fresh shocks out of 57,792 shock hours (report Table V).

Hours back to normal

PLAIN WORDS How long until a shocked coin is as calm as it was before?

HOW IT WORKS For each fresh shock, the first hour after which the coin's volatility is back at its level before the shock; report the median, with intervals from redrawing whole days.

IN OUR PROJECT 8 to 10 hours for every type; 87% within a day.

Re-shocks, followers and continuation

PLAIN WORDS After a shock: does the same coin shock again, how many other coins follow, and does the price keep going or bounce back?

HOW IT WORKS Re-shock = another shock of the same coin within 24 hours. Followers = other coins shocking within 6 hours. Continuation = the return after the shock, in units of the coin's volatility, signed so that positive means "the move continued".

IN OUR PROJECT Re-shock 55 to 65%; followers 8.1 after a market-wide crash against 4.5 to 5.1 after an isolated shock; direction mostly unpredictable.

Matched null and half-life

PLAIN WORDS Compare the hours after a shock with the hours after an ordinary calm hour of the same coin in the same month, then ask how fast the difference halves.

HOW IT WORKS Extra jumpiness over the next 72 hours minus the same profile after a random calm hour; the half-life is the first hour at which the extra is half its value at hour 1.

IN OUR PROJECT About 2.5 hours in every sector, with no sector differences.

Blast radius of a crisis

PLAIN WORDS How many coins did a crisis hit, how much more often than usual, and how fast?

HOW IT WORKS Start each crisis at the hour of its date with the most shocks; count the coins hit within 72 hours and compare the shock rate with the week before.

IN OUR PROJECT All 9 crises were market-wide and reached every coin; LUNA raised the shock rate most (+0.19), the ETF approval least (+0.01).

10. Stress tests and compression

Random input noise

PLAIN WORDS Shake the inputs a little and see if the answers wobble.

HOW IT WORKS Add random bell-curve noise of size σ to both coins' features and the event's numbers. Inputs are standardised, so σ = 0.5 means half a typical deviation.

IN OUR PROJECT σ = 0.5: TGN −5.7%, boosting −4.7%. The TGN's chances move half as much.

Missing events

PLAIN WORDS Pretend part of the data feed never arrived.

HOW IT WORKS A random share of test events never reaches the TGN's memory and recent lists; every event is still scored.

IN OUR PROJECT Up to 50% missing: under 0.5% change.

FGSM attack

PLAIN WORDS The worst small nudge: an attacker who knows the model pushes every input a little, exactly in the direction that hurts it most.

HOW IT WORKS Fast gradient sign method (Goodfellow and colleagues, 2015): new input = input + ε × (the sign of the loss gradient), using the true answers. "White-box": the attacker sees everything.

IN OUR PROJECT ε = 0.1 cuts the 1-hour score by about 35%; random noise of the same size costs under 1%.

Transfer attack

PLAIN WORDS Use inputs crafted to fool one model on a different model the attacker never saw.

HOW IT WORKS Feed the TGN's FGSM inputs to gradient boosting.

IN OUR PROJECT No harm at 1 hour, 11 to 33% lower at 6 hours.

PGD attack

PLAIN WORDS A more patient attacker: many small nudges instead of one big one.

HOW IT WORKS Projected gradient descent: 10 FGSM steps from a random start, each time pulled back inside the allowed ±ε box.

IN OUR PROJECT Within 0.002 of FGSM at every size, so FGSM was already close to the worst case.

Driver stability

PLAIN WORDS If inputs wobble a little, does the explanation still name the same reasons?

HOW IT WORKS Recompute the integrated-gradient reasons under noise; compare the top reason, and the top-5 lists by Jaccard similarity (shared items ÷ all items).

IN OUR PROJECT The top reason stays for 99, 98 and 97% of the 2,000 riskiest alerts at noise 0.05, 0.1 and 0.25.

Stress days and turbulence

PLAIN WORDS An outside yardstick for "the market went wild today", taken from the returns, not from any model.

HOW IT WORKS Stress day = at least one hour in which 70% or more of the coins shocked (20.7% of days). Turbulence = the day's average |z| across coins.

IN OUR PROJECT Turbulence spots stress days well (0.893); the T-GAE does not (0.511).

Quantisation and pruning

PLAIN WORDS Make the model file smaller by storing its numbers with fewer digits (quantisation) or by deleting the weakest connections (pruning).

HOW IT WORKS int8 quantisation stores knobs as 8-bit whole numbers instead of 32-bit decimals; 50% pruning sets the smallest half of the knobs to zero.

IN OUR PROJECT int8: 3.1 times smaller but 1.8 times slower on CPU and 0.004 to 0.007 less AUPRC; pruning: no gain. The dashboard serves the full model.

PART 12

The report, section by section

After this part you can answer anything the professor points at in the report: every section, table, figure, equation and reference.

SIMPLE VERSION

The report is the 9-page paper you submitted on 19 October 2026, written in the IEEE conference style (two columns, numbered sections, 38 references). The professor has read it and may point at any line: "explain equation 4", "what is Table VI?", "who is reference 1?". This part walks through it in order.

Title: "CryptoGraph-Secure: Predicting Cryptocurrency Shock Contagion with Temporal Graph Networks". Hardik Gulati (PRN 24070126063) and Dakshesh Chauhan, Division AIML–A, Department of AI and ML, Symbiosis Institute of Technology, Pune; CA3, Deep Learning, AY 2026–27. Keywords: cryptocurrency, financial contagion, temporal graph network, dynamic graph learning, spillover, explainability, deep learning, class imbalance.

CAREFUL

Dakshesh's PRN is left blank on the cover and on page 1 ("PRN: ____"). If you bring a printed copy, fill it in by hand or be ready to say it.

The report at a glance

Section	What it says	Be ready for
Abstract	The whole story in one paragraph, with the key numbers	"Summarise your work"
I. Introduction	Why "who is hit next" matters; the exact task; why deep learning; four research questions; four contributions	"What are your research questions?", "Why deep learning?"
II. Literature review	38 references in five groups, and the gap we fill	"Who made the TGN?", "What is new here?"
III. Methodology	Data, cleaning, equations 1 to 3 (shock and label), network analysis, the models (equations 4 to 7, Figures 1 and 2), training (equation 8, Table II), scoring (equations 9 and 10)	"Explain equation 4", "Why AUPRC?"
IV. Results	Tuning (Figure 3), test results (Table III, Figures 4 and 5, Table IV), shock types (Table V, Figure 6), explanations and network findings, robustness (Figure 7), limitations, Table VI	"Is the TGN really better than boosting?", "What is Table VI?"
V. Conclusion	What we showed, and five lines of future work	"What would you do next?"
Acknowledgment	AI use, data availability, reproducibility (197 tests passed, 5 skipped; replay within 0.0007)	"Did you use AI?"

The abstract in plain words

Most studies of crypto contagion measure how strongly coins move together; they do not predict which coin is hit next. We turned that into a prediction task: when coin i has a shock, will coin j have one within 1, 3, 6, 12 or 24 hours? We built a fair, no-peeking test from Binance hourly data (30 coins, 2020 to 2026, 1.46 million events) and raced a TGN against DyGFormer, gradient boosting, an LSTM, an independent cascade and classic finance baselines. On 2024 to 2026 the TGN scores 0.205 at 1 hour (random 0.105, cascade 0.159) and 0.739 at 24 hours. It beats the cascade everywhere and boosting at 1 to 3 hours (borderline); DyGFormer ties it; memory helps only from 6 hours. The

signal is mostly the receiving coin's current state. A deliberate attack cuts the score by about 35%; noise and missing data do little. Negative results and every test-period score are reported.

I. Introduction

- **Motivation:** crypto trades around the clock and reacts violently to news; LUNA (May 2022), FTX (November 2022) and 10 October 2025 each pulled most big coins down within hours. A trader, an exchange's risk desk or a regulator wants to know which coins will be hit next, and how soon. Classic measures (correlations, spillover indices) describe average links; they do not say who is next. Daily data mash several shocks and their order into one number; hourly data keep the order.
- **Problem statement:** coin i shocks at hour t; for every other active coin j, predict whether j shocks within k hours, $k \in \{1, 3, 6, 12, 24\}$ ("$\in$" means "is one of"). Only 10.5% of events are "yes" at 1 hour, and everything depends on time, so the model must know who interacts with whom, and the test must never leak the future.
- **Why deep learning:** a coin's risk depends on its own state, on the shocking coin's state and on their recent history together. A TGN keeps a learned memory per coin that every event updates, which matches this; an LSTM sees one series only, and trees see one fixed snapshot of features. Whether the extra machinery is worth it is tested against strong baselines with significance tests.

The four research questions and our answers

Question	Answer	Evidence
RQ1. Do temporal graph models predict hourly contagion better than a fitted network-diffusion model?	Yes	TGN and DyGFormer beat the cascade at every window: +0.046 at 1 h, +0.084 at 6 h; no bootstrap rebuild favoured the cascade
RQ2. Do they beat plain machine learning on the same inputs, and does memory matter?	Barely, and only at long windows	Against boosting +0.028 and +0.023 at 1 and 3 h (p about 0.03, borderline), tied from 6 h; memory adds +0.010 to 0.013 at 6 to 24 h, nothing at 1 h
RQ3. What drives their alerts?	The receiving coin's current state	About 65% of the attribution; volume surge 25%, 24 h volatility 12%; TempME finds no special chains of events
RQ4. How long do shocks last, and how robust is the model?	About half a day; robust to noise, not to attacks	Back to normal after a median of 8 to 10 h; 50% missing events under 0.5%; FGSM 0.1 about −35%

The four contributions, in plain words: (1) a fair, no-peeking hourly test for contagion with 1.46 million events over 30 coins and seven calendar years (2020 to 2026); (2) a careful race between the TGN, DyGFormer, boosting, the LSTM, the independent cascade and HAR-RV, with five seeds and whole-day significance tests; (3) TempME rebuilt for our TGN, plus integrated gradients, showing where the signal lives; (4) studies of how long shocks last, how robust the model is and whether an anomaly detector works, including the honest failures.

II. Literature review: the 38 references

You do not need to know every paper, but you should know why each is cited. Grouped as in the report.

#	Who, year	What it is, plainly	Why we cite it
A. Contagion and spillovers in finance			
1	Forbes and Rigobon, 2002	"No contagion, only interdependence": much apparent contagion is ordinary co-movement that looks bigger in wild times	We judge shocks against each coin's own volatility and only count strictly later shocks
2	Diebold and Yilmaz, 2012	A spillover index from a VAR's variance decomposition ("Better to give than to receive")	Our spillover network and NET ranking
3	Diebold and Yilmaz, 2014	The same idea drawn as a network of connectedness between firms	Same
10	Billio, Getmansky, Lo and Pelizzon, 2012	Networks of financial firms as a measure of systemic risk	Networks as a lens on systemic risk
11	Pesaran and Shin, 1998	The "generalized" decomposition that does not depend on the order of the variables	We use it instead of Cholesky
12	Antonakakis, Chatziantoniou and Gabauer, 2019	Crypto market contagion and changing connectedness	Spillovers among coins rise in stress
13	Ji, Bouri, Lau and Roubaud, 2019	Dynamic connectedness and integration in crypto markets	Same
4	Kammoun, 2026 (Digital Finance)	What drives return and volatility spillovers among cryptocurrencies	Recent spillover work: measures, does not predict
5	Nefzi, Kayral, Loukil and Jeribi, 2026 (Digital Finance)	Whether safe havens (Bitcoin, gold, currencies) still protect investors	Same

#	Who, year	What it is, plainly	Why we cite it
14	Zarifian, Overbeck, Gallus and Blasiak, 2026 (Digital Finance)	Spotting crises from the biggest eigenvalues of the correlation table, using the Marčenko–Pastur law	Matches our single dominant market factor
15	Vo, 2026 (Digital Finance)	The USDT premium as a crisis signal, 2019 to 2026	Why we watch the USDT peg
6	Mohamad, 2026 (Digital Finance)	Hourly data to catch contagion after cybercrime (hacks) in crypto	Recent work uses hourly data
7	Teruel-Gutiérrez and Aparicio Serrano, 2026 (Digital Finance)	How Ether reacts, hour by hour, to US inflation news	Same
B. Network science of markets			
16	Mantegna, 1999	The minimum spanning tree of a correlation distance shows the market's hierarchy	Our MST
17	Onnela, Chakraborti, Kaski, Kertész and Kanto, 2003	The tree shrinks during crashes	Our tree length
18	Blondel, Guillaume, Lambiotte and Lefebvre, 2008	The Louvain method for finding communities	Our communities
19	Kempe, Kleinberg and Tardos, 2003	The independent cascade: each newly active node gets one chance to activate each neighbour	Our network-science baseline
20	Alamsyah, Salsabila and Rizal, 2026 (Digital Finance)	A network of over seven million Ethereum transactions across six DeFi tokens	Network science applied to crypto

#	Who, year	What it is, plainly	Why we cite it
C. Temporal graph learning			
8	Rossi, Chamberlain, Frasca, Eynard, Monti and Bronstein, 2020	The Temporal Graph Network (Twitter research; arXiv 2006.10637)	Our main model
21	Cho and colleagues, 2014	Introduced the GRU, inside a model for machine translation	The TGN's memory update
22	Xu, Ruan, Korpeoglu, Kumar and Achan, 2020	TGAT: attention over a node's time-stamped neighbours, with a time encoding	The TGN's attention and time code come from TGAT
23	Yu, Sun, Du and Lv, 2023	DyGFormer, a Transformer over each node's history, and the DyGLib library	Our second model
24	Vaswani and colleagues, 2017	The Transformer ("Attention is all you need")	Basis of DyGFormer and of the TGN's attention layer
25	Peng, Khushi and Poon, June 2026	CryptoGAT: crypto price prediction as a cross-coin graph problem with graph attention; questions purely time-series models	Closest graph work, but it predicts prices, not contagion
26	Gagneja, Gupta, Batish, Saini and Rani, 2026 (Digital Finance)	ES-LSTM, a hybrid LSTM for financial forecasting	LSTMs are a common choice; ours is a baseline
D. Explaining, imbalance and scoring			
27	Chen and Ying, 2023	TempME: explains temporal graph models with motifs chosen under an information bottleneck	Our event-level explainer

#	Who, year	What it is, plainly	Why we cite it
28	Sundararajan, Taly and Yan, 2017	Integrated gradients	Our feature-level explainer
29	Amara and colleagues, 2022	GraphFramEx: how to score explanations of graph models (fidelity)	Our characterization score
30	Lin, Goyal, Girshick, He and Dollár, 2017	Focal loss, first made for spotting objects in images	Our loss for rare positives
31	Davis and Goadrich, 2006	Precision–recall curves beat ROC curves when positives are rare	Why AUPRC is the main score
32	Corsi, 2009	HAR-RV, the standard simple volatility model	Volatility baseline
33	Goodfellow, Shlens and Szegedy, 2015	Adversarial examples and the FGSM attack	Our attack test
Data, tools and the rest			
34	Binance	The public data archive (data.binance.vision)	Our hourly prices
35	Coin Metrics	Free community data	Cross-check and the USDT peg
36	Epps, 1979	Correlations look weaker over very short intervals	Our Epps check
37	Fey and Lenssen, 2019	PyTorch Geometric	The graph library (TGNMemory, TransformerConv)
38	Kingma and Ba, 2015	The Adam optimizer	How every neural model was trained

The gap (section II-E), in plain words: finance papers measure how connected coins are or how they react to events; graph-learning papers test their models on "will these two users interact?". Nobody

had joined the two: a fair hourly test for coin-to-coin contagion, a fair race between temporal graph models and finance, network and machine-learning baselines, and a look at what the models learn.

III. Methodology

A. The data and Table I

- **Which coins:** the first 30 coins in CoinGecko's market-cap ranking of 27 September 2026 (from the top 60, leaving out stablecoins and "wrapped" tokens, which are copies of other coins) that have a Binance USDT market, at least 730 days of hourly history and at least 90% coverage of the test period. LUNC and FTT are kept only as event coins; XMR was left out because Binance delisted it in February 2024. Every exclusion is recorded with its reason.
- **Checks:** 3,049 monthly files verified against their SHA-256 fingerprints, 0 failures. Coin Metrics daily prices are used only to cross-check (median difference about 0.1%) and to flag USDT depegs; that check ends on 23 May 2026, where the archive ends.
- **Traps:** the LUNAUSDT name covers two different coins (old Terra LUNA and the new LUNA) and is never stitched together; MATIC to POL is a 1:1 swap with its 79-hour gap masked; delisted coins and gaps longer than 2 hours are masked, never filled, while shorter gaps are filled with the last price and flagged; Binance's timestamps switch from milliseconds to microseconds in 2025 and are converted to UTC.

Table I, row by row

Row	Value	In plain words
Coins, period	30 coins; 1 Jan 2020 to 27 Sep 2026; hourly, UTC; about 1.61 million coin-hours	Every hour of every coin since it was listed
Train / validation / test	to 30 Jun 2022 / to 31 Dec 2023 / 1 Jan 2024 to 27 Sep 2026; 7-day embargo	Practice, mock exam, final exam, with a week's gap at each border
Shock events $i \to j$	1,463,693 (361 k / 301 k / 802 k); 795,787 test events scored at 1 h	One question per shock per other coin; the scored test events exclude the embargo week and windows that run past the end
Event vector (5)	lag-1 and same-hour correlation of i and j, return of i, volatility of i, shock flag	The 5 numbers describing each event
Node features (14)	6 market + 8 network	What the model knows about each coin
Labels (5)	y(k): j shocks within hours t+1 to t+k, for k = 1, 3, 6, 12, 24	Five yes/no answers per event
Positive rate, test	10.5%, 20.6%, 31.0%, 44.7%, 62.7%	How often the answer is "yes"; also the score of random guessing

B. Cleaning and the labels

On daily data, prices fail the stationarity test and log returns pass, so all models use log returns; returns have fat tails (excess kurtosis about 18), so inputs are robust-scaled with training medians and interquartile ranges, and number targets use Huber loss. Hourly jumpiness follows the clock (0.69 to 2.17 times its average by hour of the week), so a training-period profile is divided out. Everything fitted (scalers, rush-hour profile, graph thresholds, the shock threshold) uses training data only, and an event is scored only if its whole answer window lies inside its own period. Labels that count the same hour were rejected, because they would call simultaneous co-movement contagion. Returns barely correlate with their own past while their sizes do, so the project predicts shocks and volatility, never direction.

The ten equations in plain words

#	The equation	What it says
1	$z_{i,t} = \tilde{r}_{i,t} / \hat{\sigma}_{i,t-1}$	The shock score: this hour's rush-hour-adjusted return ($\tilde{r}$) divided by the coin's typical hourly move ($\hat{\sigma}$) over the 168 hours up to the hour before.
2	$S_{i,t} = 1[\ \lvert z_{i,t} \rvert > \theta\ ]$	The shock flag: 1 if the size of z is above $\theta = 2.29$, otherwise 0. "1[...]" means "1 if true, 0 if false".
3	$y^{(k)}_{i \to j,t} = 1[\ \Sigma_{u=t+1 \ldots t+k}\, S_{j,u} \geq 1\ ]$	The answer: 1 if coin j has at least one shock in hours t+1 to t+k. The sum starts at t+1, so the same hour never counts.
4	$m_i(t) = [\ s_i(t^-) \,\|\, s_j(t^-) \,\|\, e_{ij}(t) \,\|\, \phi(\Delta t_i)\ ]$	The message for coin i's diary: glue together i's diary just before t (64 numbers), j's diary (64), the event's 5 numbers, and a 16-number time code of how long since i's diary last changed. 149 numbers in all.
5	$s_i(t) = \mathrm{GRU}(\ s_i(t^-), m_i(t)\ )$	The diary update: the GRU reads the old diary and the message and writes the new diary.
6	$h_i = \mathrm{Attn}(\ s_i, \{s_u, e_{iu}, \phi(\Delta t_u)\}$ for u in $N_{10}(i)\ )$	The coin's current summary (64 numbers): attention over its 10 most recent partners u, using their diaries, the events and how long ago they happened.
7	$\hat{p}_{ij} = \sigma(\ W_2\ \mathrm{ReLU}(\ W_1\ [\ h_i \,\|\, h_j \,\|\, x_i \,\|\, x_j \,\|\, e_{ij}\ ]\ )\ )$	The prediction: glue both summaries (64 + 64), both coins' 14 features (14 + 14) and the event's 5 numbers (161 in all), pass them through a 64-neuron layer with ReLU (W_1) and an output layer (W_2), and squeeze the 5 outputs into chances with the sigmoid σ.

#	The equation	What it says
8	$FL(p_t) = -\alpha_t\,(1-p_t)^{\gamma}\log p_t$	Focal loss: p_t is the chance the model gave to the right answer. Easy cases (p_t near 1) are shrunk by $(1-p_t)^{\gamma}$; $\gamma = 2$, $\alpha = 0.25$ for "yes", 0.75 for "no".
9	$AP = \Sigma_n\,(R_n - R_{n-1})\,P_n$	AUPRC as average precision: walk down the ranked list; each time recall goes up, add that step times the precision at that point.
10	$\Delta^{(b)} = AP_A(D^{(b)}) - AP_B(D^{(b)})$	The bootstrap: in rebuild b of the test days ($D^{(b)}$), the difference between model A's and model B's score. The middle 95% of 1,000 such differences is the interval.

SAY IT IN THE VIVA

Equation 4 is the one most worth practising: "Each time a coin is involved in an event, we build a message from both coins' memories, the event's five numbers and a time code; equation 5 then lets a GRU write that message into the coin's 64-number memory. Equations 6 and 7 read the memories back out through attention and a small two-layer network to give five chances."

C. Network analysis

Coins are the actors; moving together, or passing on shocks, are the ties. The report lists: correlation networks with a threshold fitted for 30% density (0.75 at 1 hour), minimum spanning trees, generalized spillover networks on 30-day windows, a shock-transmission network with arrows, PageRank and eigenvector centrality, Louvain communities compared with sectors by NMI, and the independent cascade as a baseline. Eight node features per daily snapshot join the six market features to give the 14-number coin description, always taken from the latest snapshot at or before the hour. The Epps effect is weak (0.58 at 1 hour to 0.62 at 12 hours).

D. The models: Figures 1 and 2

- **Figure 1** is the pipeline from raw hourly prices to evaluated and explained predictions (the same stages as Part 13).
- **Figure 2** is the TGN: memory and temporal attention feeding a 161 → 64 → 5 head; a dashed arrow marks the memory update, which happens only *after* a batch has been scored. 70,021

parameters.

- **"Memory and neighbour lists are updated only after a batch has been scored"**: the model must give its answer before it is allowed to learn from that hour's events, otherwise it would peek. And **batches never split an hour**, so no event can see the update from another event of the same hour.
- **DyGFormer**: a compact rebuild (24,021 parameters; 2 layers, 2 heads, width 32, pieces of 2, last 16 events, neighbour co-occurrence). The official library was not used because its random "negative" examples and fraction-based splits would break the no-peeking rules.
- **Baselines**: (i) gradient boosting on the TGN head's inputs without memory or attention, plus coin ids and shock-state context; (ii) the LSTM on each coin's last 168 hours (5,665 parameters); (iii) the independent cascade; (iv) HAR-RV and simple rules; (v) the TGN with memory and attention removed.

E. Training and Table II

All neural models: Adam with weight decay, focal loss (equation 8), early stopping on validation AUPRC at 1 hour, five seeds (0 to 4). Code: Python 3.12 with PyTorch, PyTorch Geometric and scikit-learn, a layered package with automated tests and continuous integration; every run on CPU, logged with its settings, seed, git commit, library versions and scores.

Table II: optimizers, settings and sizes

Model	Settings	Parameters
TGN	Adam, learning rate 0.0005, weight decay 0.0001, focal; memory 64, time code 16, embedding 64, 10 neighbours, dropout 0.2, 200 events per batch, up to 100 epochs, patience 10	70,021
DyGFormer	same optimizer and loss; history 16, patch 2, width 32, 2 layers, 2 heads, dropout 0.2, at least 30 epochs, patience 10	24,021
LSTM	Adam, learning rate 0.001, weight decay 0.0001, focal (shocks) or Huber (volatility); 168 h window, 1 layer, 32 hidden, dropout 0.2, batch 512, up to 30 epochs, patience 5	5,665
Boosting	histogram gradient boosting, learning rate 0.05, L2 1, 31 leaves, at least 200 samples per leaf, up to 1,000 trees, early stopping on validation (patience 4)	–
Cascade	pair chances estimated on training shocks	–

F. How models are scored

Accuracy is meaningless (always saying "no" is about 90% accurate at 1 hour), so the main score is average precision (equation 9), with AUROC, the precision of the riskiest 1% and lift alongside.

Model differences use the day-block bootstrap (equation 10) over 971 test days and 1,000 rebuilds, with both models scored on exactly the same events. Neural models enter the bootstrap as five-seed ensembles (the average of the five chances), while tables of single numbers report the average over seeds. All tuning used validation only, and every configuration scored on the test period is listed in Table VI.

IV. Results

A. Tuning (Figure 3)

Figure 3 plots validation AUPRC at 1 hour for every tuning experiment (star: the adopted TGN; filled circles: other TGN versions; hollow square: no memory; triangle: DyGFormer; diamond: boosting; seed spread about ±0.005). The first TGN (up to 30 epochs, patience 4) stopped while still improving (0.138); 100 epochs with patience 10 gave 0.153 (adopted). Single-seed tries: halving the learning rate on plateaus 0.151, event context 0.151, stronger regularisation (dropout 0.4, weight decay 0.001) 0.143, a smaller 32-number network 0.133, futures data 0.131 (boosting 0.140). None beat 0.153 by more than the seed spread. DyGFormer's curve peaks early and dips (0.133 at epoch 2), so it gets at least 30 epochs (0.142); event context raised boosting from 0.153 to 0.155.

B. Test results: Table III, Figures 4 and 5, Table IV

- **Table III** is the main results table (Part 9). Neural models: average of five seeds; boosting and cascade: one fixed fit; the ensemble: weighted on validation.
- **Figure 4(a)** shows AUPRC divided by the positive rate (lift) for each window, so windows with different base rates can be compared; **4(b)** the precision of the riskiest 1% of events against the positive rate (grey bars).
- **Figure 5** shows the TGN's score minus four rivals' (cascade, boosting, no memory, DyGFormer) with 95% bootstrap intervals; filled markers mean the interval excludes zero, so the difference is reliable.
- **The careful sentences:** against boosting, one-sided p = 0.028 and 0.029 at 1 and 3 hours, but the two-sided intervals just include zero, so "borderline"; DyGFormer is better in the extreme top (43, 67, 80, 87, 92%) and a refitted boosting model reaches 34% at 1 hour and 96% at 24 hours, so **the TGN's edge is in overall ranking, not in the very top**; the 13 of 15 crash alerts are "an anecdote, not a statistic", because two of the three crashes lie in the training and validation years.

Table IV: the comparative study in one table

Question	Finding
TGN vs cascade	Better at every window, +0.046 (1 h) to +0.084 (6 h); no rebuild favours the cascade
Memory	Nothing at 1 h (−0.003); +0.010 to 0.013 at 6 to 24 h; borderline at 3 h ($p = 0.049$)
TGN vs DyGFormer	Tied everywhere
TGN vs boosting	+0.028 (1 h), +0.023 (3 h), p about 0.03, intervals just include 0; tied from 6 h; boosting overfits
Market-wide crashes, 1 h	TGN 0.245 vs boosting 0.205; after isolated shocks 0.114 vs 0.117
Ensemble	Better than the TGN only at 24 h (+0.008, $p = 0.001$); worse at 1 to 3 h
Refit on train + validation	Boosting 0.178 → 0.185; TGN 0.205 → 0.188; the training-only TGN kept

C. Shock types: Table V and Figure 6(a)

Table V: shock types (all periods for the first four columns; test events at 1 h for the rest)

Source shock	Fresh shocks	Median hours	Re-shock	Followers	Positive rate
Market-wide crash	6,275	10	0.61	8.1	0.163
Market-wide pump	3,068	9	0.58	7.2	0.118
Partial crash	1,320	9	0.61	6.2	0.088
Partial pump	1,808	9	0.65	6.2	0.070
Isolated crash	1,653	8	0.55	5.1	0.058
Isolated pump	4,047	10	0.63	4.5	0.043

Median hours: until volatility is back to its level before the shock. Re-shock: share followed by another shock of the same coin within 24 hours. Followers: other coins shocking within 6 hours. **Figure 6(a)** draws the TGN-against-boosting columns with random guessing as dotted lines. The message: the graph helps after broad shocks and adds nothing after a coin moves on its own news, so a real system could use the TGN after broad shocks and a simpler model after isolated ones.

D. Explanations and network findings: Figure 6(b)

Integrated gradients (completeness checked to 0.009): receiver about 65% (volume surge 25%, 24-hour volatility 12%, latest return 9%), source 25%, event 10%, memory and recent events about 7%. TempME, rebuilt for the TGN with motifs of up to 3 steps and a time-shuffled null: **Figure 6(b)** shows its characterization score with 3 events kept, 0.157, against random 0.152, most recent 0.192 and the per-alert best mask 0.248. Network findings: one market factor explains 64% (eigenvalue 19.3 of 30); in all nine crises correlation, density, spillover and shock lift rise and the tree shrinks; communities partly match sectors (NMI 0.47, $p = 0.003$) but contagion ignores sectors (9.3% inside and across; meme coins 13.7% against 10.9%); lead-lag is weak (0.035), so no arrows were added to the graph.

E. Robustness and negative results: Figure 7

Stress tests use the seed-0 TGN (clean 1-hour AUPRC 0.203) and boosting. **Figure 7(a)**: random noise of size σ and random removal of past events; **7(b)**: FGSM of size ε on the TGN and the same inputs on boosting. Losing half the past events costs under 0.5%; noise of 0.5 costs 5.7% (boosting 4.7%); FGSM of 0.1 costs about 35%, against under 1% for random noise of the same size. The T-GAE failed (0.511 against 0.893 for turbulence). HAR-RV beats the LSTM on volatility (5.0 against 5.5×10^{-5}), and the LSTM's next-hour shock score is only 0.109 (2.8 times random, AUROC 0.706): "deep learning earns its place on rare events and networks, not on smooth volatility".

F. Limitations, as the report lists them

- Overall discrimination is modest (AUROC 0.64 to 0.69): most shocks are news-driven and hit the whole market in the same hour.
- One market and one test period; a second, rolling test window is the clean fix.
- Survivorship bias: the 30 coins were picked from a 2026 ranking.
- The "garden of forking paths": several versions were scored on the test after earlier results were known. Table VI lists all of them, and the main model was fixed from validation evidence before the refit and never switched for a better test score.
- Early stopping looks at the 1-hour score only, though memory helps most at longer windows.
- Calibration was not checked on the test period; no money-based evaluation; no social-media or order-book data.
- The final TGN, DyGFormer, LSTM and boosting runs come from clean commits; the T-GAE, robustness and prediction runs had uncommitted changes, which the run registry flags.

Table VI: every configuration scored on the test period (1 hour)

Version	Test AUPRC	Status
TGN v1 (up to 30 epochs, patience 4), 5 seeds	0.201	superseded: validation still rising
TGN v2 (up to 100 epochs, patience 10), 5 seeds	**0.205**	**adopted main model**
No memory v1 (up to 30 epochs), 1 seed	0.206	superseded
No memory v2 (up to 100 epochs), 5 seeds	0.195	superseded: stopped on a noisy early peak
No memory v3 (at least 30 epochs), 5 seeds	0.208	reported
DyGFormer (at least 30 epochs), 5 seeds	0.208	reported
Gradient boosting, one fit	0.178	reported
TGN refit on train + validation, 5 seeds	0.188	reported, not adopted
Boosting refit on train + validation	0.185	reported
TGN on 2-hour bars (first bar), 5 seeds	0.165	reported

SIMPLE VERSION

Table VI is a confession list. Every time we looked at the final exam, we wrote it down, including the versions we threw away. That way nobody can accuse us of quietly trying versions until one scored well. Notice that the no-memory model's score moved from 0.206 to 0.195 to 0.208 between versions: that wobble is why we treat differences of about 0.01 with care.

V. Conclusion, acknowledgment and reproducibility

The conclusion repeats the main results and lists the honest negatives (no memory gain at 1 hour, no benefit from refitting, a weak anomaly detector). **Future work:** a rolling second test window and

a second market; social-media sentiment and order-book features; calibration and a money-based evaluation of alerts; adversarial training and checking inputs across several sources; a deployed early-warning dashboard.

Acknowledgment: Claude (Anthropic's AI assistant) helped with code, cross-checking numbers against the experiment logs, re-running the tests and the final model, and drafting the report and figures; the authors are responsible for the content. The price data are public; the code and run registry are in a private repository, available to evaluators on request. Reproducibility: the test suite gave 197 passed and 5 skipped with pinned library versions, and replaying the final TGN reproduces its stored test AUPRC within 0.0007.

CAREFUL

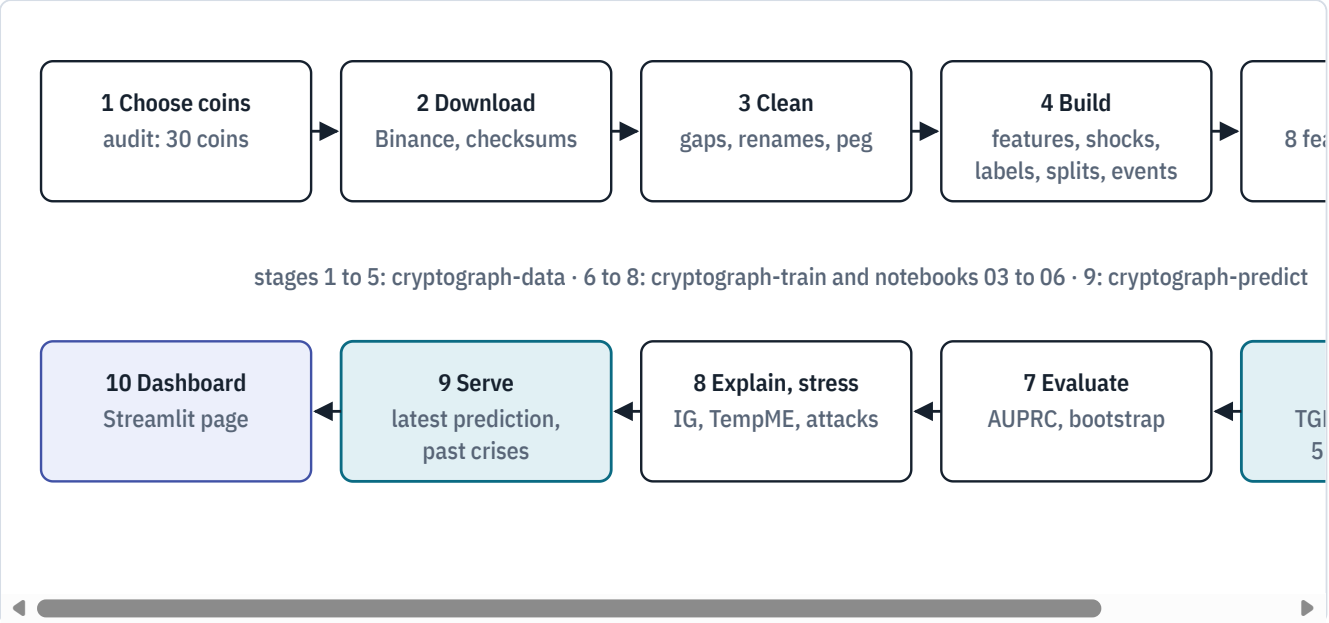
Small differences between the report and the code, with the true story:

- **Events:** Table I says 1,463,693, counted while the data were built. The stream the models read holds 1,463,147 (360,590 for training): the 546 shocks in the first week of January 2020 are skipped, because an event's correlation numbers need a full week of history.
- **Centrality:** section III-C says PageRank and eigenvector centrality were computed on the directed networks. PageRank was; eigenvector centrality is computed on the correlation network, which has no arrows.
- **Tests:** the report's 197 passed and 5 skipped were counted at submission; the deployed dashboard branch has 210 tests after later changes.
- **"About 90% accurate":** exactly 89.5% at 1 hour (100% minus the 10.5% positive rate).
- **MATIC to POL:** the report says the 79-hour gap is masked; notebook 01 says the two series are joined. Both are true: the gap hours are masked and the coin continues as one series.

PART 13

How the whole system runs, end to end

After this part you can walk through the pipeline, name the tools, and explain how results stay trustworthy.



The pipeline. Every stage reads only what earlier stages wrote, so any stage can be re-run on its own.

The stages in plain words

1. **Choose coins** (`cryptograph-data --stage audit`): apply the four rules from Part 3 to the top 60 coins and keep 30.
2. **Download** (`--stage fetch`): monthly and daily files from the Binance archive, each checked against its checksum, kept as write-once snapshots.
3. **Clean** (`--stage clean`): quality checks, short gaps filled and flagged, long gaps masked, renames joined, USDT peg flagged, 2-hour bars made, cross-check against the daily data.
4. **Build** (`--stage build`): features, time-of-day pattern, scaling, shocks, labels, splits, correlation graphs and the 1.46 million-event stream.
5. **Network** (`--stage network`): the 8 daily network features, coin rankings and the independent-cascade baseline.
6. **Train** (`cryptograph-train --config configs/experiment/<name>.yaml --seed n`): every run writes its own folder in `runs/`: the exact settings, environment, git commit, log, scores, saved model and per-event predictions.
7. **Evaluate** (notebook 03): tables, bootstrap tests, alert quality.

8. **Explain and stress-test** (notebooks 04 to 06): integrated gradients, TempME, case studies, persistence, attacks, the anomaly model.

9. **Serve:** `cryptograph-predict` scores the latest shocks in the data and saves each coin's chances and reasons; a second run (`configs/experiment/predict_events_1h.yaml`) does the same for the 9 catalogued crises, for the Past crises tab.

10. **Dashboard:** `streamlit run dashboard/app.py` shows it all (Part 14).

Tools

- **Python 3.12**; **PyTorch 2.11** for neural networks; **PyTorch Geometric 2.8** for graph models.
- **pandas** and **NumPy** for data; **scikit-learn** for gradient boosting, logistic regression, isotonic calibration and metrics; **statsmodels** for the VAR and statistical tests; **NetworkX** for network measures.
- **Streamlit** for the website and **Plotly** for its interactive charts.
- **YAML** files for every setting (`configs/`), checked automatically for typos in the data settings.
- **Git** and a private GitHub repository; **GitHub Actions** runs the tests on every push.
- Everything ran on a laptop CPU. No GPU was needed.

Where things live

Folder	What is inside
`src/cryptograph/data/`	downloading, cleaning, features, labels, graphs, datasets
`src/cryptograph/network/`	centrality, communities, MST, spillover, the independent cascade
`src/cryptograph/models/`	TGN, DyGFormer, LSTM, T-GAE
`src/cryptograph/training/`	training loops, losses, metrics, run tracking
`src/cryptograph/explain/`	integrated gradients, TempME port
`src/cryptograph/robustness/`	noise, missing data, FGSM and PGD attacks
`src/cryptograph/serving/`	predict (chances and reasons), past-crises replays, compression, dashboard data
`dashboard/app.py`	the Streamlit website
`configs/`	every setting as YAML: data, models, experiments
`notebooks/`	01 data and EDA, 02 networks, 03 models, 04 explanations, 05 persistence, 06 robustness
`runs/`	one folder per run, plus `results.csv`, the registry of every run
`tests/`	210 automated tests

Why the results can be trusted

- **210 automated tests** check the no-peeking rules, the label maths, the models, the explanations and the dashboard. They run on every push (tests that need the large data files run only on the laptop, since data is never stored on GitHub).
- The **run registry** records every run with its git commit and a "dirty" flag if the code had unsaved changes. Final models came from clean commits.
- **Checksums** for the raw data and for every processed file.
- **Safe model loading:** saved files are opened in a mode that cannot run hidden code.

- **Reproducible:** replaying the final saved model reproduces its stored test score (to within 0.0007) and its predictions.

PART 14

The dashboard, screen by screen

After this part you can demo the deployed dashboard and say where each number comes from.

SIMPLE VERSION

The dashboard is a website for people who are not data scientists. It answers three questions: what just happened, which coins are likely to move next and why, and how the model did in past crises.

This part describes the version on GitHub (branch `feat/hourly-network-models`, last updated on 19 October 2026), which is the one deployed on Streamlit. It is built with **Streamlit**, a Python library that turns a Python script into a web page, and its charts are drawn with **Plotly**. On a laptop it starts with `streamlit run dashboard/app.py` and opens at `localhost:8501`. The page does no heavy computing: it reads prediction runs that `cryptograph-predict` computed earlier and draws them. Nothing is retrained when you open it.

Left sidebar

- **Advanced:** pick which saved prediction run to show (the newest by default).
- **Look ahead:** the time window every chart uses: 1, 3, 6, 12 or 24 hours (6 by default).
- The model's run name (the final TGN, `tgn_1h_final_v2`, seed 0) and "A research tool, not financial advice."

Top of the page

- **What just happened:** the newest hour with big moves in the data and, for each coin that moved: up or down, how many times its usual hourly move, whether other coins moved in the same hour, how many big moves it made in the last 24 hours, and how long such moves usually take to calm down (from notebook 05).
- **Three numbers:** the coin most likely to move next and its chance; a typical coin's chance (the base rate); how many coins are at higher-than-usual risk.

With the research data, which ends on 27 September 2026, the page opens on **27 Sep 2026, 23:00 UTC**: QNT dropped sharply, about 4.8 times its usual hourly move, on its own, and had made 9 big moves in the previous 24 hours. Within 6 hours the riskiest coin is JUP at 37% against 27% for a typical coin (1.4 times), and 1 of 29 coins is at higher-than-usual risk. Open the deployed page once before the viva: if it was given a different prediction run, the date and coins will differ.

The five tabs

Tab	What it shows	Where the numbers come from
Who's at risk	a bar chart of every coin's chance of a big move within the chosen window, most likely first, coloured against a typical coin (much higher: 1.5 times or more; higher: 1.15 times or more; about normal; lower), with a dotted line for the typical coin. "Details: every time window" adds a heatmap of every coin at every window and a table (chance, × typical coin, raw model score, the coin whose move triggered it, the time).	the TGN's calibrated chances for the newest hour with big moves
Why	pick a coin (riskiest first): its chance, its level, "× typical coin", which coin's move triggered it, and the five inputs that pushed its score up (▲) or down (▼), in plain words with real values, such as "JUP: trading at 1.7× its usual volume (raises the risk)"	integrated gradients on the TGN's prediction head; real values from the data at that hour
Past crises	the 9 catalogued crises inside the hourly data. For each: a banner saying whether it is a fair test; a description with a source link; the worst hour (the hour when most coins moved), coins hit in that hour and within a week; "Did the model see it coming?": the model's 5 riskiest coins at the worst hour, how many made a big move within the window, how many 5 random coins would have caught, and a verdict; a table of every coin's chance and what happened; an animated network replay of the crisis hour by hour (▶ Play under the picture); a bar chart of coins moving per hour; per-coin details with a sector filter	a separate prediction run over the crises (`predict_events_1h`)
Market map	coins on a circle grouped by what they are used for; lines join coins whose prices moved closely together over the past week (correlation of 0.75 or more); bigger dots have a higher chance within the chosen window; a star marks the coin that just moved. "Details" lists network rankings (spillover NET, shock PageRank, how often others follow)	the daily correlation graph at that hour; rankings from notebook 02

Tab	What it shows	Where the numbers come from
How it works	what a big move is, how the model works and what the reasons mean, in plain words; "How good is it?": the served model's test score at each window against random guessing and "times better than random"; "How long do big moves last?": the typical hours until things calm down for six kinds of move, with the kind that just happened in red	the trained run's own test metrics; notebook 05

What the Past crises tab says about each crisis

At the default 6-hour window, the model's 5 riskiest coins at each crisis's worst hour:

Crisis	Period	Model's 5 that moved	5 random coins	Verdict on the page
COVID, 12 Mar 2020	train	3	3.7	about as well as random
China ban, 19 May 2021	train	4	4.1	about as well as random
LUNA collapse, 9 May 2022	train	3	2.5	about as well as random
Celsius, 13 Jun 2022	train	2	1.9	about as well as random
FTX collapse, 8 Nov 2022	validation	5	5.0	almost every coin moved
SVB failure, 10 Mar 2023	validation	2	1.2	about as well as random
Bitcoin ETF approval, 10 Jan 2024	**test**	**3**	**1.6**	**clearly better than random**
Yen carry-trade unwind, 5 Aug 2024	test	5	5.0	almost every coin moved
Liquidation cascade, 10 Oct 2025	test	5	5.0	almost every coin moved

At the 1-hour window the bitcoin ETF approval is again the clear case (2 of the 5 moved against 0.9 for random picks); all other crises read "about as well as random" or "almost every coin moved".

CAREFUL

- The tab opens on COVID, a training-period crisis, with an amber "Not a fair test" banner and a so-so verdict. In a demo, switch straight to the bitcoin ETF approval (test period, green banner).
- In big crashes nearly every coin moves, so any pick is right; the page says so itself. Present it as honesty built into the tool, not as a failure.
- The opening screen (QNT dropping on its own) is the model's weakest situation: after isolated shocks the TGN only ties boosting. Use it to explain the page, then move on.
- "How it works" shows the served model's own test scores. The dashboard serves one model (seed 0), with a test AUPRC of 0.203 at 1 hour; the report's 0.205 is the average of five seeds.
- The copy of the dashboard on your laptop currently has extra, unreleased tabs from later work. For the presentation, use the deployed link.

How a "chance" on the page is made

1. The TGN gives a raw score for each event pointing at a coin.
2. **Calibration** turns raw scores into real chances: validation events (2022 to 2023) are sorted by score, grouped into bins of at least 500, and an always-rising curve (isotonic regression) maps each score to how often such events really came true. The test years are never used for this.
3. Chances are made never to decrease with the window: the chance within 6 hours is at least the chance within 3 hours.
4. A coin's chance on the page is the highest over the events pointing at it in that hour (one per shocking coin), and "after the move of" names the coin behind it.
5. "× typical coin" (the **lift**) is the chance divided by the base rate, the share of coins that moved after a shock in the validation years.

SAY IT IN THE VIVA

"The dashboard reads tracked prediction runs. Risk is the TGN's score calibrated by binned isotonic regression on validation predictions, made monotone in the horizon, and aggregated per coin as the maximum over the events pointing at it. The reasons are integrated gradients, translated into plain language with the coin's real values."

A demo flow for the deployed dashboard (about 5 minutes)

1. **Opening screen (1 min).** Read "What just happened", the three numbers and the bar chart in Who's at risk. Explain "chance of a big move within 6 hours" and "× typical coin".

2 **Why (1 min).** Pick the riskiest coin and read its reasons out loud. Say they come from integrated gradients and explain the model, not the market.

3 **Past crises (2 min).** Switch to "Jan 2024 · US spot BTC ETF approval" (test period). At 6 hours, 3 of the model's 5 riskiest coins moved against 1.6 for random picks: "clearly better than random". Press ▶ Play under the network to replay the hours. Then open 10 Oct 2025 and point out that the page itself says almost every coin moved, so that crisis cannot judge the model.

4 **Market map (30 s).** Coins grouped by sector, lines for coins that moved together last week, bigger dots for higher chances.

5 **How it works (30 s).** The model's test scores against random guessing, and how long big moves usually last.

PART 15

Is it live? Where the dashboard's numbers come from

After this part you can answer "is this real time?" honestly and explain the crisis replays.

SAY IT IN THE VIVA

"No, the deployed dashboard is not live. It shows the newest prediction on our research data, which ends on 27 September 2026, plus replays of nine past crises. Refreshing it means running the data pipeline on the new Binance hours and then `cryptograph-predict`; the model itself does not change. Automatic hourly updates are future work."

How the numbers are produced

1. `cryptograph-predict` replays the whole event stream through the final TGN (about 1.46 million events, under a minute on a CPU), so each coin's memory holds exactly what it would in real use.
2. It scores the events of the newest hour with big moves, turns the scores into chances (calibration), keeps each coin's highest chance per window, and works out the reasons with integrated gradients.
3. It saves small result files (chances, reasons, real values, alerts, a summary) in its own folder under `runs/`; the dashboard reads the newest one.
4. A second run does the same at the worst hour of each of the 9 catalogued crises and stores what happened afterwards, for the Past crises tab.

Why the crisis replays are honest

- Every crisis is labelled with its period. Only the three test-period crises (bitcoin ETF approval, yen unwind, 10 October 2025) are genuine forecasts; the others were used for training or for choosing the model, and the page says so.
- The model's 5 riskiest coins are always compared with what 5 random coins would have caught.
- When almost every coin moved, the page says the crisis cannot show whether the model is useful.

CAREFUL

The report mentions that 13 of the 15 top alerts in three crashes (LUNA, FTX, 10 October 2025) came true. In those crashes almost every coin moved, so random picks would have scored nearly as many. If it comes up, say that yourself, and point to the bitcoin ETF approval instead, the one crisis where the model clearly beat random picks.

PART 16

Limitations and future work

Name the weaknesses yourself. It shows you understand the work.

Limitations

- **Modest discrimination** (AUROC 0.64 to 0.69): most shocks are news-driven and hit the whole market in the same hour.
- **One market, one test period:** Binance, prices in USDT, 2024 to 2026.
- **Survivorship bias:** coins were chosen by their 2026 ranking, so coins that died earlier are missing (apart from LUNC and FTT as case studies).
- **Several versions touched the test period** (all disclosed in the report's Table VI); the main model was fixed from validation evidence and never switched for a better test score.
- **Model choice used the 1-hour score only**, although memory helps most at longer windows.
- **The label depends only on the receiving coin** (Part 4), so "who spread it to whom" is not learned directly.
- **A simple model is competitive** at hourly resolution (Part 9).
- **No money-based test** yet: we have not measured whether acting on alerts reduces losses.
- **No news, social-media or order-book data.**
- **The dashboard is not live:** it shows the newest prediction on the research data, which ends on 27 September 2026.

Future work

- A second, rolling test period and other exchanges.
- Order-book, funding and minute-level data, where "who moves first" may be visible.
- Direction-aware labels ("does j crash after i crashes?").
- News and social-media sentiment as inputs.
- A money-based backtest: for example, step out of coins flagged "much higher than usual" for 6 hours and measure the reduction in losses and volatility.
- Defences against manipulation: adversarial training and cross-checking inputs from several sources.
- Automatic hourly updates for the dashboard: fetch each new Binance hour, rebuild it with the frozen pipeline, check nothing in the past changed, predict, and keep a running record of how the alerts turned out.

PART 17

Viva question bank

Questions the professor is likely to ask, with model answers. Turn on quiz mode, answer out loud, then reveal.

A. The basics

A1 What does your project do, in one sentence?

After one cryptocurrency makes an unusually big hourly move, it predicts which other coins will make one of their own within 1, 3, 6, 12 or 24 hours, explains why, and shows it on a plain-language dashboard.

A2 What exactly is a "shock"?

An hour in which a coin's time-of-day-adjusted return is more than 2.29 times its typical hourly move over the previous 168 hours ($|z| > 2.29$), up or down. The threshold was set on training data so that about 4% of hours are shocks.

A3 What do you mean by contagion?

A shock in coin i at hour t followed by a shock in another coin j in a strictly later hour, within k hours. Same-hour moves are excluded because they are a common reaction to the same news, not spreading.

A4 Why is this useful?

Traders and funds can cut risk on the coins most likely to be hit next, exchanges can raise margins before forced selling spreads, and regulators can watch systemic risk. Classic spillover measures describe average connectedness; they do not say who is hit next and how soon.

A5 Why cryptocurrencies?

They trade 24/7, so hourly data has no gaps for nights or weekends; there are many tightly linked coins; crashes are frequent; and Binance publishes free, checksum-verified hourly data. Contagion here moves real money within hours.

A6 What are T1, T2 and T3?

T1, the main task: pairwise contagion, solved by the TGN. T2: will a single coin shock in the next hour (LSTM against simple rules). T3: a coin's future volatility at 1, 6 and 24 hours (LSTM against HAR-RV).

A7 Why don't you predict whether prices go up or down?

Our analysis showed hourly returns barely relate to past returns, so direction is close to unpredictable, while the size of moves clusters strongly in time. We predict what has signal. Big moves in either direction matter for risk.

A8 What is your contribution?

A leakage-controlled hourly contagion benchmark (30 coins, 1.46 million events); a fair comparison of temporal graph models with network-science, econometric and machine-learning baselines (five seeds, day-block bootstrap); a TempME port and integrated-gradient analysis showing where the signal lives; persistence, robustness and anomaly analyses including negative results; and a plain-language dashboard with past-crisis replays.

B. Data

B1 Where does the data come from?

The Binance public archive: hourly bars for 30 coins, January 2020 to 27 September 2026. Every file was verified against its SHA-256 checksum (3,049 files, no failures). A separate daily dataset (Yahoo Finance and Coin Metrics) is used only for cross-checking.

B2 Why hourly data?

Contagion in crypto happens within hours, and daily bars merge several shocks and their order into one observation; that was also your review comment on our first version. We tested 2-hour bars and 1 hour was better. Minute data is far bigger and noisier, and is listed as future work.

B3 How did you choose the 30 coins?

From the largest coins by market value (CoinGecko, 27 September 2026) we kept the first 30 that are not stablecoins, wrapped or real-world-asset tokens, trade against USDT on Binance, have at least 730 days of hourly history and at least 90% coverage in the test period. Every exclusion is recorded with its reason.

B4 How big is the dataset?

59,088 hours per coin, about 1.6 million coin-hours, and 1.46 million shock events: 360,590 for training, 300,998 for validation and 801,559 for test.

B5 How did you handle missing hours?

Gaps of 1 or 2 hours are filled with the last price and flagged; longer gaps are left empty and the coin counts as inactive there. Nothing is filled before a coin was listed, and returns are never clipped.

B6 What went wrong with LUNA?

Binance reused the ticker LUNAUSDT for a brand-new coin after the old Terra LUNA collapsed in May 2022. Joining the two would create a fake price jump, so they are never joined; the old coin is kept as LUNC, an event coin for case studies.

B7 What does the model know about each coin?

14 numbers per hour: 6 market features (last return, volatility over 24 and 72 hours, de-seasonalised volume, volume against its past week, time since listing) and 8 network features updated daily (strength, eigenvector centrality, MST degree, community size, spillover PageRank and NET, shock PageRank, shock out-lift). Each event adds 5 numbers.

B8 How did you scale the features?

Robust scaling per coin, using the median and interquartile range from the training years only, clipped at ±5. Volume and volatility are first divided by an hour-of-week profile fitted on training data, because activity follows the clock (0.69 to 2.17 times its average).

B9 Why check the USDT peg?

Prices are quoted in USDT. If USDT itself lost its one-dollar value, every coin would appear to move together. We flag days when it was more than 0.5% off its peg, using Coin Metrics data until that archive ends in May 2026.

C. Labels, splits and leakage

C1 What is data leakage, and how did you prevent it?

Leakage is future information slipping into the inputs, which gives great but fake results. We use trailing windows only; fit every scaler and threshold on training data only; split by dates with a 7-day embargo; drop samples whose windows cross a split; update the TGN's memory only after a batch is predicted, with batches that never split an hour; and run automated tests on these rules.

C2 Why not a random train/test split?

Shuffling time would let the model learn from the future and be tested on the past. In time series you train on earlier years and test on later ones.

C3 Why a 7-day embargo?

Features use windows of up to a week, and market conditions carry over between neighbouring days. A gap at the start of validation and test stops the end of one period from informing the start of the next.

C4 Why the threshold 2.29?

It is the level at which about 4% of training hours are shocks, comparable with the 3.9% rate of our daily data. It was fitted on training data only, so the test period could not influence it.

C5 Why the windows 1, 3, 6, 12 and 24 hours?

They run from immediate spread to a full day. Shocks typically calm down in 8 to 10 hours, so 24 hours covers a whole episode, and several windows show where memory starts to help.

C6 How imbalanced is the task?

On the test period the share of "yes" answers is 10.5% within 1 hour, 20.6% within 3, 31.0% within 6, 44.7% within 12 and 62.7% within 24. That is why we use focal loss and AUPRC.

C7 Does the label depend on which coin shocked first?

No: the answer for coin j depends only on j and the hour, so if several coins shock at once, j gets several events with the same answer. The model effectively asks which coins move next after a shock somewhere, and the dashboard shows one chance per coin. A direction-aware label is future work.

D. The TGN

D1 Which model did you use?

A Temporal Graph Network, TGN. Our version, ContagionTGN, uses PyTorch Geometric's TGN memory module, a graph-transformer attention layer over each coin's recent events, and our own five-output prediction head.

D2 Who proposed the TGN, and when?

Emanuele Rossi, Ben Chamberlain, Fabrizio Frasca, Davide Eynard, Federico Monti and Michael Bronstein, researchers at Twitter with Imperial College London: "Temporal Graph Networks for Deep Learning on Dynamic Graphs", first published in June 2020 and presented at an ICML 2020 workshop.

D3 Explain the TGN simply.

Every coin keeps a 64-number memory, like a diary. When a coin shocks, the model reads the diaries of that coin and each other coin, uses attention over each one's 10 most recent events, adds both coins' current features and the event's details, and outputs five chances. Only after the whole hour is scored does it write the hour's events into the diaries.

D4 How many parameters does it have?

70,021 trainable parameters: 41,280 in the GRU that updates memory (59%), 17,984 in the attention layer (26%), 10,693 in the prediction head (15%) and 64 in two time encoders. The memory itself (30 × 64 numbers) is state, not parameters.

D5 Why so small?

30 nodes and about 360 thousand training events. A small model overfits less and trains on a laptop CPU. A smaller 32-number variant did worse on validation, and stronger regularisation did not help either.

D6 How is the memory updated?

Each event creates a message for both coins: their two memories, the event's 5 numbers and a 16-number time code (149 numbers). The latest message per coin goes through a GRU, which decides what to keep and what to overwrite, giving the new 64-number memory. This happens after the batch is predicted.

D7 What does the attention layer do?

It is a graph-transformer layer (TransformerConv) with 2 heads of 32. For each coin it weighs its 10 most recent events, using their timing and messages, with weights that add up to 1, and produces a 64-number embedding.

D8 What goes into the prediction head?

161 numbers: both embeddings (64 + 64), both coins' features at hour t (14 + 14) and the event message (5). A dense layer to 64 with ReLU and 20% dropout, then a layer to 5 outputs, each passed through a sigmoid.

D9 Why five sigmoids and not a softmax?

The windows are not mutually exclusive: "within 6 hours" includes "within 3 hours". Each window is its own yes/no question (a multi-label problem). When serving, we also force the chances to never decrease with the window.

D10 How does the TGN avoid seeing the future?

Events are processed in time order. For each batch the model predicts using memory and neighbour lists built only from earlier batches, then updates them. A batch never splits an hour, so events in the same hour cannot see each other. Features at hour t use data up to t.

D11 Why did you choose the TGN?

Our data is a stream of timed events, which is exactly what the TGN was built for. Its per-coin memory captures recent turbulence. It combines each coin's own state with its relationships. Its predict-then-update design matches our no-peeking rule. It is small, fast and well established, with a maintained implementation in PyTorch Geometric, and explainers such as TempME work on it.

D12 Why not a GCN, a GAT or a snapshot GNN?

A GCN or GAT on a single graph ignores time. A snapshot GNN (one graph per day plus a GRU) loses the order of events within the day. We planned the snapshot model but dropped it, because the TGN, DyGFormer and boosting comparison already answers whether the graph adds value.

D13 Why not a large Transformer or a language model?

A large model needs far more data than 30 coins provide and would overfit. DyGFormer is the Transformer-style dynamic-graph model, and we tested it. Language models work on text, not on numeric event streams.

D14 Does the memory help?

Not within 1 hour: without memory the score is the same (−0.003, interval −0.014 to +0.010). It does help from 6 to 24 hours, by 0.010 to 0.013 AUPRC (p at most 0.025). Memory captures the slower, multi-hour spread.

D15 What were the training settings?

Adam with learning rate 0.0005 and weight decay 0.0001; focal loss (γ 2, α 0.25); batches of about 200 events; up to 100 epochs with early stopping after 10 epochs without improvement in validation AUPRC at 1 hour; five seeds.

D16 How long did training take, and on what?

About 30 to 50 minutes per seed on a laptop CPU, with no GPU; the best epochs were 45 to 98. A full pass over the 1.46 million events takes one to two minutes.

D17 What is focal loss and why use it?

A loss from Lin and colleagues (Facebook AI, 2017) that down-weights easy, well-classified examples, so the rare positives get more attention. It was chosen over a class-weighted cross-entropy by validation AUPRC.

D18 What is a seed, and why five?

The seed fixes the random starting values. Five trainings show how much of a result is luck: the spread is about ±0.005 AUPRC, and we report the average. Comparisons use the average of the five models' predictions.

D19 Why are there two final TGN versions?

Version 1 (up to 30 epochs, patience 4) stopped while its validation curve was still rising (0.138). Looking only at validation curves, we raised the budget to 100 epochs with patience 10: version 2, validation 0.153, which is the main model. Both are reported.

E. The other models

E1 Why did you choose these models only?

Each one tests a claim. Random and simple rules show there is signal. The independent cascade is the network-science baseline. Gradient boosting tests whether plain machine learning on the same inputs is enough. The LSTM tests a single coin's history without the graph. DyGFormer checks that the result does not depend on one architecture. The no-memory TGN isolates what memory adds.

E2 What is the independent cascade model?

An epidemic-style diffusion model (Kempe, Kleinberg and Tardos, 2003). Each shocked coin gets one chance to "infect" each other coin, with a probability estimated from training shocks and shrunk towards that coin's base rate. With several shocked coins, the chance for *j* is one minus the product of all the "miss" chances.

E3 Why did gradient boosting lose?

It matched the TGN on validation (0.155 against 0.153) but had memorised its training years (training score 0.64), so it generalised worse to the test period: 0.178 against 0.205 at 1 hour. From 6 hours on the two are tied.

E4 What is DyGFormer, and why re-implement it?

A Transformer over each coin's last 16 events, cut into patches of 2, with an encoding of how often the two coins appeared together (Yu and colleagues, NeurIPS 2023). The official library trains with random fake links and splits time by fractions, which would break our leakage rules, so we wrote a compact version: 24,021 parameters. It ties the TGN.

E5 Why did the LSTM lose to HAR-RV on volatility?

Volatility is smooth and well described by a simple linear formula of recent volatility (last hour, day and week). A neural network adds noise there. Deep learning earns its place on rare events and networks.

E6 What is T-GAE, and why did it fail?

A temporal graph autoencoder that learns what normal hours look like and flags hours it cannot reproduce. Stress days are extreme market-wide moves that a simple turbulence measure already catches (AUROC 0.89 against 0.51). It is reported as a negative result, though it ranked 10 October 2025 as the top anomaly.

E7 What did you try that did not work?

Futures-market data (validation fell to about 0.13 for the TGN and 0.140 for boosting), retraining the TGN on train plus validation (0.205 down to 0.188), 2-hour bars, a learning-rate schedule, extra context features, stronger regularisation and a smaller model. All are reported.

F. Evaluation and results

F1 Why not accuracy?

89.5% of one-hour events are "no", so a model that always says "no" is 89.5% accurate and useless. We use AUPRC, AUROC and the precision of the top 1% of alerts.

F2 What is AUPRC, and what does random score?

Average precision over the ranked list of events: each time a real positive is reached, take the precision so far, then average. A perfect ranking scores 1; random scores the positive rate (0.105 within 1 hour). The TGN's 0.205 is about twice random.

F3 Why not F1 score?

F1 needs one fixed threshold. AUPRC summarises every threshold and rewards a good ranking, which is what alerts need: the riskiest coins first.

F4 What AUROC did you get?

0.64 to 0.69 across windows, where 0.5 is random. Clearly above chance but modest, because most shocks are news-driven and arrive across the market in the same hour.

F5 What is a day-block bootstrap?

We rebuild the test period 1,000 times by drawing its 971 days with replacement, compute model A's AUPRC minus model B's each time, and take the middle 95% as the interval. Whole days, because one crash creates hundreds of dependent events; resampling single events would make us overconfident.

F6 What are your main results?

Test AUPRC for the TGN: 0.205, 0.352, 0.464, 0.588 and 0.739 at 1, 3, 6, 12 and 24 hours, against random 0.105 to 0.627 and the independent cascade 0.159 to 0.672. It beats the cascade at every window, beats boosting at 1 to 3 hours (borderline), ties DyGFormer, and memory helps at 6 to 24 hours.

F7 Is the TGN significantly better than the cascade?

Yes, at every window: +0.046 at 1 hour (interval 0.019 to 0.074), +0.084 at 6 hours, and no bootstrap resample favoured the cascade.

F8 Why does DyGFormer tie the TGN?

Both pick up the same signal: the receiving coin's turbulence and the market's breadth. Two different designs agreeing makes the result more credible, not less.

F9 Where does the graph help most?

After market-wide crashes (TGN 0.245 against boosting 0.205 at 1 hour). After isolated shocks they tie (0.114 against 0.117): when one coin moves on its own news, the network adds nothing.

F10 How good are the top alerts?

Of the 1% riskiest test events, 32% came true within 1 hour (random 10.5%), 61% within 3, 76% within 6 and 89% within 24 (random 63%).

F11 Did a simple model do as well?

In our final review, a four-feature logistic regression (the coin's volume against its past week, hours since its last shock, market breadth over the last day, size of its current move) matched the TGN on the test period. It confirms the explanations: at hourly resolution most of the signal is the receiving coin's own turbulence. The TGN learns this from the raw stream and clearly beats the network baseline; finer data is where a graph model could pull ahead.

F12 How many times did you use the test set?

Each final configuration scored it once, and every one is listed in the report's Table VI: TGN version 1 and 2, three no-memory versions, DyGFormer, boosting, the two refits and the 2-hour TGN. The main model was fixed from validation evidence and never switched for a better test score.

F13 Would this make money?

We have not tested that. A money-based backtest (for example stepping out of flagged coins for 6 hours and measuring losses avoided) is future work. The project predicts risk, not direction.

G. Explanations and robustness

G1 How do you explain a prediction?

With integrated gradients on the prediction head: how much each input pushed the score up or down compared with an average situation. The pushes add up exactly to the score, which we check every run. TempME was ported to look for chains of past events.

G2 What do the explanations show?

About 65% of the attribution comes from the receiving coin's own state (volume against its past week 25%, 24-hour volatility 12%, last return 9%), about 25% from the source coin, 10% from the event, and memory about 7%. The model flags coins that are already stretched when another coin crashes.

G3 Why was TempME weak?

Its fidelity was barely above random (0.157 against 0.152) and below taking the most recent events (0.192). The TGN's attention is almost uniform and shocks spread across the market in the same hour, so there are no chains of events to find. It is a finding about the market.

G4 How robust is the model?

Removing up to half of the past events changes the score by under 0.5%; random noise of 0.5 standard deviations costs 5.7%. A targeted FGSM attack of only 0.1 standard deviations costs about 35%, while random noise of the same size costs under 1%. The risk is deliberate manipulation, not noise.

G5 What is FGSM?

The fast gradient sign method (Goodfellow and colleagues, 2015): nudge every input a small step in the direction that increases the model's error the most. We also ran PGD, a 10-step version, which was barely stronger, so FGSM is close to the worst case here.

H. The dashboard

H1 What does the dashboard show?

Five tabs. Who's at risk: every coin's chance of a big move within the chosen window, against a typical coin. Why: the inputs behind one coin's risk, in plain words. Past crises: nine catalogued crises with the model's top picks against random picks and an hour-by-hour replay. Market map: coins grouped by sector and linked by how closely they moved. How it works: plain explanations, the model's test scores and how long big moves last. Above the tabs, "What just happened" describes the newest big moves.

H2 How is a chance on the page calculated?

The TGN's raw score is calibrated by binned isotonic regression fitted on validation predictions (never the test years), made non-decreasing across windows, and for each coin the highest value over the events pointing at it is shown.

H3 What does "× typical coin" mean?

The lift: the coin's chance divided by the base rate, how often any coin moved after a shock in the validation years. 1.5 or more is shown as "much higher than usual", 1.15 or more as "higher than usual".

H4 Is it real time?

No. It shows the newest prediction on the research data, which ends on 27 September 2026, plus nine crisis replays. Refreshing it means running the data pipeline on new Binance hours and then `cryptograph-predict`; the model does not change. Automatic hourly updates are future work.

H5 Does the page run the model when I open it?

No. `cryptograph-predict` replays the whole event stream through the trained TGN, scores the newest hour with big moves and saves the chances and reasons as small files. The page only reads and draws them, so it opens fast and nothing is retrained.

H6 Why does the dashboard show 0.20 when the report says 0.205?

The dashboard serves one trained model (seed 0), whose test AUPRC at 1 hour is 0.203. The report's 0.205 is the average of five seeds.

H7 Why Streamlit?

It turns a Python script into an interactive web page with no separate web stack, works directly with our pandas and Plotly code, and suits a research team.

H8 Why do most crisis replays look no better than random?

In a market-wide crash nearly every coin moves, so any pick is right; the page says so itself. The bitcoin ETF approval (January 2024, test period) is where the model clearly beat random picks: 3 of its 5 riskiest coins moved within 6 hours, against 1.6 for random picks.

I. The project and the team

I1 What was the hardest problem?

Getting the question right without leakage. The first version's label counted same-hour moves and used whole-sample statistics, which made results look far better than they were. We redefined shocks with trailing windows, made the label strictly later, moved to hourly data and fixed date splits, and rebuilt everything.

I2 What would you do with more time?

A second rolling test period and other exchanges; order-book and minute data; direction-aware labels; news sentiment; a money-based backtest; defences against manipulation; and automatic hourly updates for the dashboard, with a running record of its alerts.

I3 Did you use AI tools?

Say it plainly, as the report's acknowledgment already does: an AI assistant (Claude) helped with code, cross-checking numbers against the run logs, re-running tests, and drafting the report and figures. Then show ownership: you can explain every choice, the numbers and the weaknesses. See the note at the end of this guide.

I4 Who did what?

Agree this with your teammate before the viva. Per the project notes: Dakshesh built the prediction command, the Streamlit dashboard and its plain-language redesign, the crisis explorer, the model compression, the PGD and driver-stability tests and part of the persistence analysis; the data pipeline, the network analysis, the models and evaluation, the explanations, and the robustness and anomaly work were on the main branch.

J. The notebooks

J1 What is in each of your six notebooks?

01 checks the data and how crypto behaves (daily and hourly); 02 studies the coin network, spillovers, communities, the FTX collapse hour by hour, and builds the cascade baseline and the network features; 03 races the models with bootstrap tests, plus the two side tasks; 04 explains the TGN's alerts; 05 measures how long shocks last by type; 06 tests robustness and the T-GAE anomaly detector. The notebooks read the pipeline's saved outputs; they do not train models.

J2 Why is notebook 01 called "market dynamics and spillover baseline" when it is about data?

History. It was planned to hold the spillover baseline too; that analysis moved to notebook 02, section 2b, next to the hourly spillover network. Notebook 01 was rebuilt on the audited data with the archived draft's layout, and the file name stayed.

J3 Why does notebook 01 have a daily part and an hourly part?

The project started on ten years of daily prices (Yahoo Finance, cross-checked with Coin Metrics). After the review we moved the models to hourly Binance data, because contagion happens within hours: a shock makes another coin's shock 2.3 times more likely in the very next hour. Part I keeps the basic facts on the longest history; Part II covers the data the models actually use.

J4 How did you check data quality?

Daily: nine checks on every coin-day; 28 bad prices corrected; 123 of 156 moves above 40% confirmed by a second source or a known crisis; missing days left empty, never invented. Hourly: 3,049 files verified by SHA-256; maintenance gaps of up to 2 hours filled and flagged (13 hours), longer ones masked; ticker changes handled; daily totals match the daily dataset to about 0.1%; four USDT off-peg days flagged.

J5 How did you handle MATIC to POL and the LUNA collapse?

MATIC to POL was a 1:1 swap: a 79-hour gap, no price jump, so the gap is masked and the series continues. LUNA: the LUNAUSDT name was reused for a new coin after the May 2022 collapse, so the old coin (now LUNC) and the new LUNA are never stitched together; the 2,863-hour gap stays masked. FTT's 7,467-hour gap is its delisting after FTX.

J6 Why do you never feed prices to the model?

Prices drift and grow, so patterns found in them are often fake. The ADF test could not reject "wandering" for 15 of 20 coins' log prices, while returns were stationary for all 20. So all inputs are built from returns and volatility.

J7 Why predict shocks and not whether prices go up or down?

Because direction is not predictable from the past: yesterday's direction correlates −0.024 with today's, and the sign repeats only 47% of days. Jumpiness is predictable: +0.184 for squared returns, and 55 to 65% of shocks are followed by another within a day. After a shock, the direction of the next move is also mostly unpredictable (notebook 05).

J8 Why remove intraday seasonality?

Jumpiness follows the clock: Wednesday 12:00 UTC is 2.17 times an average hour, Sunday 06:00 is 0.69 times. Without the correction, ordinary Wednesday afternoons would be labelled shocks and Sunday mornings never. The profile is fitted on training data and divided out before features and shocks are computed.

J9 Does Bitcoin lead the market?

No. Coins move together in the same hour (correlation 0.65 with BTC), but one hour later the link is tiny (0.035 at most, and that peak is a 24-hour daily rhythm). After correction only 8 of 45 BTC/ETH-to-coin Granger tests pass, the effects are about 0.02% of variance, and the pattern does not repeat on validation. So the graph has no "who leads" arrows; direction comes from which coin shocked first.

J10 How did you choose the graph threshold of 0.75?

It was fitted on training data to give the same density as the daily graph, about 30% of possible links. With week-long windows almost every pair clears 0.5 (75% of pairs), a hairball with no structure. 0.75 at 1 hour, 0.77 at 2 hours, plus each coin's two strongest links so no coin is alone.

J11 What does the eigenvalue of 19.3 mean?

The 30 eigenvalues of the correlation table add up to 30; the biggest, 19.3, says one common direction, "the whole market moving", explains 64% of hourly movement. It is the only one above the 1.08 that pure noise could produce. Most of a coin's hourly move is the market.

J12 Your hourly spillover ranking equals eigenvector centrality. Is that a problem?

It is a finding. With one lag and tiny lagged effects, the hourly decomposition is driven by same-hour correlation, so NET measures how tied a coin is to the market, not who moves first. That is why we did not treat it as ground truth: explanations were also compared with the daily ranking and with the time-ordered shock-transmission PageRank.

J13 Do sectors matter?

For grouping, partly: communities match sectors with NMI 0.47 against 0.32 for shuffled labels ($p = 0.003$). For contagion, no: another coin shocks within the hour equally often inside and across sectors (9.3% and 9.3%), except meme coins (13.7% against 10.9%). So the sector is a coin feature, not the shape of the graph.

J14 What does the minimum spanning tree show? Why is DOT the hub?

It is the market's skeleton: the 29 strongest links joining all 30 coins. DOGE–SHIB and BTC–ETH are the strongest; ETH is the hub of the big coins. DOT has the most links (11), but to loners such as TRX, ZEC and QNT whose nearest neighbour happens to be DOT, so it is a meeting point for outsiders, not a leader. The tree shrinks in every crisis.

J15 Walk me through FTX hour by hour.

6 November 15:00 UTC: Binance's CEO says he will sell FTT; FTT falls 14 standard deviations and no other coin shocks. 8 November 02:00: FTT falls 10 and 18 of 30 coins shock in the same hour; 16:00, the takeover news, 20 coins jump together; 17:00 to 18:00, as the deal unravels, FTT falls 17 and all 24 coins trading then shock down. A random hour has about 1.2. Contagion was simultaneous and news-driven, not a chain.

J16 Why is test AUPRC (0.205) higher than validation (0.153)?

The test years (2024 to 2026) had more shocks and more moving together than 2022 to 2023: the positive rate is 10.5% against 8.6% at 1 hour, and even the cascade scores higher on test. Compared with random, the gap is smaller: lift 1.95 on test against 1.78 on validation. It is a property of the period, not a sign of leakage.

J17 Why do the tables count 795,787 test events, not 801,559?

The stream holds 801,559 test events; only events after the 7-day embargo and with their whole answer window inside the test period are scored. That leaves 795,787 at 1 hour and slightly fewer at 12 and 24 hours, whose windows run past 27 September 2026 at the very end.

J18 Is your TGN overfitting?

Somewhat, at short windows. Its lift drops from training to validation by a ratio of about 1.5 at 1 hour, against about 1.1 for the cascade, which cannot memorise. At 24 hours all models are close to the cascade (1.08 to 1.14). Stronger regularisation or a smaller model narrowed the gap only by losing validation accuracy, so early stopping keeps it under control.

J19 Did you train on a GPU?

No. Every final run logged a CPU-only PyTorch build, as the report says. A TGN epoch over the 360,590 training events took about half a minute. Notebook 01 mentions a Colab GPU as the original plan for the full stream, but the CPU was enough.

J20 Why is the TGN's attention perfectly uniform? Is that a bug?

No. The captured inputs reproduce the model to 2.4×10^{-7}, removing half of the past events changes the score by under 0.5%, and the no-memory model is as good at 1 hour. The model learned that no single recent event matters more than the others, which fits the market: shocks hit many coins in the same hour, so event patterns look the same as in a time-shuffled stream.

J21 What did notebook 05 find, and what does it mean for the model?

Shocks fade fast: a coin is back to normal after a median of 8 to 10 hours, and the extra jumpiness against an ordinary hour halves in about 2.5 hours. 55 to 65% re-shock within a day; market-wide crashes pull in about 8 other coins within 6 hours against about 5 for isolated shocks. So the 1 to 24-hour labels cover a shock's whole life, and the TGN helps most after broad shocks.

J22 Which days did the T-GAE flag?

Its top test day is 10 October 2025, the liquidation cascade. The next nine cluster in early 2024 and mid 2026; most were not stress days and none is near a known event, so we report them as "unclassified": the score says a day's structure was unusual, not why, and we claim no manipulation.

J23 What is in notebooks/archive, and why keep it?

Our first two draft notebooks, kept unchanged with a README listing every error. Three examples: the shock was measured against a window that included the same day (leakage); "38.71% contagion" counted coins shocking on the same day, which is co-occurrence, not contagion; and the spillover numbers came from the order-dependent Cholesky method, which made Bitcoin look like a giant. All three were fixed in the rebuilt pipeline. Keeping the drafts shows the corrections were honest.

J24 Notebook 02 shows a cascade score of 0.160 and the report 0.159. Which is right?

0.159. Notebook 02's saved output dates from 1 October; on 2 October the cascade was switched to score exactly the TGN's evaluation events (commit 07f4c27). Same model, final event set.

K. Techniques and statistics

K1 What is a p-value, in plain words?

How surprising the data would be if nothing real were going on. A small p-value means "hard to explain by luck". It is not the probability that our claim is true.

K2 What is the null hypothesis of the ADF test?

That the series has a unit root: it wanders and never returns to a stable level. Rejecting it (a small p-value) means the series is stationary.

K3 What does an excess kurtosis of 18 mean?

Extreme moves are far more common than a bell curve allows (a bell curve has 0). A squared-error loss would be dominated by a few crash days, so we use Huber or focal loss, clip inputs, and score with AUPRC.

K4 Is Granger causality real causality?

No. It only asks whether one series' past improves forecasts of another beyond its own past. A common cause, or a coin whose price updates late (thin trading, as with QNT), can produce it without any real influence.

K5 Why Benjamini–Hochberg rather than Bonferroni?

With 90 tests, Bonferroni guards against even one fluke and would hide real effects. Benjamini–Hochberg controls the share of flukes among the results called significant (about 5%), the right balance for a screening question like "is there any lead-lag?".

K6 Why a circular shift rather than a random shuffle for the lead-lag null?

A shuffle destroys each coin's own calm and stormy spells, which makes the null too easy to beat. Rotating the series keeps those habits and only breaks the timing link. It also keeps the 24-hour rhythm, so it is an upper limit, which is why Granger tests were run as well.

K7 What is the Marchenko–Pastur bound?

The biggest eigenvalue that pure noise could produce for a given number of coins and hours, $(1 + \sqrt{(N/T)})^2$, here 1.08. Eigenvalues above it are real structure; only the market factor (19.3) is.

K8 Generalized against Cholesky decomposition: what is the difference?

Cholesky assumes the first variable can affect the others within the same period but not the reverse, so results depend on the order of the coins. The generalized version (Pesaran and Shin, 1998) is order-free. Our first draft used Cholesky and Bitcoin, listed first, looked like a giant transmitter.

K9 What does Louvain optimise, and why remove the market mode first?

Modularity: links inside groups minus what chance would give. Without removing the market factor every coin is strongly tied to every other and Louvain finds one big group.

K10 Why the distance $\sqrt{(2(1 - \rho))}$ for the MST?

It turns a correlation into a proper distance (Mantegna, 1999): 0 for identical coins, 1.41 for unrelated ones, 2 for opposites, and it obeys the triangle rule, which a tree algorithm needs.

K11 Why is PageRank run on a "depends on" network?

With the arrow pointing from the driven coin to the driving coin, importance flows to the coins that others depend on. With the arrows the other way it would collect at the receivers.

K12 What is the "shrinkage" in your independent cascade?

Each pair's chance is pulled towards the receiver's normal shock rate, as if we had 10 extra observations at that rate: (hits + 10 × base rate) ÷ (tries + 10). Pairs with little data get sensible numbers instead of 0% or 100%.

K13 What is a GCN, and where did you use one?

A graph convolution: each node takes a weighted average of its neighbours' numbers and its own, then applies a learned transformation. Two GCN layers are the first part of the T-GAE.

K14 In integrated gradients, what is the baseline and what is completeness?

The baseline is the "average situation", the training mean of every input. Completeness means the attributions add up exactly to the score at the real input minus the score at the baseline; we check it every run and the largest gap was 0.009.

K15 What is the characterization score?

One number for how faithful an event explanation is: it combines "the chosen events alone keep the prediction" and "removing them changes it" (a harmonic mean of fidelity+ and 1 − fidelity−). 1 would be perfect; TempME reached 0.157 against 0.152 for random choice.

K16 FGSM against PGD?

FGSM takes one step of size ε in the most harmful direction. PGD takes 10 smaller steps from a random start, staying within ε, so it should find worse inputs. Here it was within 0.002 of FGSM, which means the model behaves almost like a straight line at these sizes and FGSM was already near the worst case.

K17 Why was int8 quantisation slower, not faster?

Most of the time goes into looking up each coin's recent events and updating memories, not into multiplying numbers; int8 only speeds up the multiplications and adds conversion work. So it is 3.1 times smaller but 1.8 times slower, and we serve the full model (about 6 microseconds per event).

K18 Spearman or Pearson: when do you use which?

Pearson for returns, where the values matter. Spearman, which uses only the order, for comparing rankings (spillover, PageRank, explanations) and for checking redundant features, because it ignores extreme values and scale.

K19 Why rank-average models in the ensemble?

Different models' chances are not on the same scale (boosting's scores are spread differently from the TGN's). Ranks put them on one scale; the weights are then chosen per window on validation only.

K20 What exactly is HAR-RV?

A straight-line formula (Corsi, 2009): future volatility = a + b × last hour's + c × last day's + d × last week's volatility, fitted by least squares on training data. It mixes short and long memory, which is why it is so hard to beat.

L. The report

L1 What are your research questions, and what did you find?

RQ1, do temporal graph models beat a fitted diffusion model? Yes, at every window. RQ2, do they beat plain machine learning, and does memory matter? Only borderline against boosting at 1 to 3 hours; memory helps only from 6 hours. RQ3, what drives alerts? The receiving coin's current state. RQ4, how long do shocks last and how robust is the model? About half a day; robust to noise and gaps, vulnerable to deliberate attacks.

L2 Explain equation 4.

It builds the message that updates coin i's memory: its memory just before the event (64 numbers), the other coin's memory (64), the event's 5 numbers and a 16-number code of the time since i was last updated, glued together (149 numbers). Equation 5 feeds it to a GRU, which writes the new memory.

L3 Explain equation 3. Why "t + 1"?

The label is 1 if coin j has at least one shock in the hours t + 1 to t + k. Starting at t + 1 means the same hour never counts: a same-hour label would call simultaneous co-movement contagion, which the report explicitly rejects.

L4 What did Forbes and Rigobon argue, and how do you respect it?

That much apparent contagion is just normal interdependence, which looks bigger in volatile times. We define shocks relative to each coin's own recent volatility and only count shocks that come strictly later.

L5 What gap does your work fill?

Finance studies measure how connected coins are; graph-learning studies test models on "will these two interact?". Nobody had a fair hourly test for pairwise contagion, a fair race between temporal graph models and finance, network and machine-learning baselines, and an analysis of what the models learn.

L6 How did you choose the 30 coins? Why is XMR missing?

The first 30 in CoinGecko's market-cap ranking of 27 September 2026 (top 60, no stablecoins or wrapped tokens) with a Binance USDT market, at least 730 days of hourly history and at least 90% test coverage. XMR was delisted by Binance in February 2024. Every exclusion is recorded. The weak spot is survivorship bias, which the report admits.

L7 Why did you not use the official DyGFormer library?

It draws random "negative" examples for link prediction and splits data by fractions, both of which would break our rules (fixed dates, an embargo, the same real events for every model). So we rebuilt a compact version (24,021 parameters) on exactly our events.

L8 What does "memory is updated only after a batch has been scored" mean, and why do batches never split an hour?

The model must answer before it learns from that hour's events; updating first would let it peek at the answers. Keeping each hour in one batch means no event can benefit from another event of the same hour.

L9 The event vector has a "shock flag". Isn't it always 1?

Yes: every event in our stream starts with a shock, so the flag is constant and carries no information. It marks the event type, so the same format could take other kinds of events later; it does no harm. The four informative numbers are the two correlations, the source's return and its volatility.

L10 Explain Table VI. Why list versions you did not use?

It is the full list of everything we scored on the test period, including superseded versions (TGN v1 0.201, no-memory v1 and v2) and rejected ones (TGN refit 0.188). Looking at the test several times risks the "garden of forking paths", quietly keeping whatever scored best. Listing everything, and fixing the main model from validation evidence, makes that visible and checkable.

L11 The no-memory model scored 0.206, 0.195 and 0.208 in three versions. What does that tell you?

That differences of about 0.01 can come from training choices alone (here, version 2 stopped on a noisy early peak, fixed by a minimum of 30 epochs). It is why we use five seeds, a bootstrap, and call the boosting comparison borderline.

L12 Why is accuracy "meaningless" here, and why AUPRC instead of AUROC?

Always saying "no" is 89.5% accurate at 1 hour and catches nothing. With rare positives, AUROC can look good while the alerts are mostly false; precision–recall curves are more informative (Davis and Goadrich, 2006), and AUPRC's random level is simply the positive rate.

L13 Why do bootstrap tests use five-seed ensembles while tables report seed means?

A test compares one prediction per event for each model, so the five seeds are averaged into one ensemble prediction. Tables report the mean score over seeds, the usual way to describe a model's typical performance. The two differ by about 0.001.

L14 What do Figures 4 and 5 show?

Figure 4(a): each model's AUPRC divided by the positive rate, per window, so windows with different base rates can be compared; 4(b): precision of the riskiest 1% against random. Figure 5: the TGN minus each of four rivals with 95% bootstrap intervals; a filled marker means the interval excludes zero.

L15 The report says the TGN's edge is "in overall ranking, not in the very top". Explain.

Across all events the TGN ranks real contagion higher than boosting. But among the top 1% most confident alerts, DyGFormer (43% at 1 hour) and a refitted boosting model (34%) are as precise or more precise than the TGN (32%).

L16 Why call the 13 of 15 crash alerts "an anecdote, not a statistic"?

Fifteen alerts are too few for a reliable rate, and two of the three crashes (LUNA, FTX) lie in the training and validation years, which the model had seen or was tuned on. Only October 2025 is a true test case.

L17 What is survivorship bias in your study?

We chose coins from a 2026 ranking, so coins that collapsed or were delisted before then are missing. The market may look calmer and more connected than it really was for an investor in 2020.

L18 The report says calibration was not checked on the test period. Did you calibrate at all?

Yes, for the dashboard: chances are calibrated on validation data (binned isotonic, at least 500 events per bin, rising with the window). What the report admits is that we did not measure on the test period how well those chances match reality.

L19 Why monitor the USDT peg? (reference 15)

All Binance prices are quoted in USDT, so if USDT itself slips from $1, every price moves for a reason that has nothing to do with the coins. And the USDT premium is itself a crisis signal (Vo, 2026). We flagged the four off-peg days rather than removing them.

L20 What is CryptoGAT, and how is your work different?

A June 2026 paper that predicts crypto prices with graph attention across coins and questions purely time-series models. It predicts prices; we predict pairwise shock contagion hour by hour, with a temporal graph model and a no-peeking event stream.

L21 What is TGAT, and how does it relate to the TGN?

TGAT (Xu and colleagues, ICLR 2020) applies attention over a node's time-stamped neighbours with a learned time encoding. The TGN reuses that attention and time encoding and adds a memory per node updated by a GRU.

L22 What do "197 passed and 5 skipped" and "within 0.0007" mean?

At submission, the automated test suite ran 197 checks successfully and skipped 5 (integration tests that only run when the full built dataset is on the machine). Replaying the saved final TGN on the test events reproduced its stored AUPRC to within 0.0007, so the reported number can be regenerated. (The deployed branch has 210 tests now.)

L23 Table I says 1,463,693 events but your splits add up to 1,463,147. Why?

The first number was counted while the data were built; the stream skips the 546 shocks in the first week of January 2020, because each event's correlation numbers need a full week of history. Both are "about 1.46 million".

L24 Why did you cite so many 2026 Digital Finance papers?

To show the gap in current work: recent studies (Kammoun; Nefzi and colleagues; Zarifian and colleagues; Vo; Mohamad; Teruel-Gutiérrez and Aparicio Serrano; Alamsyah and colleagues) still measure spillovers, connectedness or event responses, some with hourly data, but none predicts for each pair of coins whether a shock will be followed by another within hours.

PART 18

Cheat sheet: numbers to memorise

Read this the morning of the viva.

Topic	Number
Coins	30 model coins (+ LUNC, FTT for case studies)
Period	1 Jan 2020 to 27 Sep 2026, hourly, UTC; 59,088 hours
Splits	train to Jun 2022 · validation Jul 2022 to Dec 2023 · test Jan 2024 to Sep 2026 · 7-day embargo
Files verified	3,049 SHA-256 checksums, 0 failures
Shock	|z| > 2.29 against the previous 168 hours; about 4% of hours
Events	1.46 million: 1,463,693 in the report (361 k · 301 k · 802 k); the stream the models read: 360,590 · 300,998 · 801,559
Positive rate (test)	10.5% · 20.6% · 31.0% · 44.7% · 62.7% (1, 3, 6, 12, 24 h)
Inputs	14 features per coin (6 market + 8 network), 5 per event
TGN size	70,021 parameters: GRU 41,280 · attention 17,984 · head 10,693 · time 64
TGN shape	memory 64 · time code 16 · embedding 64 · 10 recent events · 2 heads × 32 · head 161 → 64 → 5
Training	Adam 0.0005 · weight decay 0.0001 · focal γ 2, α 0.25 · batch 200 · up to 100 epochs · patience 10 · 5 seeds · 30 to 50 min per seed on CPU
TGN paper	Rossi et al., Twitter, June 2020
Other sizes	DyGFormer 24,021 · LSTM 5,665
TGN test AUPRC	**0.205 · 0.352 · 0.464 · 0.588 · 0.739**
Random	0.105 · 0.206 · 0.310 · 0.447 · 0.627
Cascade	0.159 · 0.279 · 0.381 · 0.507 · 0.672
Boosting	0.178 · 0.330 · 0.451 · 0.581 · 0.736
DyGFormer	0.208 · 0.351 · 0.463 · 0.587 · 0.736
AUROC	0.64 to 0.69

Topic	Number
TGN vs cascade	+0.046 at 1 h (0.019 to 0.074); +0.084 at 6 h
Memory	no effect at 1 h; +0.010 to 0.013 at 6 to 24 h
Top 1% alerts	32% · 61% · 76% · 83% · 89% (random 10.5% · 21% · 31% · 45% · 63%)
Market-wide crash, 1 h	TGN 0.245 vs boosting 0.205; isolated 0.114 vs 0.117
Explanations	receiver ~65% · source ~25% · event ~10% · memory ~7%
TempME fidelity	0.157 vs random 0.152 vs recent 0.192 vs best 0.248
Robustness	50% events missing: under 0.5% · noise 0.5 sd: −5.7% · FGSM 0.1 sd: −35%
Persistence	back to normal in 8 to 10 h (median) · 55 to 65% re-shock within 24 h
Market	one factor explains 64% of hourly moves · lag-1 lead-lag 0.035
T2 / T3	LSTM next-hour AUPRC 0.109 (2.8× random) · HAR-RV beats LSTM on volatility
Dashboard	5 tabs · serves seed 0 (test AUPRC 0.203 at 1 h) · bitcoin ETF approval: 3 of top 5 vs 1.6 random at 6 h
Tests	210 automated tests now; 197 passed and 5 skipped at submission (report)

From the report

Topic	Number
Report	9 pages, IEEE style, 19 Oct 2026, 38 references, 10 equations, 6 tables, 7 figures
Coin rules	CoinGecko top 60 on 27 Sep 2026, no stablecoins or wrapped tokens, Binance USDT market, ≥ 730 days of history, ≥ 90% test coverage; XMR out (delisted Feb 2024)
Table I	≈ 1.61 M coin-hours · 1,463,693 events · 795,787 test events scored at 1 h
Message (eq. 4)	64 + 64 + 5 + 16 = 149 numbers
Head (eq. 7)	64 + 64 + 14 + 14 + 5 = 161 → 64 → 5
Focal loss (eq. 8)	γ 2 · α 0.25 for positives, 0.75 for negatives
Tuning (Fig. 3)	v1 0.138 → v2 0.153 · LR halving 0.151 · context 0.151 · strong regularisation 0.143 · small 0.133 · futures 0.131 (boosting 0.140) · DyGFormer 0.133 → 0.142 · boosting 0.153 → 0.155
Table VI	TGN v1 0.201 · v2 0.205 · no memory 0.206 / 0.195 / 0.208 · DyGFormer 0.208 · boosting 0.178 · refits 0.188 / 0.185 · 2 h 0.165
TGN vs boosting	one-sided p 0.028 (1 h) and 0.029 (3 h): borderline
Top 1%, others	DyGFormer 43 · 67 · 80 · 87 · 92% · boosting refit 34% (1 h), 96% (24 h)
Table V	fresh shocks 6,275 · 3,068 · 1,320 · 1,808 · 1,653 · 4,047 (market-wide crash, pump; partial crash, pump; isolated crash, pump)
Reproducibility	197 passed, 5 skipped · final TGN replay within 0.0007 · every run on CPU

From the notebooks

Topic	Number
01 · Data	prices wander 15/20, returns stationary 20/20 · kurtosis 18 · direction −0.024, jumpiness +0.184 · rush hour 2.17×, quiet 0.69× · lead-lag ≤ 0.035 · Epps 0.58 (1 h) to 0.62 (12 h) · threshold 0.75 · contagion jump 2.3× / 2.4× / 2.7×
02 · Network	eigenvalue 19.3 (64%), noise bound 1.08 · TSI 91.6% hourly, 86.0% daily · NET = eigenvector (1.00), daily vs hourly 0.21 · NMI 0.47 (p 0.003) · tree length 0.66 (lowest 0.36) · sectors 9.3% = 9.3%, meme 13.7% · FTX 0 → 18 → 24 coins · cascade 0.159 vs 0.106
03 · Models	seed ensemble +0.001 · DyGFormer tied · boosting train 0.64 vs validation 0.155 · ensemble +0.008 at 24 h · refit TGN −0.017 to −0.040 · lift ratio 1.5 (TGN) vs 1.1 (cascade) · 2 h: 0.168 vs 0.148
04 · Explanations	rebuild 2.4×10^{-7} · completeness 0.009 · attention entropy 1.000 · motifs 83% vs 86% · 65 / 25 / 10% · TempME vs rankings about −0.8 · crashes 13 of 15
05 · Persistence	18,171 fresh shocks · 8 to 10 h · 28 / 87 / 99% within 6 / 24 / 72 h · re-shock 55 to 65% · followers 8.1 vs 4.5 to 5.1 · half-life 2.5 h · LUNA +0.19
06 · Robustness	T-GAE 3,646 parameters, 0.511 vs turbulence 0.893 · stress days 20.7% · 50% missing: under 0.5% · FGSM 0.05 / 0.1 / 0.25: −19 / −35 / −61% · PGD within 0.002 · top driver kept for 97 to 99%

PART 19

Glossary in plain words

Every technical word in this guide, explained in one or two sentences.

Active mask

1 when a coin exists and has data that hour, 0 otherwise; the models ignore the zeros.

ADF test

Tests whether a series wanders off for good (a unit root) or keeps returning to a stable level.

Attention

Giving different weights to different inputs (adding up to 1), so the important ones count more.

AUROC

The chance a real positive is ranked above a real negative; 0.5 is random, 1 is perfect.

Autoencoder

A network trained to rebuild its input; what it rebuilds badly is unusual.

Batch

A group of examples processed together; about 200 events for the TGN.

Beta

How strongly a coin swings when the market swings.

Bootstrap (day-block)

Adam

The rule used to adjust the knobs during training; it picks a step size for each knob on its own.

Assortativity

Whether nodes link mostly to their own kind (positive) or to others (negative).

AUPRC (average precision)

How well a model ranks the real positives at the top; random scores the positive rate.

Autocorrelation (ACF)

How much a series looks like itself a few steps earlier.

Base rate (positive rate)

How often the answer is "yes"; the score of random guessing.

Benjamini–Hochberg

A correction for running many tests, so that few of the results called significant are flukes.

BIC

A score for choosing a model's size: rewards a good fit, charges for every extra number.

Breadth (of a shock)

Rebuilding the test period many times from randomly drawn whole days, to see how much a result could be luck.

Calibration

Making scores mean real chances: a 30% risk should happen about 30% of the time.

Characterization score

One number for how faithful an event explanation is; combines fidelity+ and fidelity−.

Circular-shift null

Fake data made by rotating a series in time, keeping its habits but breaking any timing link.

Contagion

A shock in one coin followed by shocks in other coins within hours.

Data leakage

Future information sneaking into a model's inputs, giving fake good results.

Diversification

Spreading money over many coins so one bad coin hurts less; fails when everything falls together.

DyGFormer

A Transformer model for event streams (NeurIPS 2023); our second-opinion model.

Eigenvalue (market mode)

The share of other coins shocking in the same hour: market-wide (30% or more), partial, or isolated (under 10%).

Candle (OHLCV)

One hour of trading in five numbers: open, high, low, close and volume.

Checksum (SHA-256)

A fingerprint of a file; any change to the file changes the fingerprint.

Confidence interval

A range of believable values for the true answer; if it excludes 0, a difference is reliable.

Correlation

How closely two series move together, from −1 to +1.

Diebold-Yilmaz spillover

A finance method that measures how much of each asset's volatility comes from the others.

Dropout

Switching off random neurons during training so the network does not lean on any one of them.

Early stopping

Stopping training when the validation score stops improving, and keeping the best version.

Embargo

A gap (7 days) between the training, validation and test periods.

How much of all the movement one common direction explains; the biggest is “the whole market” (64%).

Embedding

A short list of numbers that summarises something complicated.

Epoch

One full pass over the training data.

Epps effect

Correlations looking weaker when measured over very short intervals; weak in crypto.

ETF

A fund traded on the stock market that holds an asset, such as Bitcoin since January 2024.

Event

"Coin i shocked at hour t; what about coin j?", one per other coin per shock.

Fat tails (kurtosis)

Extreme moves far more common than a bell curve allows; kurtosis measures it (18 for crypto, 0 for a bell curve).

FGSM

A deliberate attack that nudges inputs in the direction that hurts the model most.

Focal loss

A loss that focuses training on rare, hard examples.

Fresh shock

A shock with no shock of the same coin in the previous 24 hours.

Funding rate

The small fee traders of perpetual futures pay each other to keep the futures price near the real price.

GCN

Graph convolution: each node averages its neighbours' numbers, then applies a learned transformation.

GFEVD

Generalized forecast-error variance decomposition: splits a coin's forecast uncertainty into shares from each coin, order-free.

GNN

A graph neural network: each node updates itself by listening to its neighbours.

Gradient boosting

Many small decision trees added one after another, each fixing the last ones' mistakes.

Granger causality

Graph

Dots (nodes) joined by lines (edges); here, coins and their relationships.

Whether one series' past improves forecasts of another beyond its own past; prediction, not proof of cause.

GRU

A small memory cell with gates that decide what to keep and what to overwrite.

HAR-RV

A simple formula that forecasts volatility from the last hour, day and week.

Huber loss

Squares small errors but counts big ones linearly, so a few crash days cannot dominate training.

Information bottleneck

Explaining with as little as possible: keep the prediction while keeping few inputs.

Interdependence

Coins always moving together anyway, which is not the same as contagion (Forbes and Rigobon).

Jaccard similarity

Overlap of two lists: shared items divided by all items.

Lead-lag

One series moving first and another following later; almost absent between coins at hourly speed.

Lift

A score divided by the random score; "× typical coin" on the dashboard.

Half-life

The time until an extra effect has halved; about 2.5 hours for a shock's extra jumpiness.

Horizon (window)

How far ahead we look: 1, 3, 6, 12 or 24 hours.

Independent cascade

An epidemic-style model: each shocked coin gets one chance to infect each other coin.

Integrated gradients

An explanation method that splits a prediction into each input's push up or down.

Isotonic regression

Fitting an always-rising curve; used to turn scores into calibrated chances.

Jarque–Bera test

Tests whether returns could come from a bell curve, using their lopsidedness and tail weight.

Leverage

Trading with borrowed money, which multiplies both gains and losses.

Liquidation cascade

Forced selling of borrowed-money positions pushing prices down and forcing

more selling, like dominoes.

Liquidity
How easily a coin trades without moving its price; thinly traded coins update late.

Logistic regression
One weighted sum of the inputs squeezed into a probability; the simplest learned classifier.

Logit
The raw score before the sigmoid; 0 means 50%.

Log return
ln(price now ÷ price before); almost the percentage change, and it adds up neatly over time.

Louvain
An algorithm that finds groups (communities) in a network by maximising modularity.

LSTM
A neural network with memory that reads a sequence, such as one coin's past week.

Marchenko–Pastur bound
The biggest eigenvalue pure noise would give; anything above it is real structure (1.08 here).

Market capitalisation
Price × number of coins in existence: a coin's total size.

Memory (TGN)
A 64-number diary per coin, updated after each hour's events.

Message
The 5 numbers describing an event, used to update memory and to predict.

Modularity
How cleanly a network splits into groups; 0 means no group structure.

Motif (temporal)
A small pattern of events in time order, such as a → b then b → c.

MST (minimum spanning tree)
The skeleton of the strongest links that connects all coins without loops.

NMI
Normalised mutual information: how much two groupings agree, from 0 to 1.

Null hypothesis
The boring starting assumption of a test, such as “there is no effect”.

Overfitting
Memorising the training data instead of learning patterns that carry over.

PageRank
Google's importance score for web pages, used here for how important a coin is as a

Parameter
One adjustable knob in a model, learned during training. The TGN has 70,021.

spreader.

PGD

A stronger, step-by-step version of the FGSM attack (10 small steps).

p-value

How surprising the data would be if the null hypothesis were true; small means hard to explain by luck.

Quantisation (int8)

Storing a model's numbers as 8-bit integers instead of 32-bit decimals, to make it smaller.

ReLU

A simple bend in a network: keep positive numbers, turn negatives into zero.

Seasonality (intraday)

Regular patterns by hour of the week, such as busy US mornings and quiet Sunday mornings.

Shock

An hour with a move above 2.29 times the coin's typical hourly move, up or down.

Spearman correlation

Correlation between rankings rather than raw values.

Stablecoin

A coin built to stay worth one US dollar, such as USDT or USDC.

Stationary

Precision / recall

Of the alarms, how many were real / of the real events, how many raised an alarm.

PyTorch / PyTorch Geometric

The deep-learning library we use, and its add-on for graph models.

R^2

The share of the ups and downs a formula explains, from 0 to 1.

Robust scaling

Centre on the median and divide by the interquartile range, fitted on training data only.

Seed

The number that fixes a training run's random starting point.

Sigmoid

Squeezes any number into 0 to 1, read as a probability.

Spillover index (TSI)

The average share of each coin's forecast uncertainty that comes from other coins (91.6% hourly).

Standard deviation

The typical distance from the average; for returns, it is the volatility.

Streamlit

Behaving the same way over time: returns are, prices are not.

Stress day

A day with at least one hour in which 70% or more of active coins shocked.

Systemic risk

The risk that trouble in one place brings down the whole system.

Temporal graph

A graph whose lines appear at specific times, so order matters.

TGAT

Temporal graph attention (2020): attention over time-stamped neighbours with a time code; the TGN reuses it.

Transformer

A model built on attention (2017); the basis of DyGFormer and of language models.

USDT

A coin designed to stay worth one US dollar; Binance prices are quoted in it.

VAR

Vector autoregression: predicts each coin from the recent past of all coins.

z-score

How many "normal moves" this hour's move is; above 2.29 counts as a shock.

A Python library that turns a script into an interactive web page.

Survivorship bias

Studying only today's survivors, which hides the coins that died along the way.

TempME

An explainer (NeurIPS 2023) that looks for chains of past events behind a prediction.

T-GAE

Temporal graph autoencoder: learns what a normal day's network looks like and flags surprising days; 3,646 parameters.

TGN

Temporal Graph Network (Twitter, 2020): memory per node, attention over recent events, a prediction head.

Turbulence

A day's average |z| across coins: how wild the market was.

Validation set

The "mock exam" years used to choose settings, kept apart from the final test years.

Volatility

How jumpy a price is; we measure it from recent hourly returns.

LAST PART

If they ask how you built it

How to talk about AI help without losing marks or credibility.

Your report already states it: an AI assistant (Claude) helped with code development, cross-checking the reported numbers against the run logs, re-running the tests and the final model, and drafting the report and figures, and the authors are responsible for the content. Professors generally accept tool use when it is disclosed and the students clearly understand the work. What loses marks is being unable to explain it, or claiming otherwise.

- **Be plain about it.** "We used an AI coding assistant, as the report says. We set the questions and the rules, chose between options, checked the results, and we can explain every part."
- **Show ownership through understanding.** Draw the TGN diagram, rebuild the 70,021 parameters, and explain the leakage rules and the weak spots without notes. That is what this guide is for.
- **Know your decisions:** hourly instead of daily (after the professor's review), a strictly-later label, fixed date splits, AUPRC instead of accuracy, five seeds and a day-block bootstrap, keeping the train-only TGN after the refit was worse, reporting negative results.
- **Agree the split of work with your teammate** before the viva, so your answers match (see question I4).

CryptoGraph Viva Guide · prepared on 27 October 2026 (words, notebooks, techniques and report parts added 28 October) from the project's code, configuration files, run registry and CA3 report. Keep it out of the repository's commits, like the earlier viva guide.